Table 1. Metadata for acoustical monitoring sites at Gates of the Arctic National Park and Preserve (GAAR). Dates reflect each site's recorded NVSPL deployment period(s).

Site	Site Name	Dates	Vegetation	Wilderness	Elevation (m)	Latitude	Longitude
GAARAMLA	NA		NA	NA	NA	NA	NA
GAARANAK	NA		NA	NA	NA	NA	NA
GAARARRI	NA		NA	NA	NA	NA	NA
GAARFLOR	NA		NA	NA	NA	NA	NA
GAARHAMM	NA		NA	NA	NA	NA	NA
GAARITKI	NA		NA	NA	NA	NA	NA
GAARKURU	NA		NA	NA	NA	NA	NA
GAARMWAL	NA		NA	NA	NA	NA	NA
GAARNARV	NA		NA	NA	NA	NA	NA
GAARNNUT	NA		NA	NA	NA	NA	NA
GAARNUTU	NA		NA	NA	NA	NA	NA
GAARNWAL	NA		NA	NA	NA	NA	NA
GAARPALA	NA		NA	NA	NA	NA	NA
GAARSELB	NA		NA	NA	NA	NA	NA
GAARUPNO	NA		NA	NA	NA	NA	NA
GAARWALK	NA		NA	NA	NA	NA	NA

Table 2. Summary of sound levels at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2014. For each site, includes the 90th percentile (LA10), existing (LA50), natural (LAnat), and 10th percentile (LA90) ambient sound levels (dB re 20 uPa, A−weighted broadband, 12.5−20,000 Hz) during the day (0700−1900) and night (1900−0700).

Time	SiteID	LA10	LA50	LAnat	LA90
Day	GAARMWAL	36.9	30.0	29.8	26.9
Night	GAARMWAL	35.3	30.0	30.0	28.1
Day	GAARNARV	33.6	30.6	30.6	28.9
Night	GAARNARV	34.3	31.5	31.5	29.7
Day	GAARNNUT	28.5	22.0	21.9	19.3
Night	GAARNNUT	27.1	21.2	21.1	19.2
Day	GAARNUTU	39.0	36.8	NA	36.2
Night	GAARNUTU	39.1	37.0	NA	36.6
Day	GAARNWAL	37.5	34.7	34.6	31.7
Night	GAARNWAL	39.7	37.6	37.6	35.5
Day	GAARSELB	33.9	31.3	31.2	29.5
Night	GAARSELB	34.1	31.8	31.8	30.1

Table 3. Number of days of audio analyzed for the Listening Center and SPLAT detection methods at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2014 − the basis for the analysis−period text reported in the Methods section.

Type	DaysText
ListeningCenter	
SPLAT	19−34

Table 4. Executive summary for Gates of the Arctic National Park and Preserve (GAAR), Summer 2014: mean overall noise audibility, top contributing noise sources, and ambient sound levels (LA50/LAnat, day and night) by site.

SiteID	Noise	Aircraft	Watercraft	NA 4	LA50 Day	LAnat Day	LA50 Night	LAnat Night
GAARMWAL	1.5	1.5	NA	NA	30.0	29.8	30.0	30.0
GAARNARV	0.7	0.7	NA	NA	30.6	30.6	31.5	31.5
GAARNNUT	2.7	2.1	0.6	NA	22.0	21.9	21.2	21.1
GAARNUTU	NA	NA	NA	NA	36.8	NA	37.0	NA
GAARNWAL	0.5	0.5	NA	NA	34.7	34.6	37.6	37.6
GAARSELB	1.6	1.5	NA	NA	31.3	31.2	31.8	31.8

Table 5. Noise impact (dB, existing minus natural ambient) and Listening Area Reduction (%) by site at Gates of the Arctic National Park and Preserve (GAAR), Summer 2014, for day, night, and the combined 24−hour period.

SiteID	DayImpact	DayLAR	NightImpact	NightLAR	AllImpact	AllLAR
GAARMWAL	0.2	4.5	0.0	0.0	0.1	2.3
GAARNARV	0.0	0.0	0.0	0.0	0.0	0.0
GAARNNUT	0.1	2.3	0.1	2.3	0.1	2.3
GAARNUTU	NA	NA	NA	NA	NA	NA
GAARNWAL	0.1	2.3	0.0	0.0	0.1	1.1
GAARSELB	0.1	2.3	0.0	0.0	0.1	1.1

Table 6. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2014.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARMWAL	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.1	0.1	0.0	0.0	0.0
GAARMWAL	1.1	1	Aircraft	1.1	Jet	0.2	0.9	1.2	191.8	217.9	212.5
GAARMWAL	1.2	1	Aircraft	1.2	Prop	2.0	0.2	2.2	398.8	426.5	401.3
GAARMWAL	1.3	1	Aircraft	1.3	Helicopter	0.6	0.1	0.6	291.1	164.0	281.3

Table 7. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2014.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARNARV	1.1	1	Aircraft	1.1	Jet	0.7	1.3	2.0	194.4	197.3	196.3
GAARNARV	1.2	1	Aircraft	1.2	Prop	0.6	0.2	0.8	292.2	162.3	266.3

Table 8. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2014.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARNNUT	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.1	0.1	0.0	0.0	0.0
GAARNNUT	1.1	1	Aircraft	1.1	Jet	1.0	0.8	1.8	259.3	257.6	258.5
GAARNNUT	1.2	1	Aircraft	1.2	Prop	2.0	0.1	2.1	448.3	365.2	443.7
GAARNNUT	1.3	1	Aircraft	1.3	Helicopter	0.8	0.0	0.8	532.1	160.0	518.8
GAARNNUT	3.0	3	Watercraft	3.0	Watercraft	1.2	0.7	1.9	305.9	223.9	275.6

Table 9. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2014.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARNWAL	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.1	0.1	0.0	0.0	0.0
GAARNWAL	1.1	1	Aircraft	1.1	Jet	0.4	0.4	0.8	192.8	175.2	184.0
GAARNWAL	1.2	1	Aircraft	1.2	Prop	0.6	0.1	0.8	300.7	338.2	307.3
GAARNWAL	1.3	1	Aircraft	1.3	Helicopter	0.1	0.1	0.2	352.5	264.7	299.8

Table 10. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2014.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARSELB	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.1	0.1	0.0	0.0	0.0
GAARSELB	1.1	1	Aircraft	1.1	Jet	0.8	0.8	1.6	218.7	192.2	205.2
GAARSELB	1.2	1	Aircraft	1.2	Prop	3.8	0.1	3.9	259.4	207.0	257.8
GAARSELB	1.3	1	Aircraft	1.3	Helicopter	0.1	0.0	0.1	99.0	509.0	235.7

Table 11. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2014.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3600	476	302	3600	494	1580	822	1438	2818	383	701	386	587	336	107	430	1003	1006	2627	712	3600	3331	2671	3600	GAARMWAL
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	2693	3600	3600	3600	3600	3600	3600	3600	3600	GAARMWAL
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARMWAL
Mean	3600	2507	2331	3600	2401	2991	2555	3086	3297	2115	2492	1960	2142	2114	1381	1834	2802	2524	3304	2438	3600	3059	3307	3600	GAARMWAL

Table 12. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2014.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3355	491	1249	2088	701	587	1260	393	422	2115	2259	3600	797	609	2187	899	2290	1305	3600	1659	3600	3428	1151	3600	GAARNARV
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARNARV
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARNARV
Mean	2949	2366	2868	3343	2727	2472	2928	2320	2095	3333	3320	3600	2612	2320	3333	2778	3333	3044	3600	3113	3596	2870	3015	3600	GAARNARV

Table 13. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2014.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	269	696	1124	3600	577	3600	1747	1221	601	111	163	313	541	146	465	353	208	236	207	371	329	3030	305	1668	GAARNNUT
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3271	2153	2768	3600	2722	3600	3185	3600	3507	2306	3600	3600	3600	3600	3600	GAARNNUT
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARNNUT
Mean	2263	2837	2875	3600	2518	3594	3064	2767	2235	1452	1269	1568	2514	1401	1945	1682	1669	1778	1435	2040	2187	3218	2189	3122	GAARNNUT

Table 14. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2014.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3222	3600	3394	3600	730	3600	3600	1714	3600	3600	212	3600	2125	1608	1276	1648	3600	1539	3051	1078	1555	3600	3600	3600	GAARNWAL
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARNWAL
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARNWAL
Mean	3018	3600	3411	3600	2621	3257	3423	3138	3402	3426	2015	3348	2966	2923	2658	2847	3383	3073	2906	2742	2917	3154	3600	3600	GAARNWAL

Table 15. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2014.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	705	831	1326	3600	798	3600	3600	1971	497	788	448	256	364	184	362	275	357	306	269	1355	2190	3600	3600	3600	GAARSELB
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	2341	2690	2468	3600	3186	2816	3600	3270	3159	3600	3600	3600	3600	3600	GAARSELB
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARSELB
Mean	2772	2748	2801	3600	2449	3435	3437	3311	2047	2298	1700	1637	1696	1825	1777	1815	2060	1705	1680	2764	3305	3399	3600	3597	GAARSELB

Table 16. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2014.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARMWAL	No Sound Audible	0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0.0	0.0	0
GAARMWAL	Jet	0	1.4	1.3	0	1.1	0.5	0.3	0.2	0.3	0.0	0.4	0.0	0.0	0.0	0.0	0.2	0.0	0.4	0.0	0.4	0	0.3	0.4	0
GAARMWAL	Prop	0	0.0	0.0	0	0.0	0.0	0.4	0.5	0.0	2.2	2.8	1.5	1.1	4.0	2.7	3.8	1.8	1.4	0.4	2.0	0	0.0	0.0	0
GAARMWAL	Helicopter	0	0.0	0.0	0	0.0	0.0	0.2	0.0	0.0	0.3	0.0	1.5	1.9	0.4	0.7	0.0	0.0	0.0	0.0	0.0	0	0.0	0.0	0

Table 17. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2014.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARNARV	Jet	0.3	1.8	1.2	0.2	0.8	1.1	0.5	0.6	1.0	0.3	0.4	0	0.2	0.3	0.0	0.4	0.2	0.5	0	0.2	0.0	0.4	0.3	0
GAARNARV	Prop	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	1.5	0.0	0.0	0	1.8	1.1	0.3	0.0	0.0	0.0	0	0.3	0.1	0.0	0.3	0

Table 18. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2014.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARNNUT	No Sound Audible	0.0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
GAARNNUT	Jet	1.2	0.8	1.2	0	1.3	0.2	0.5	0.8	2.3	0.8	0.7	0.9	0.0	0.0	0.3	0.2	0.0	0.5	0.3	0.8	0.0	0.0	0.0	0.0
GAARNNUT	Prop	0.0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.9	3.7	3.8	1.7	3.5	2.1	3.5	2.3	2.5	1.1	0.9	0.3	0.0	0.0	0.0
GAARNNUT	Helicopter	0.0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.0	2.3	2.1	0.4	0.7	0.7	0.7	1.1	1.4	0.3	2.2	0.0	0.1	0.0	0.0	0.0
GAARNNUT	Watercraft	0.4	0.1	0.0	0	0.0	0.0	0.0	0.0	0.0	0.8	1.1	1.3	0.4	1.4	1.3	0.5	0.5	0.8	2.2	1.7	1.2	0.2	0.6	0.2

Table 19. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2014.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARNWAL	No Sound Audible	0.0	0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
GAARNWAL	Jet	0.3	0	0.4	0	0.8	0.1	0.2	0.5	0.2	0.2	0.4	0.0	0.0	0.0	0.0	0.3	0.0	0.4	0.2	0.2	0.0	0.0	0	0
GAARNWAL	Prop	0.0	0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	1.6	0.0	0.6	0.6	1.5	0.1	0.2	0.3	0.4	0.5	0.8	0.0	0	0
GAARNWAL	Helicopter	0.0	0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.3	0	0

Table 20. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2014.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARSELB	No Sound Audible	0.0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0.0
GAARSELB	Jet	0.8	1.1	0.8	0	0.8	0.2	0.1	0.3	1.3	0.7	0.2	0.5	0.0	0.0	0.4	0.3	0.0	0.4	0.4	0.4	0.0	0.0	0	0.1
GAARSELB	Prop	0.0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.5	1.6	4.7	3.2	3.8	1.5	1.7	3.1	1.8	2.4	2.7	0.3	0.3	0.1	0	0.0
GAARSELB	Helicopter	0.0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.0	0.0	0	0.0

Table 21. Percent time above sound level thresholds for acoustic records collected at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2014. Values show the percent of time ANS−weighted sound levels (20−1,250 Hz) and the full measured frequency range (12.5−20,000 Hz) exceeded 35, 45, 52, and 60 dB (LAeq,1s), by site and day/night.

Site ID	Frequency (Hz)	35 dB Day	45 dB Day	52 dB Day	60 dB Day	35 dB Night	45 dB Night	52 dB Night	60 dB Night
GAARMWAL	Full (12.5−20,000)	31.0	1.2	0.1	0	18.3	1.0	0.0	0
GAARMWAL	ANS (20−1,250)	15.1	0.2	0.0	0	7.5	0.0	0.0	0
GAARNARV	Full (12.5−20,000)	14.2	0.5	0.1	0	9.9	0.4	0.0	0
GAARNARV	ANS (20−1,250)	4.8	0.1	0.0	0	1.9	0.0	0.0	0
GAARNNUT	Full (12.5−20,000)	7.7	0.5	0.1	0	6.2	0.3	0.0	0
GAARNNUT	ANS (20−1,250)	4.2	0.3	0.0	0	1.4	0.0	0.0	0
GAARNUTU	Full (12.5−20,000)	98.4	1.3	0.2	0	100.0	1.5	0.1	0
GAARNUTU	ANS (20−1,250)	32.0	0.1	0.0	0	51.6	0.0	0.0	0
GAARNWAL	Full (12.5−20,000)	92.2	0.7	0.0	0	82.7	0.1	0.0	0
GAARNWAL	ANS (20−1,250)	91.4	0.2	0.0	0	81.5	0.0	0.0	0
GAARSELB	Full (12.5−20,000)	9.7	0.2	0.0	0	14.7	0.1	0.0	0
GAARSELB	ANS (20−1,250)	5.8	0.1	0.0	0	7.6	0.0	0.0	0

Figure 1. GAARMWAL: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2014.

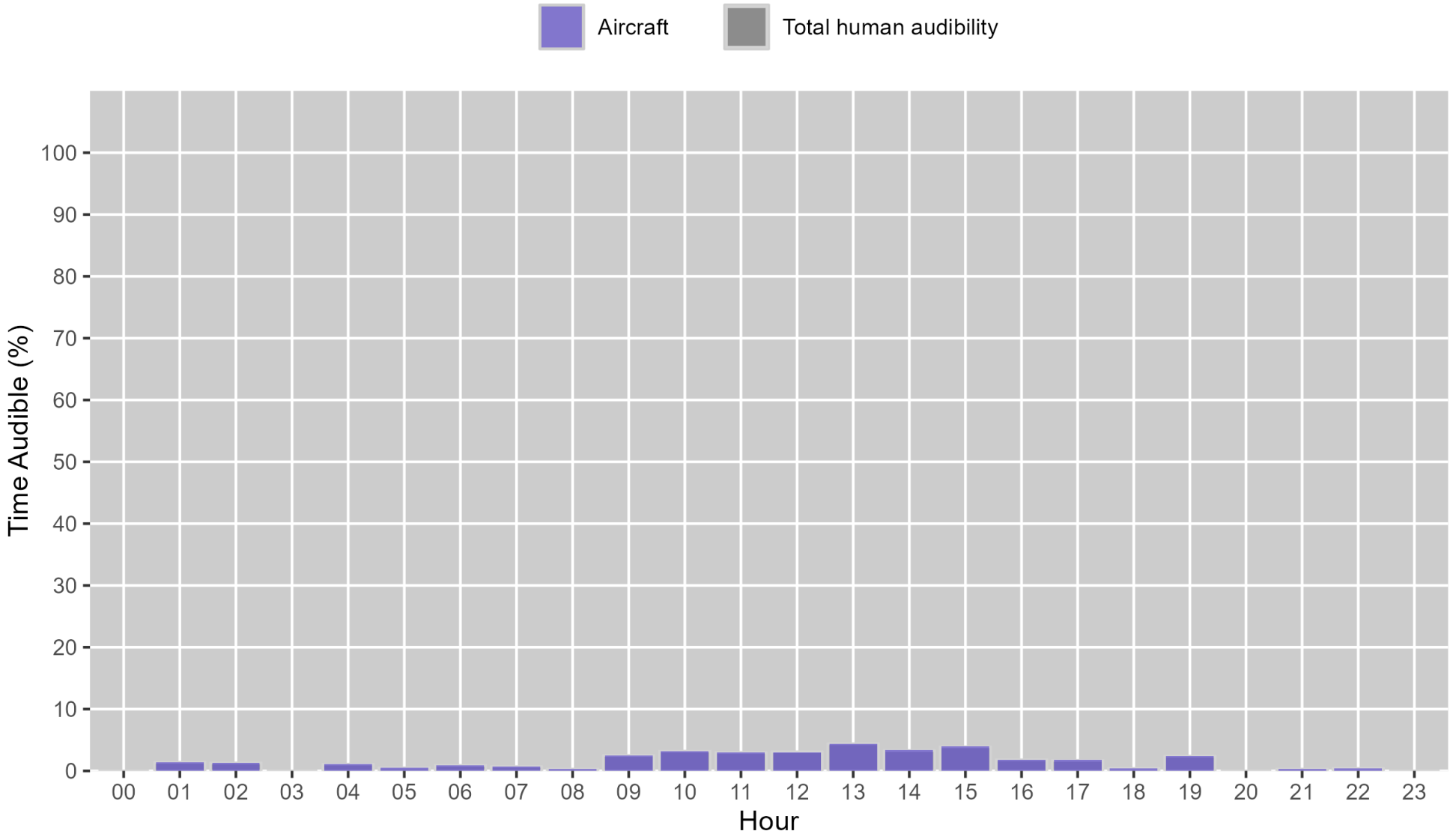


Figure 2. GAARMWAL: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2014.

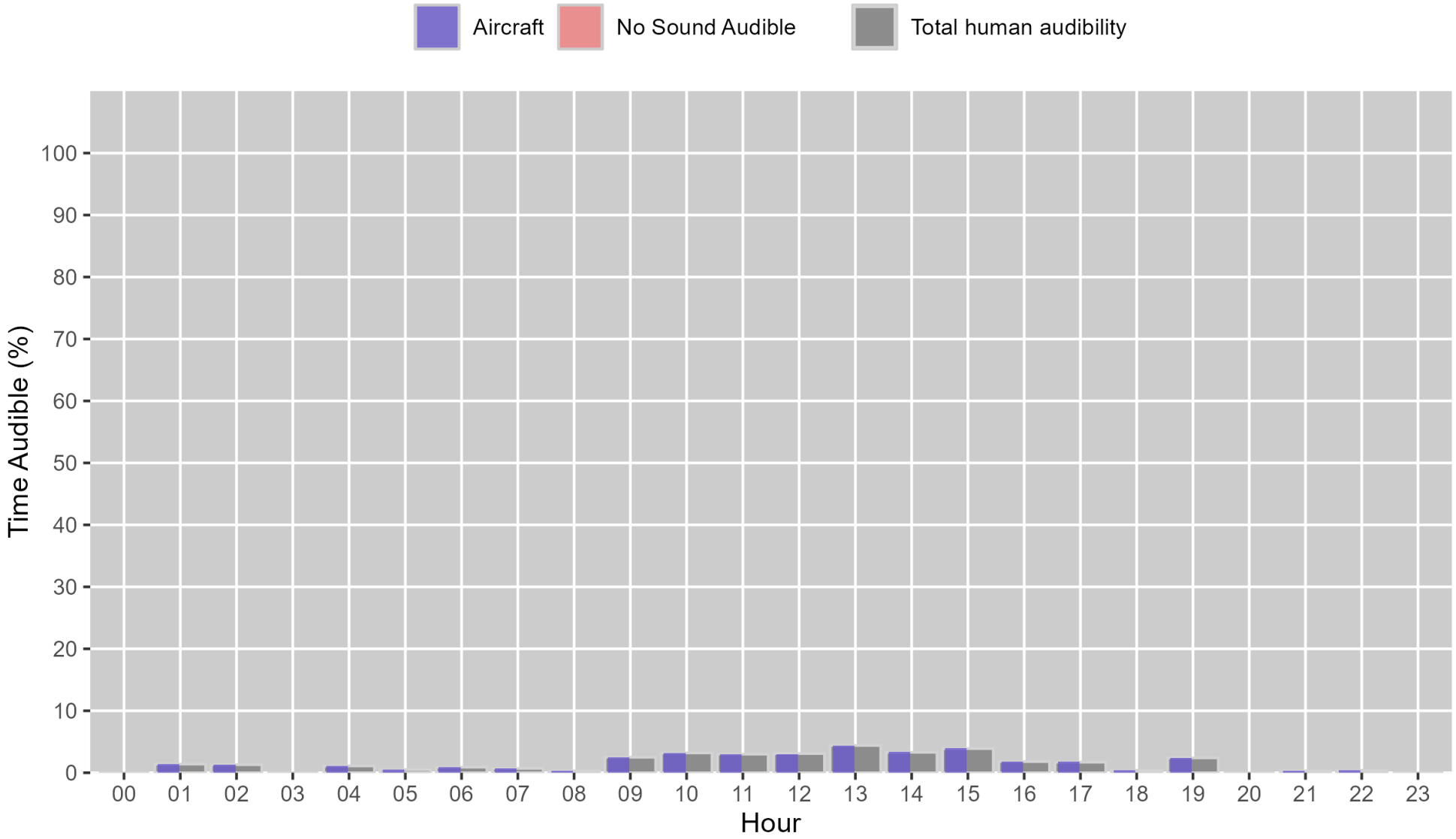


Figure 3. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARMWAL in Summer, 2014, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

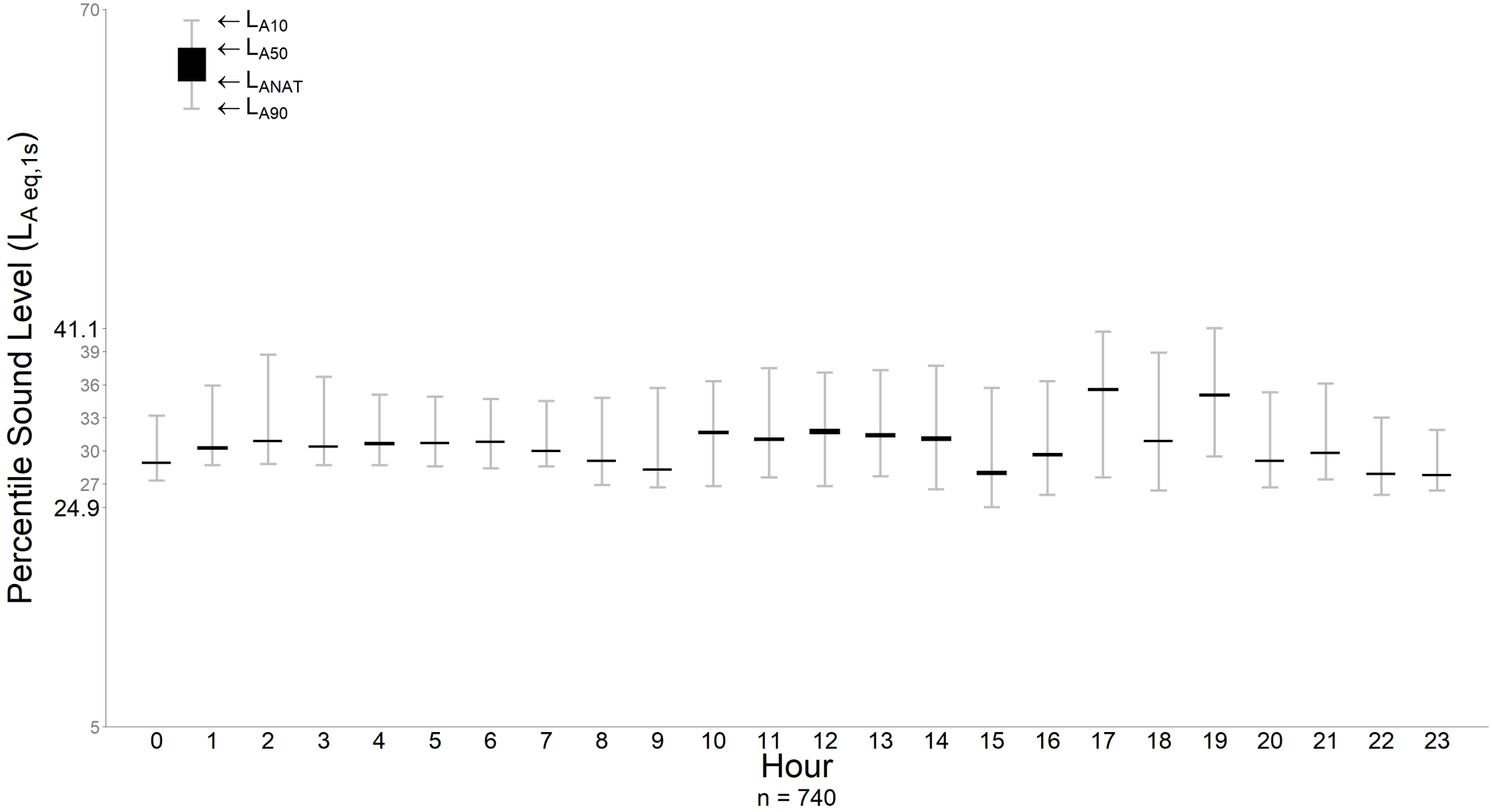


Figure 4. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARMWAL in Summer, 2014.

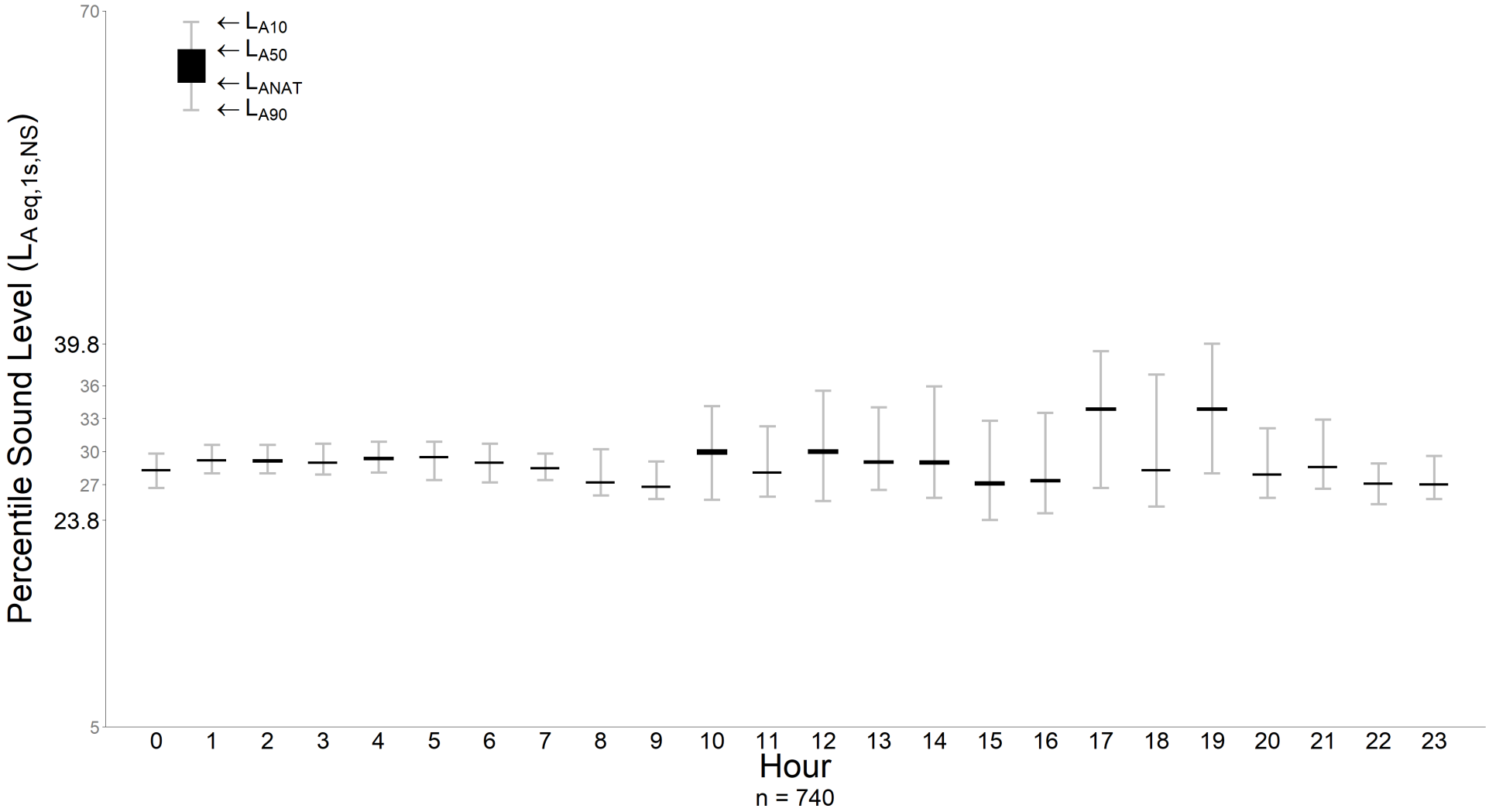


Figure 5. GAARMWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

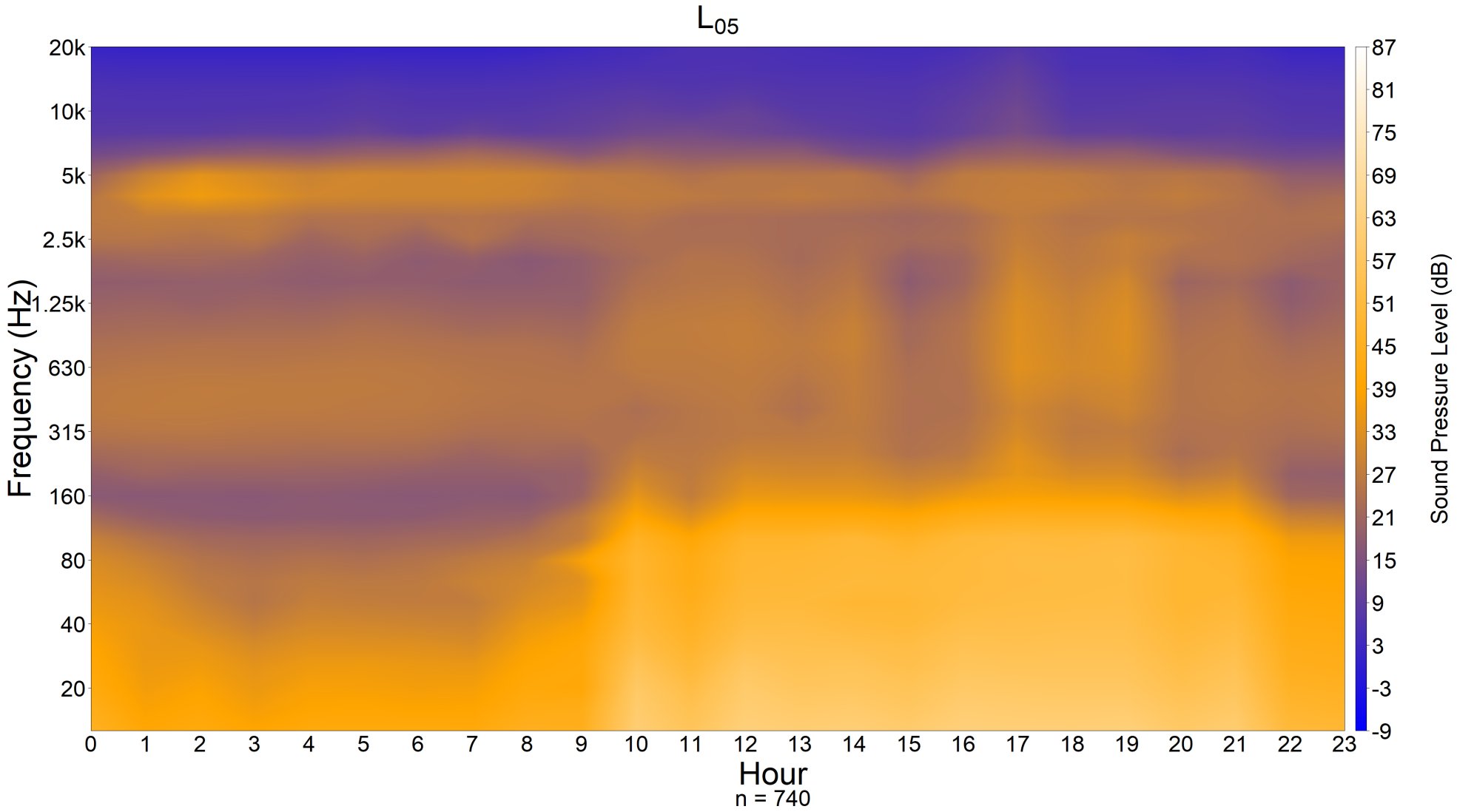


Figure 6. GAARMWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

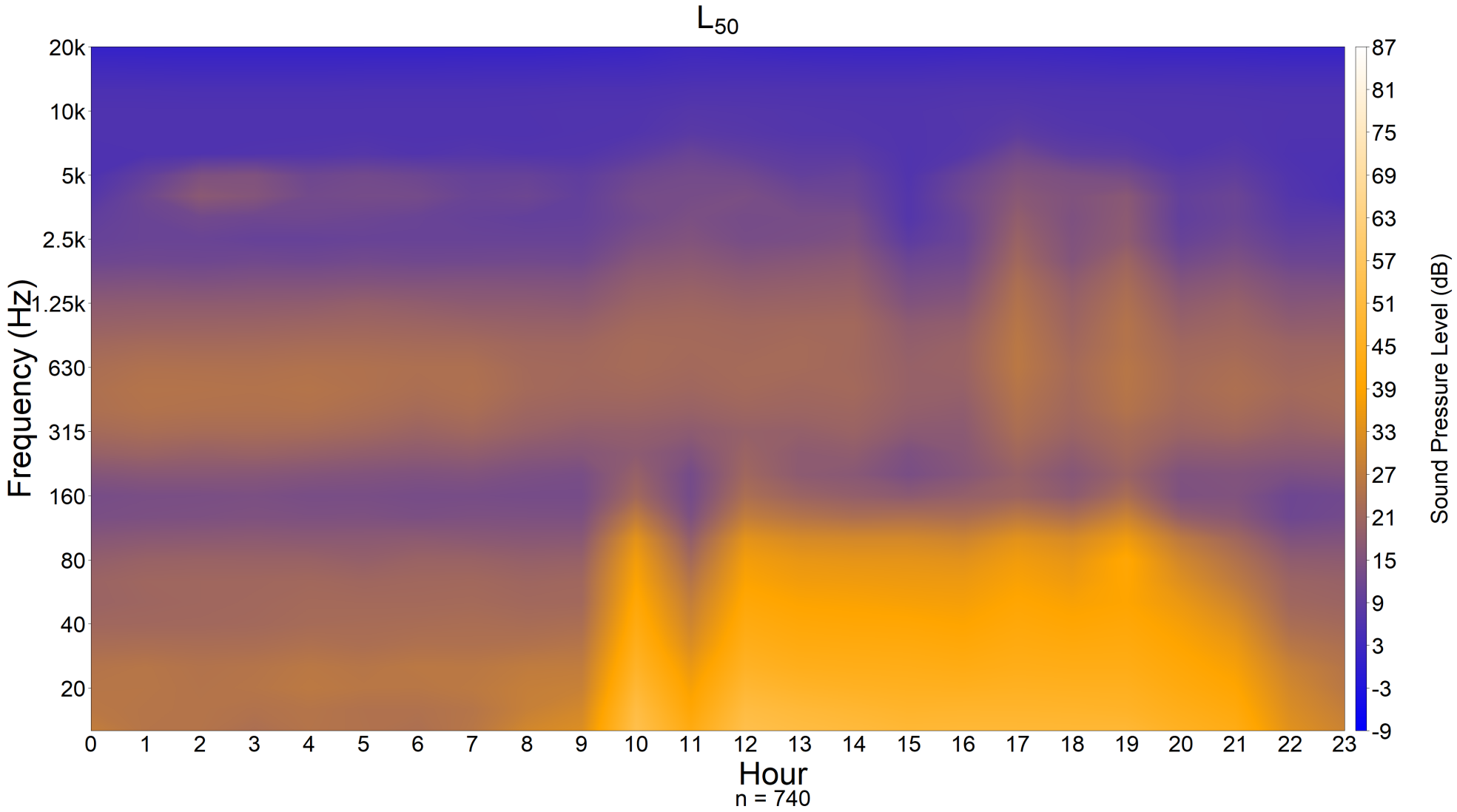


Figure 7. GAARMWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

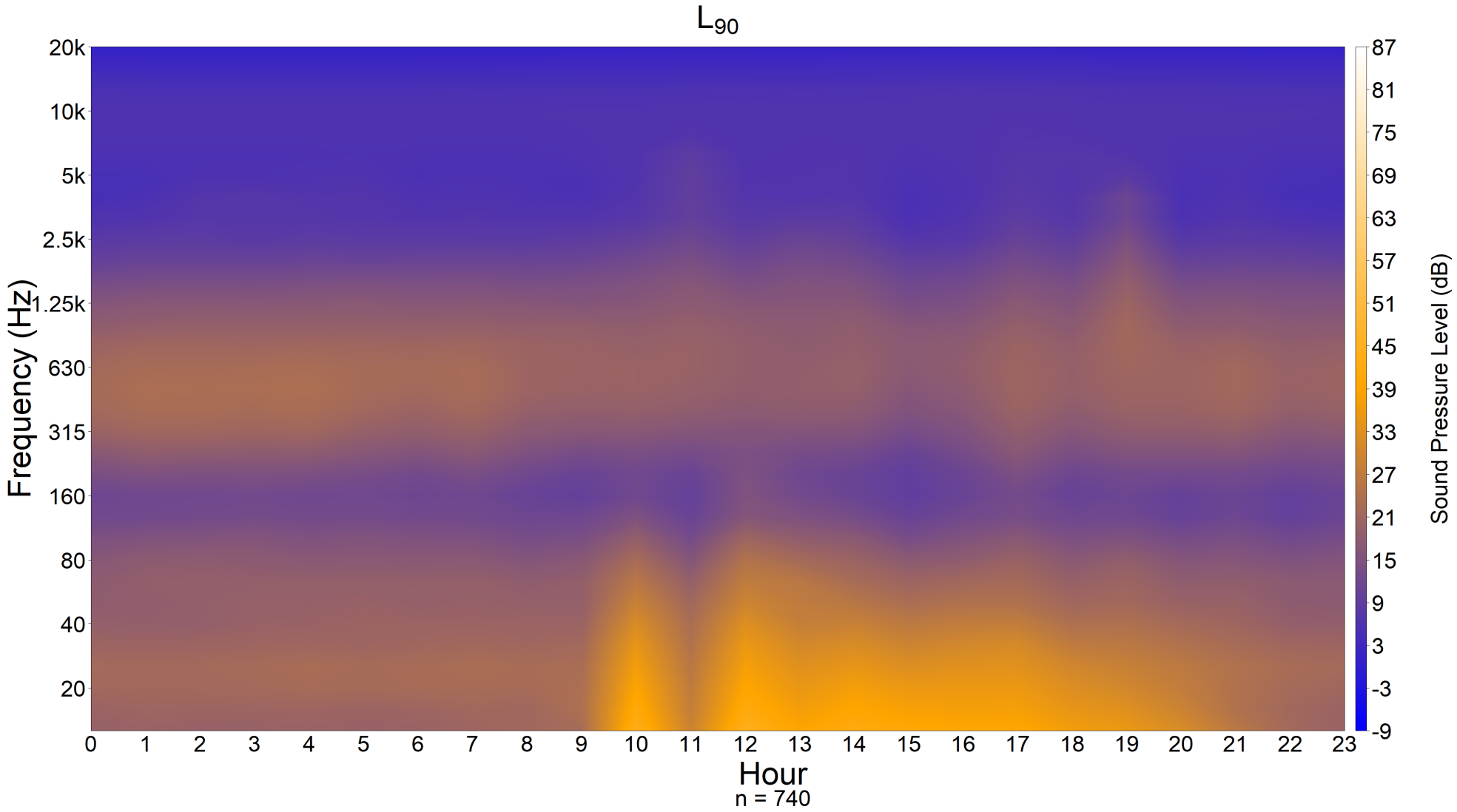


Figure 8. GAARMWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

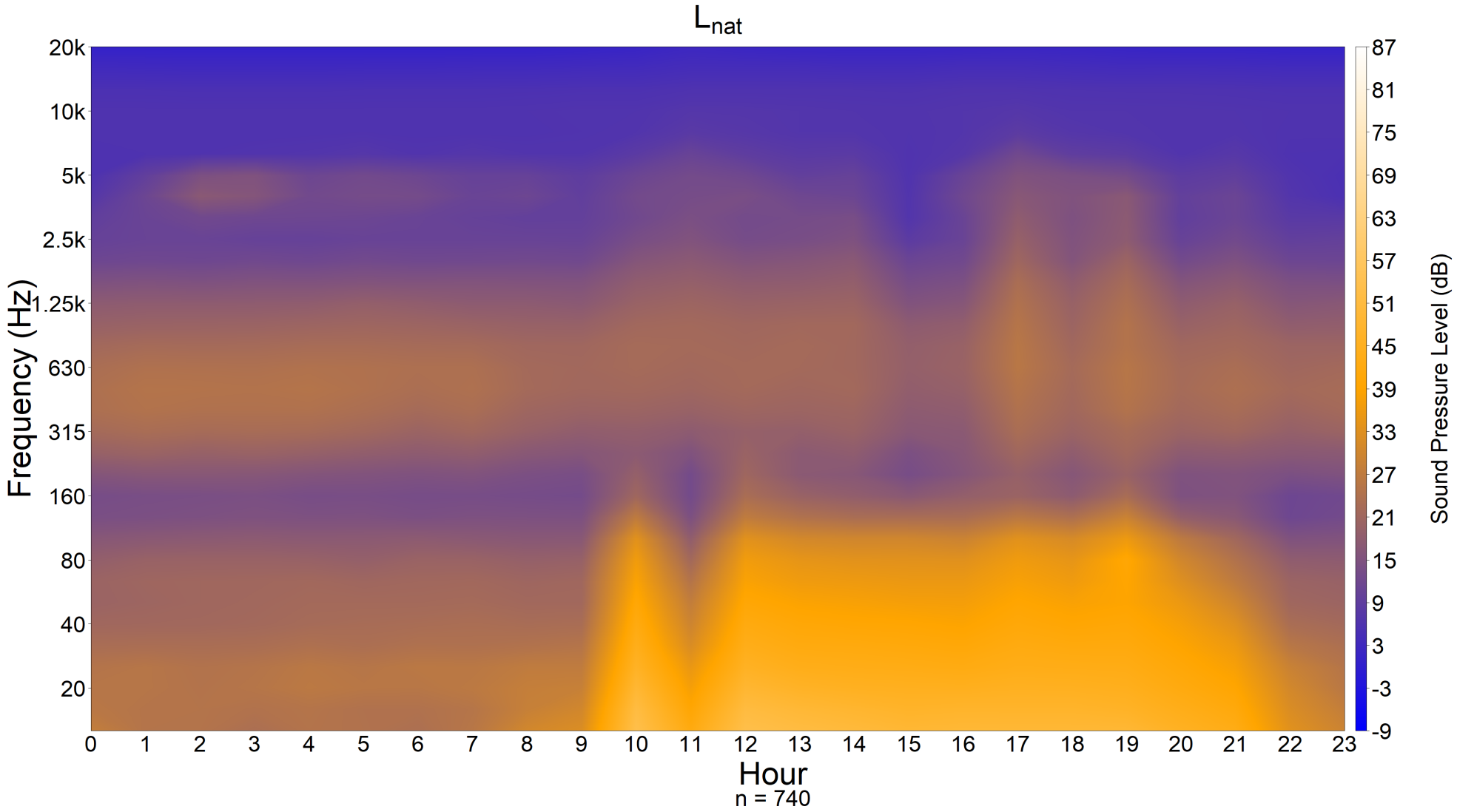


Figure 9. GAARMWAL: median and mean duration of noise−free intervals by hour of day, in Summer 2014.

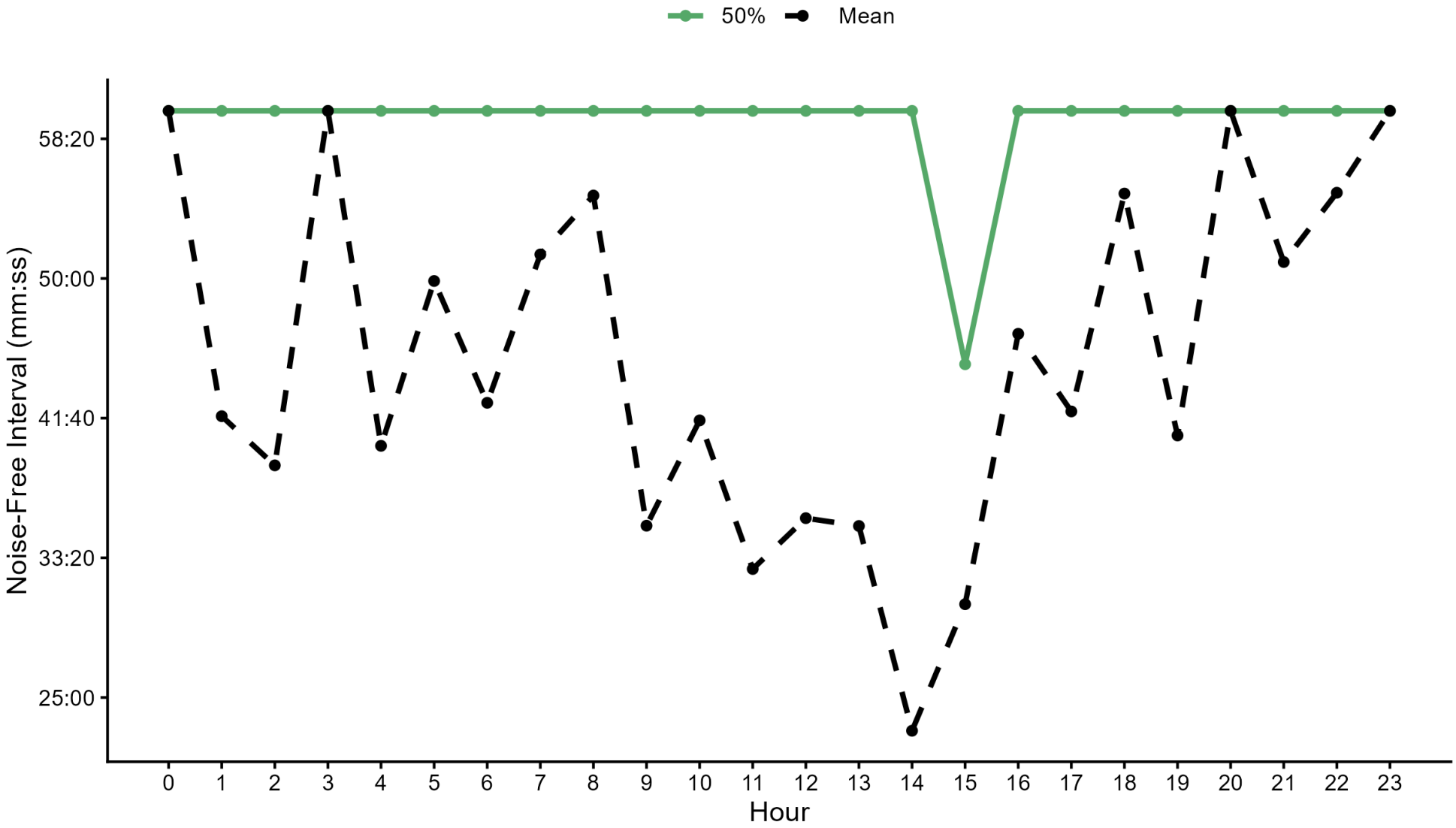


Figure 10. GAARMWAL: percent of time audible by sound source category (noise/natural), in Summer 2014.

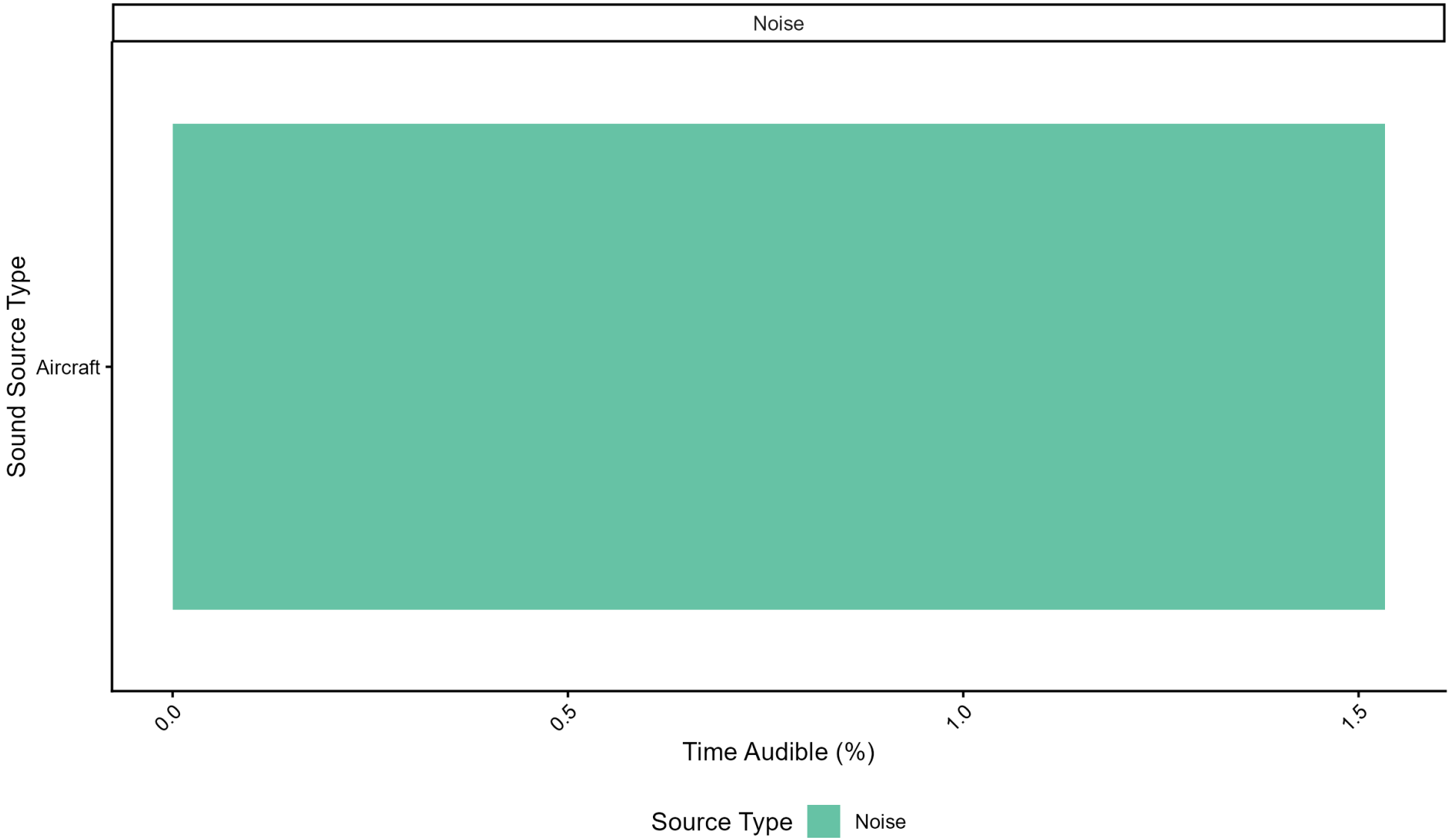


Figure 11. GAARMWAL: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2014.

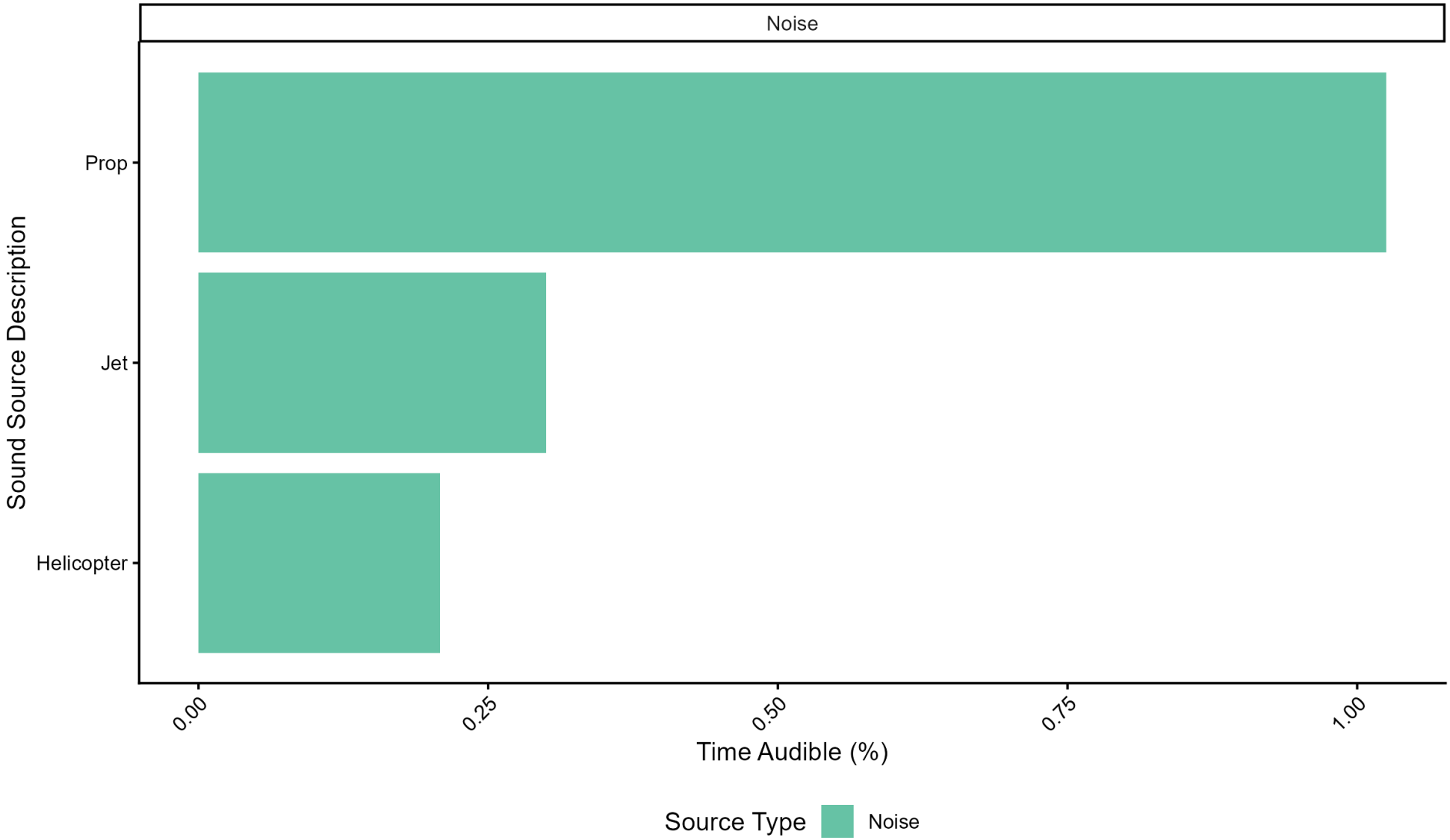


Figure 12. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARMWAL in Summer, 2014. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

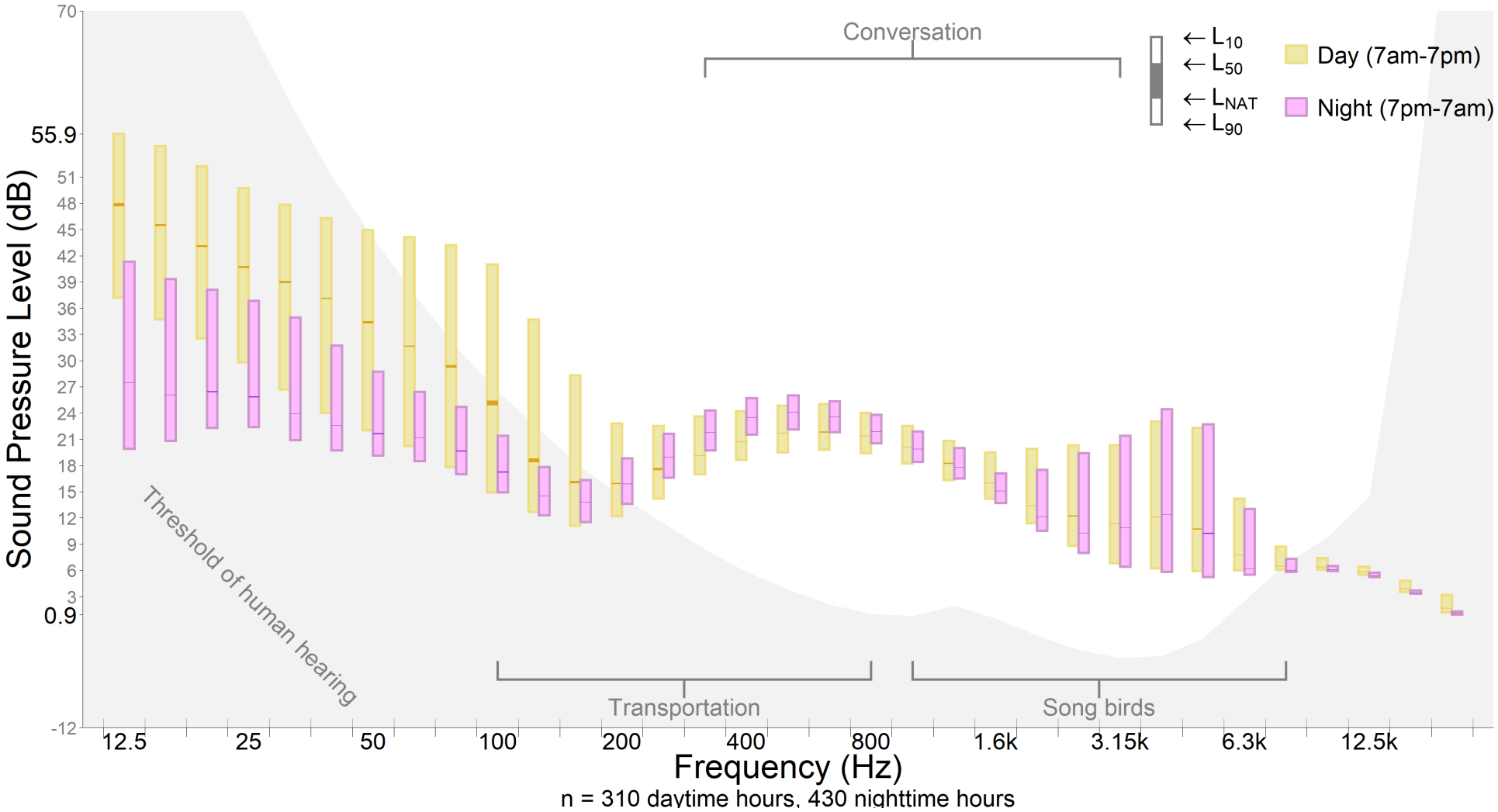


Figure 13. GAARNARV: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2014.

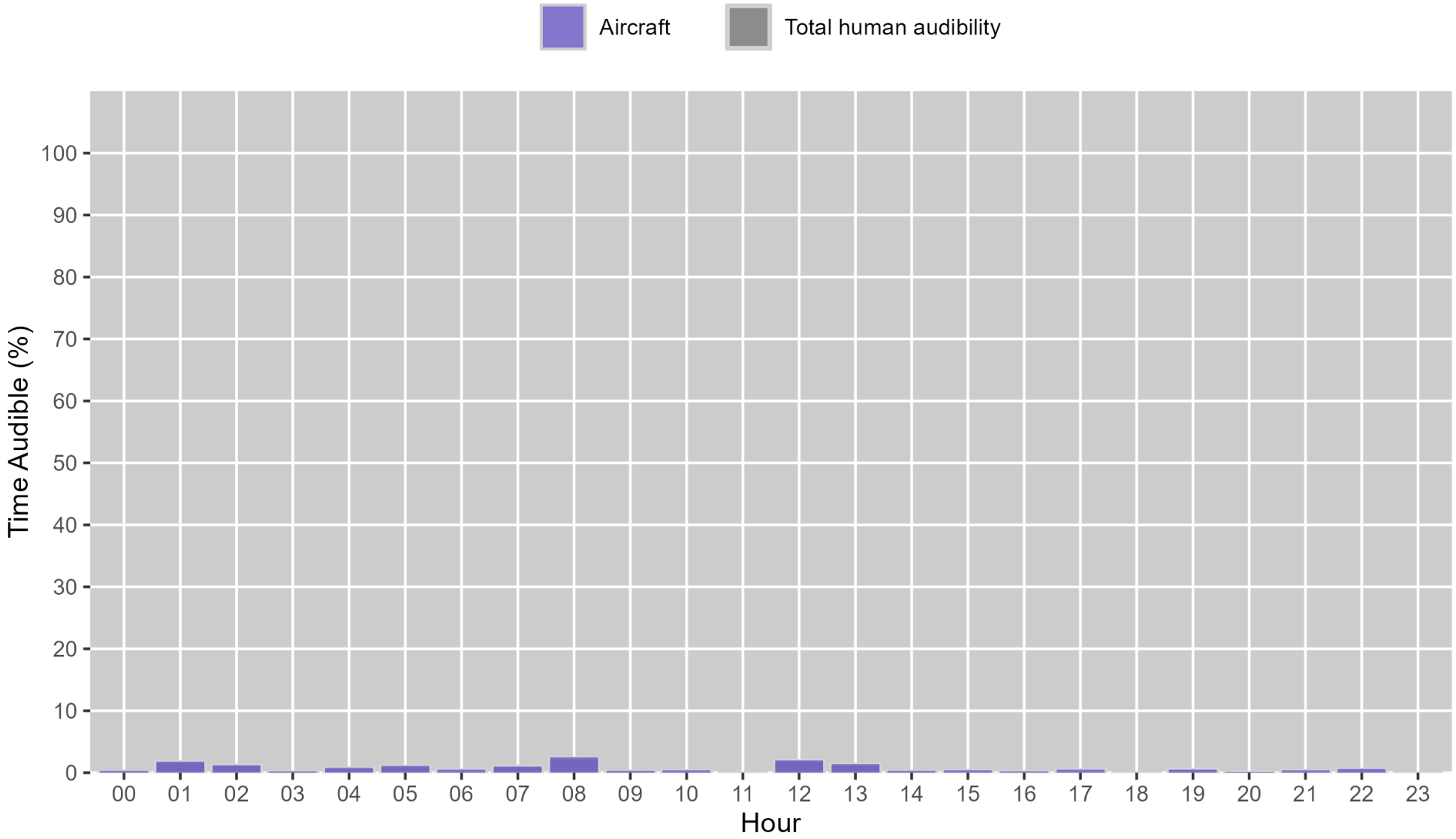


Figure 14. GAARNARV: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2014.

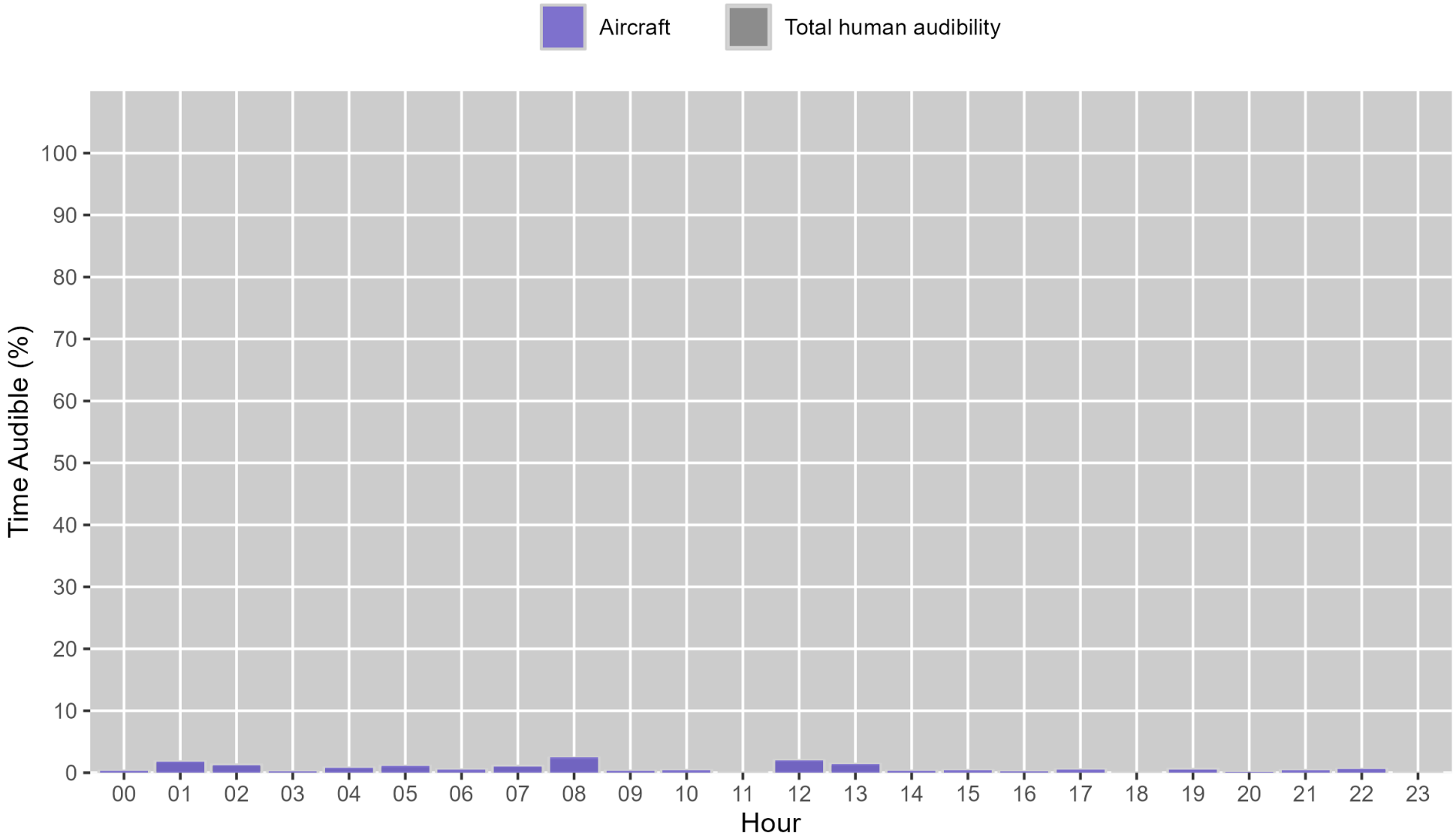


Figure 15. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARNARV in Summer, 2014, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

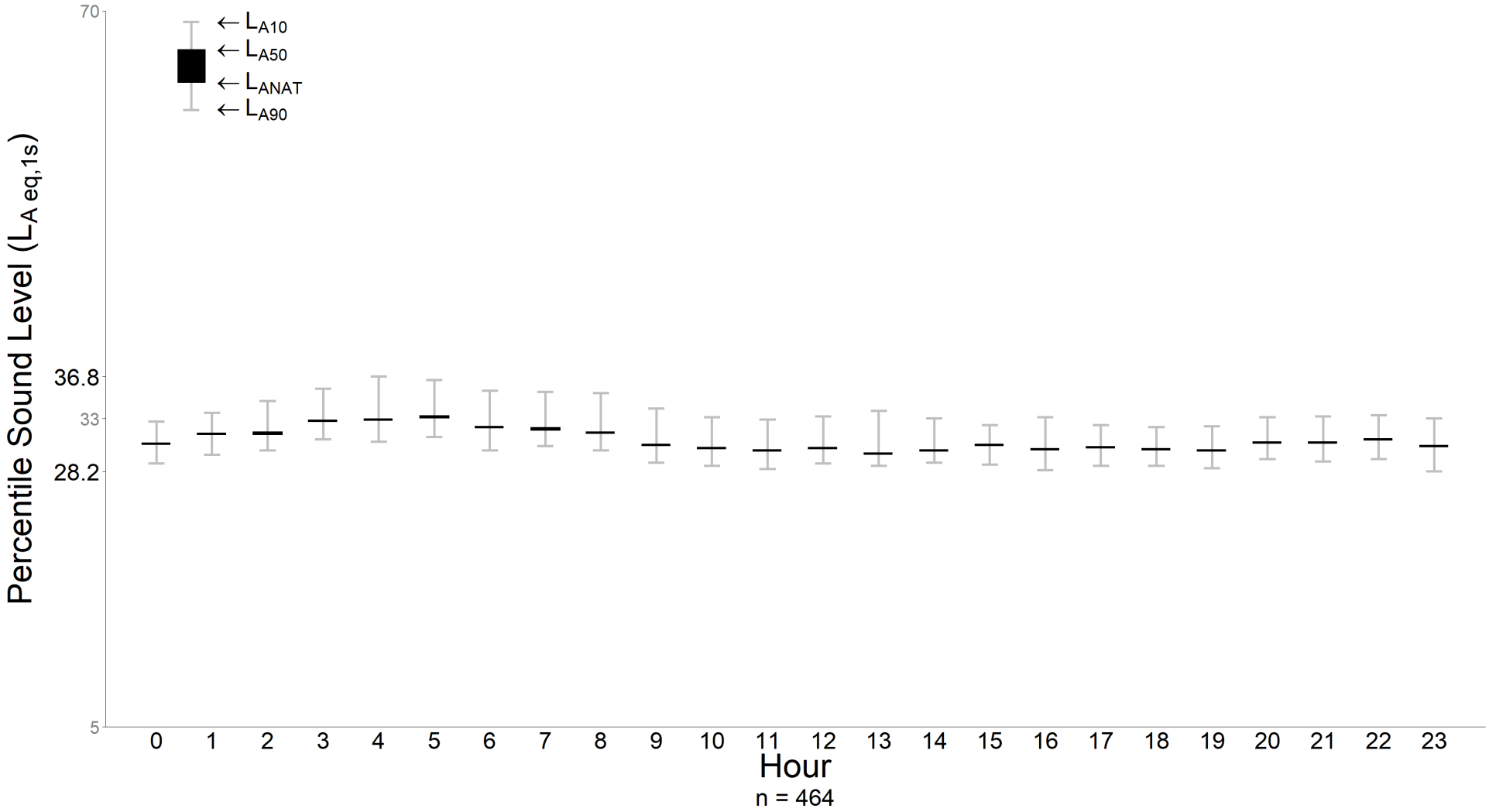


Figure 16. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARNARV in Summer, 2014.

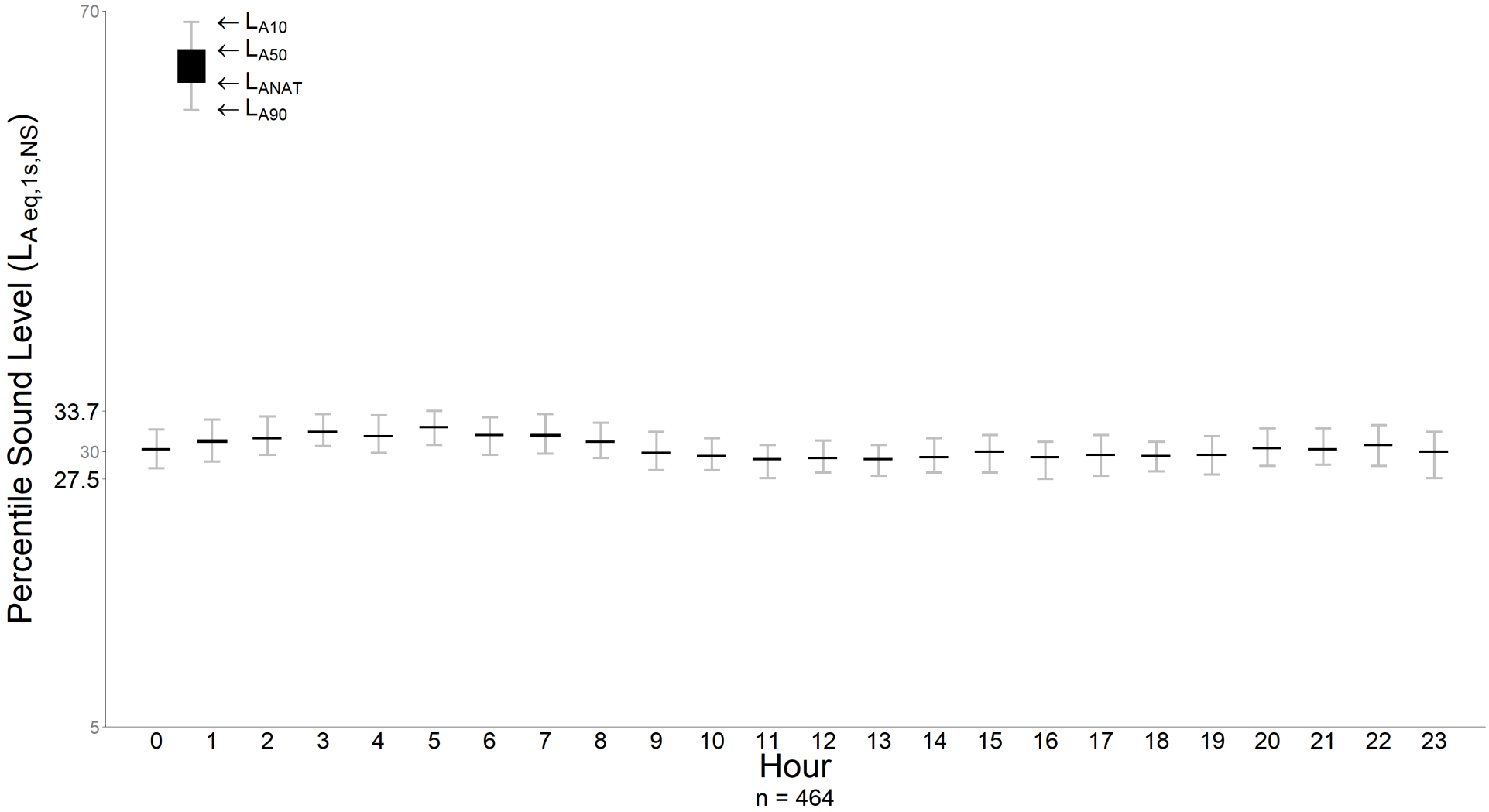


Figure 17. GAARNARV: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

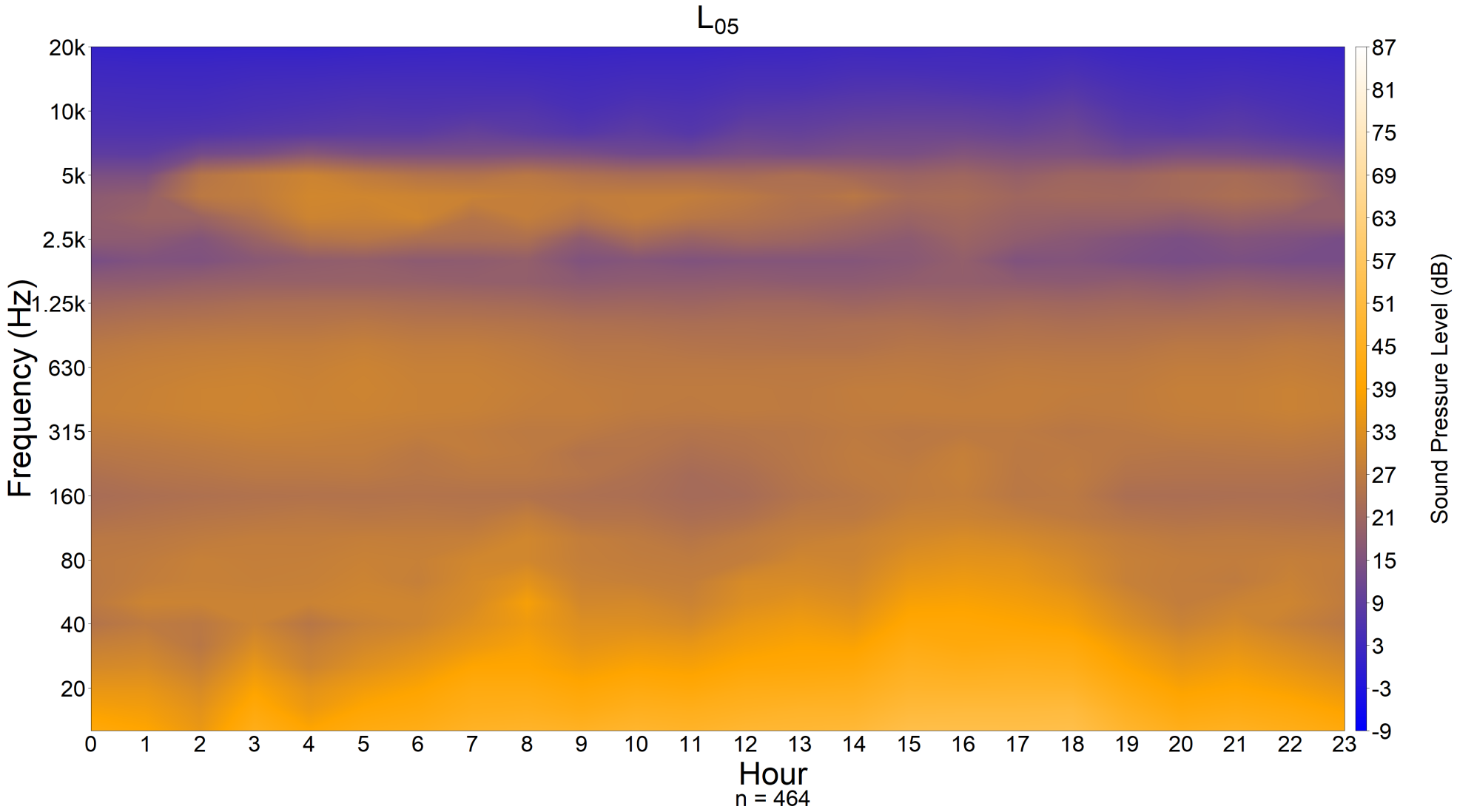


Figure 18. GAARNARV: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

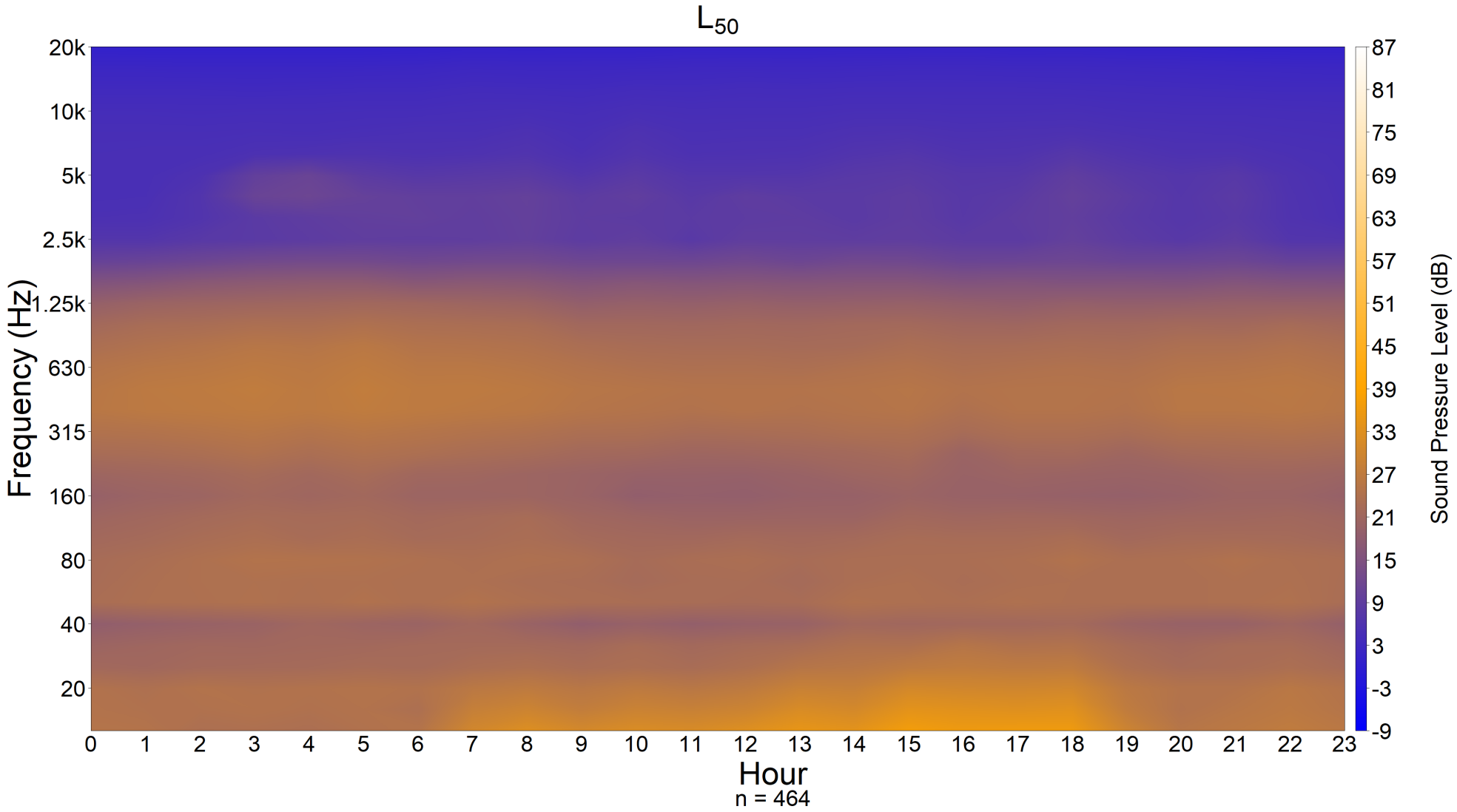


Figure 19. GAARNARV: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

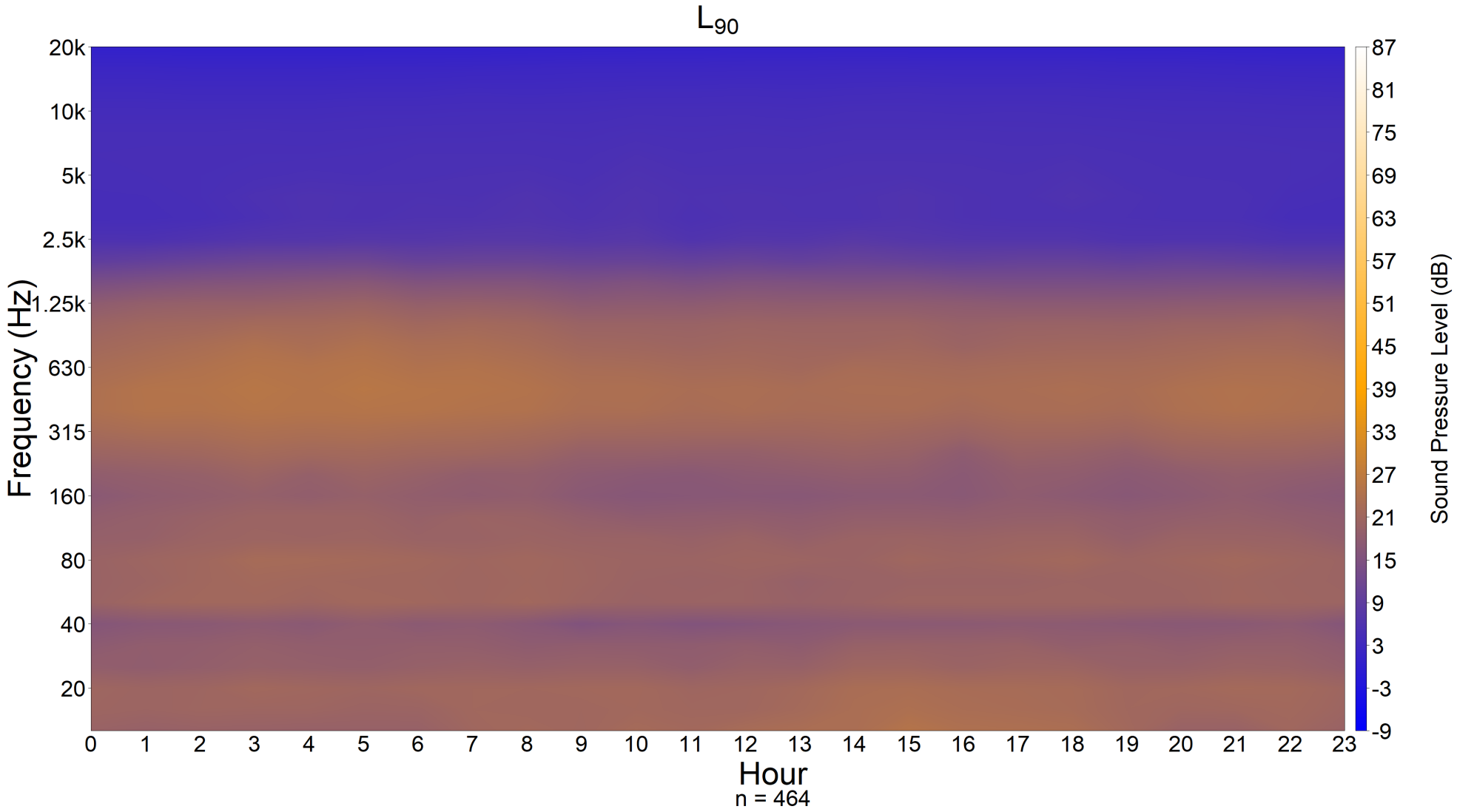


Figure 20. GAARNARV: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

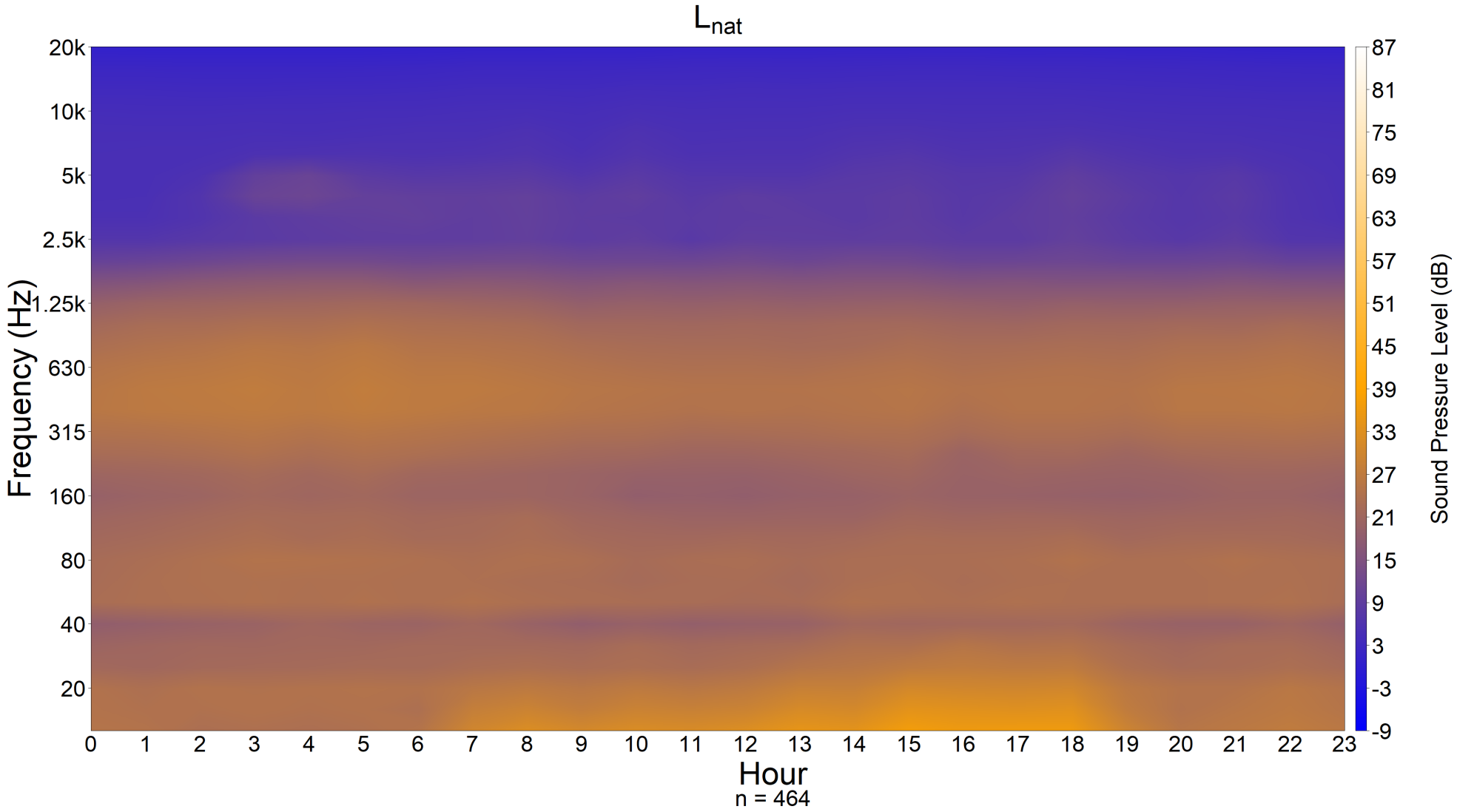


Figure 21. GAARNARV: median and mean duration of noise−free intervals by hour of day, in Summer 2014.

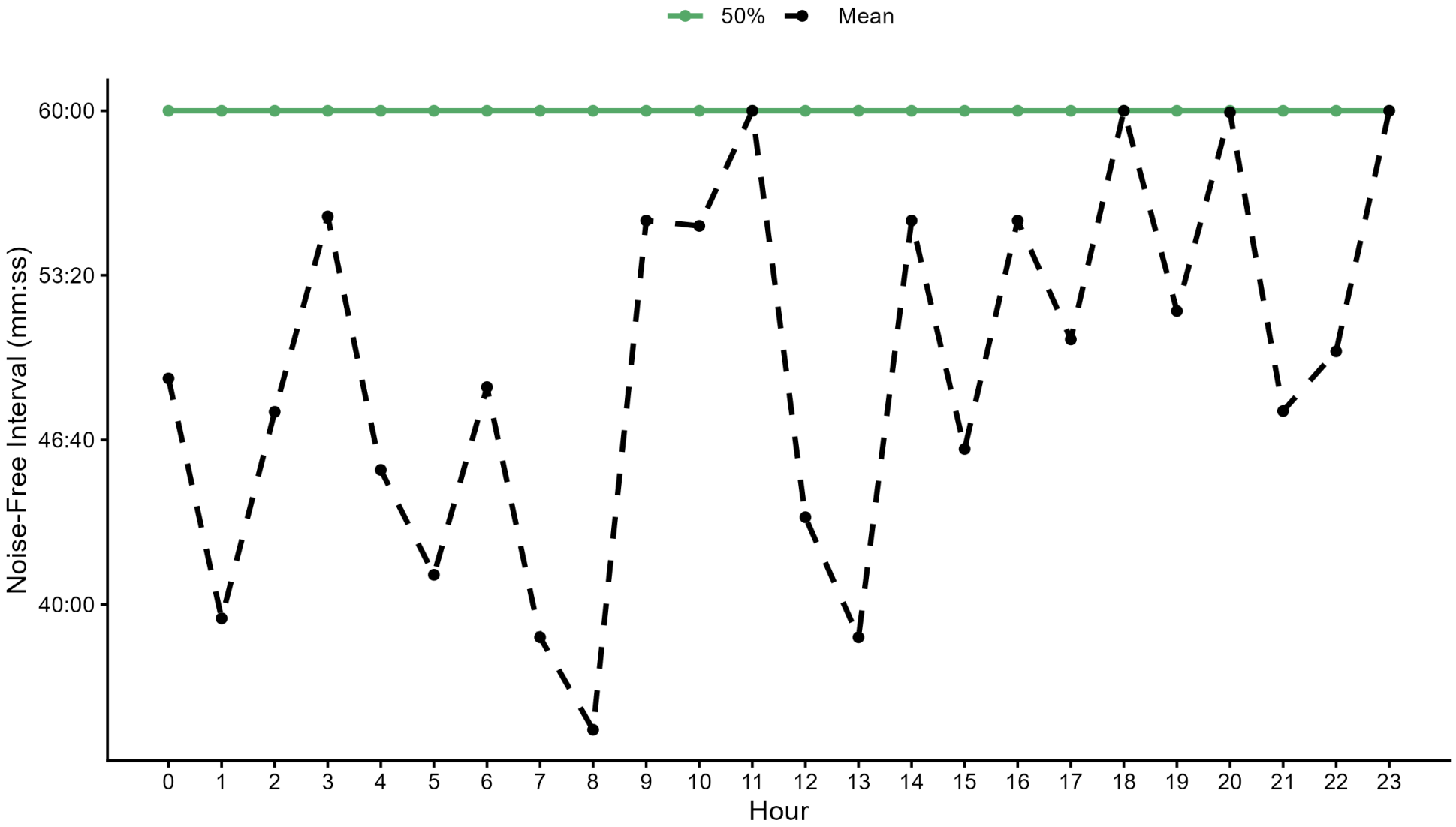


Figure 22. GAARNARV: percent of time audible by sound source category (noise/natural), in Summer 2014.

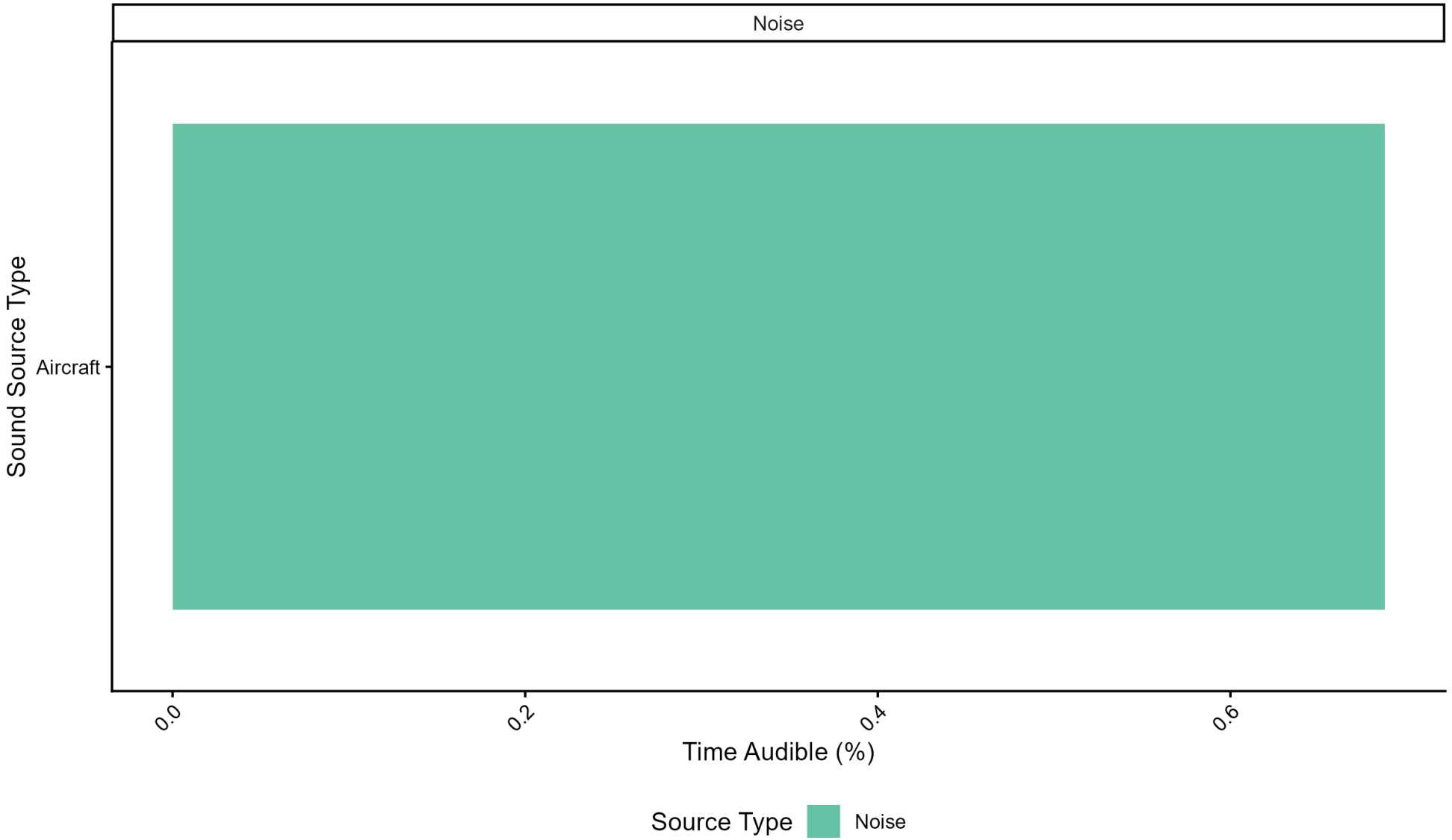


Figure 23. GAARNARV: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2014.

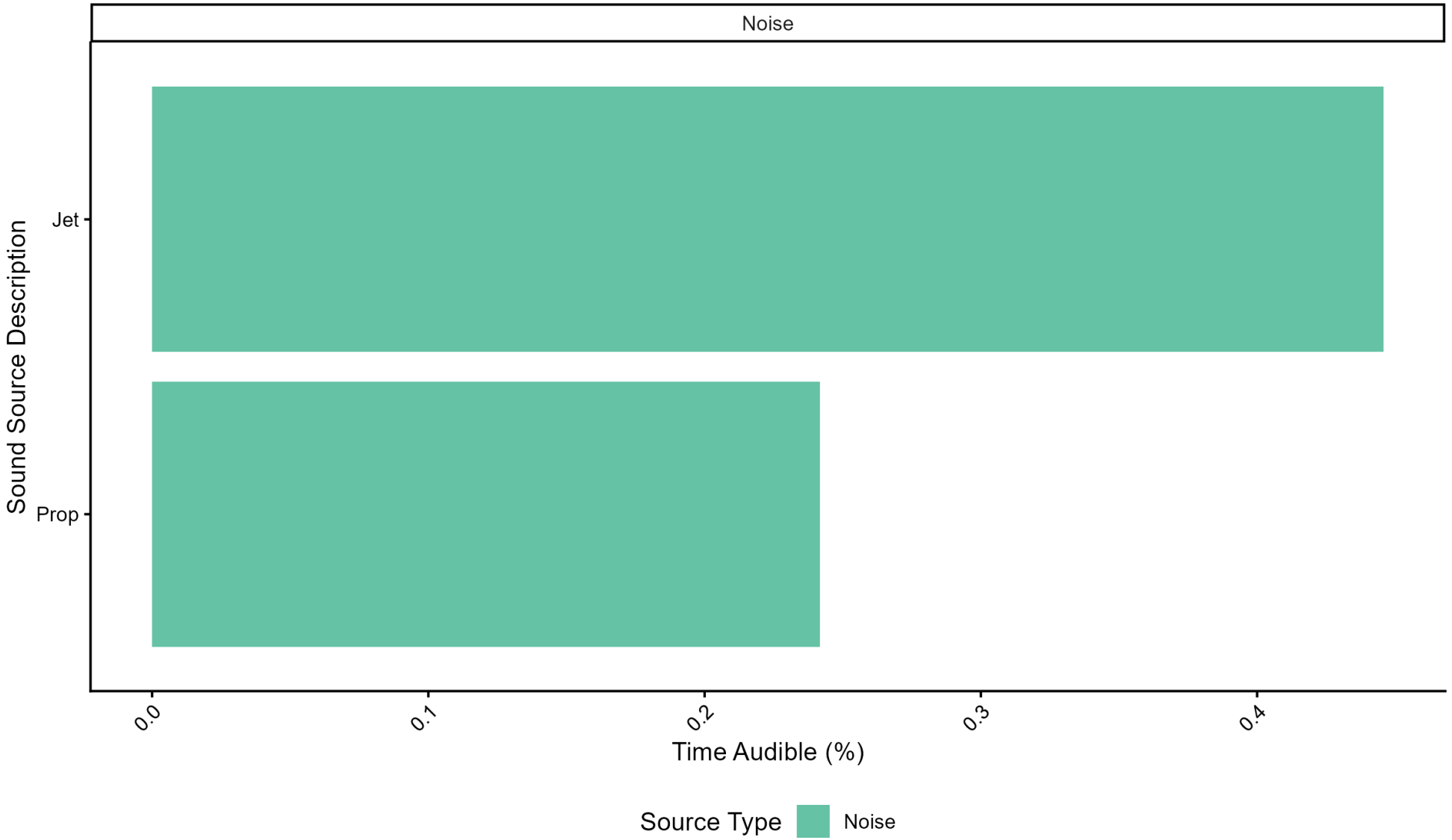


Figure 24. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARNARV in Summer, 2014. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

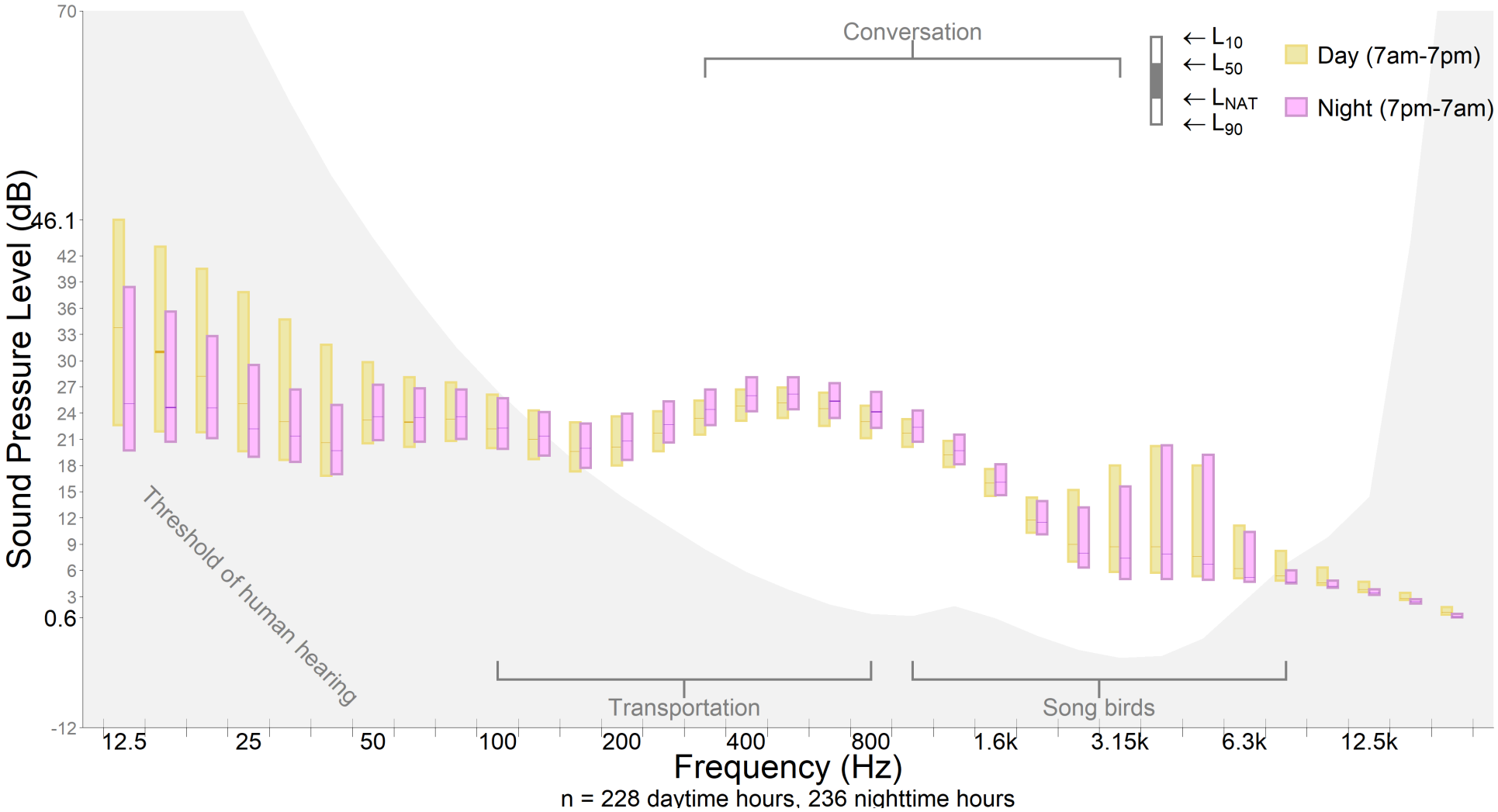


Figure 25. GAARNNUT: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2014.

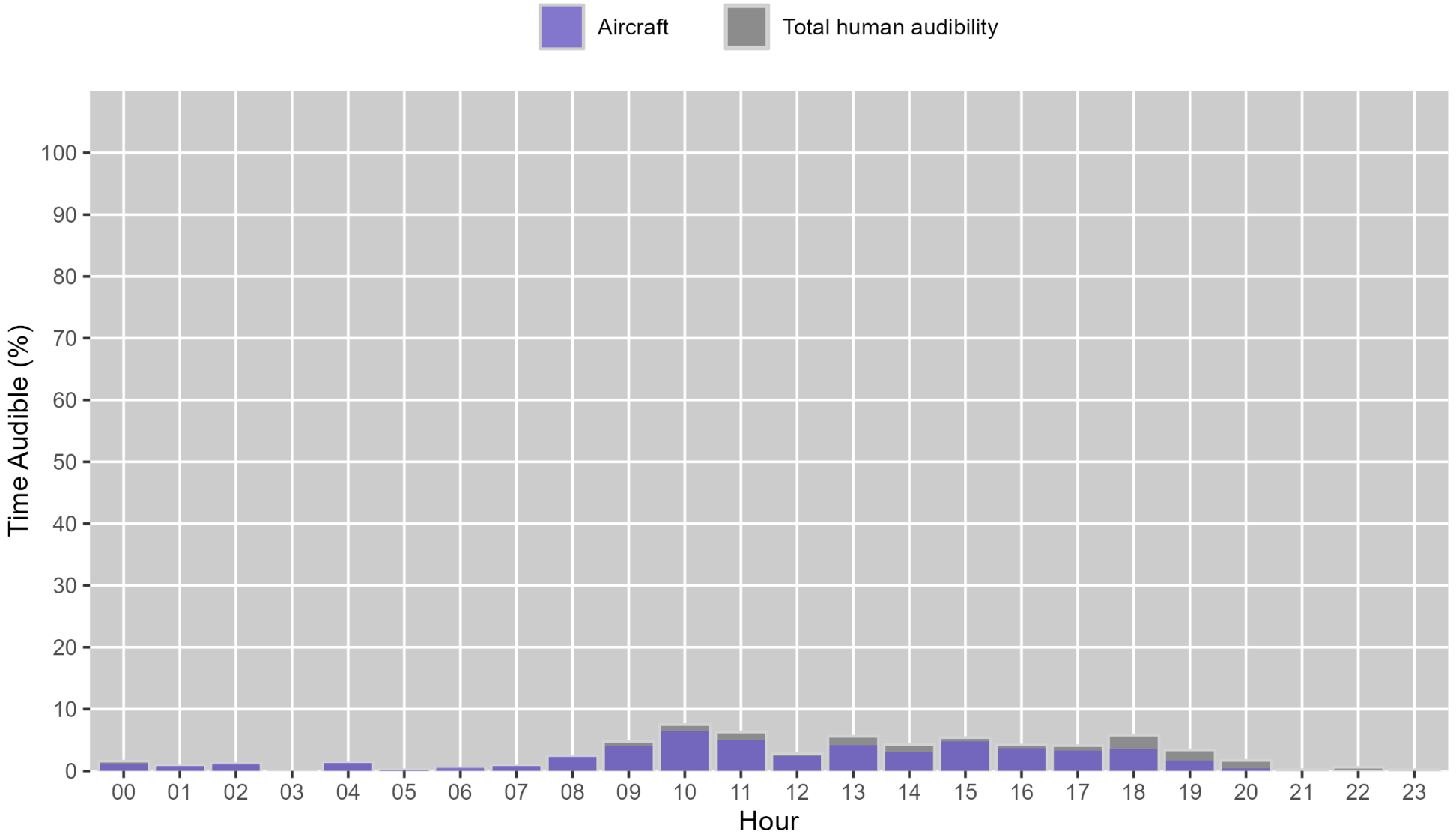


Figure 26. GAARNNUT: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2014.

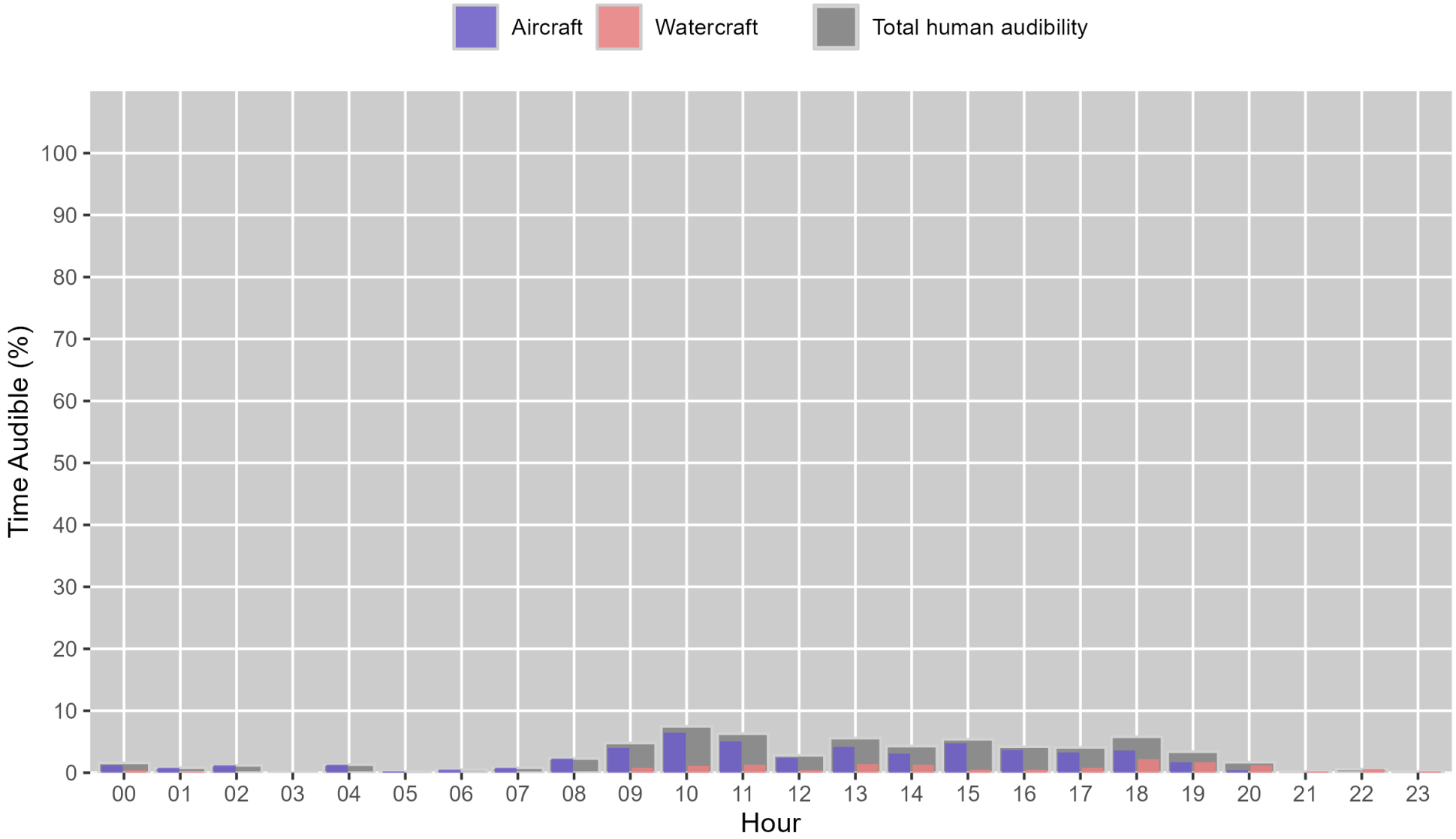


Figure 27. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARNNUT in Summer, 2014, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

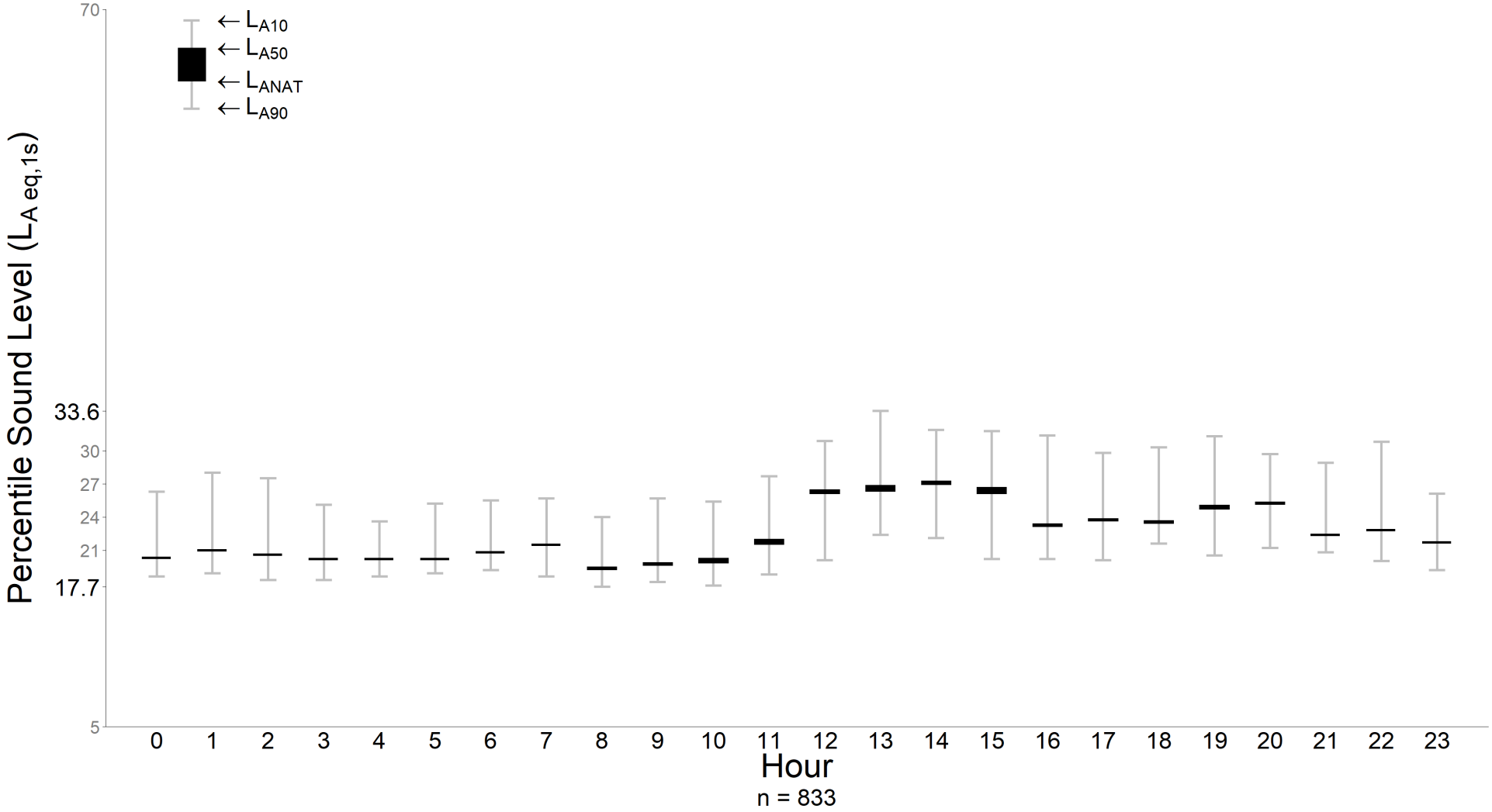


Figure 28. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARNNUT in Summer, 2014.

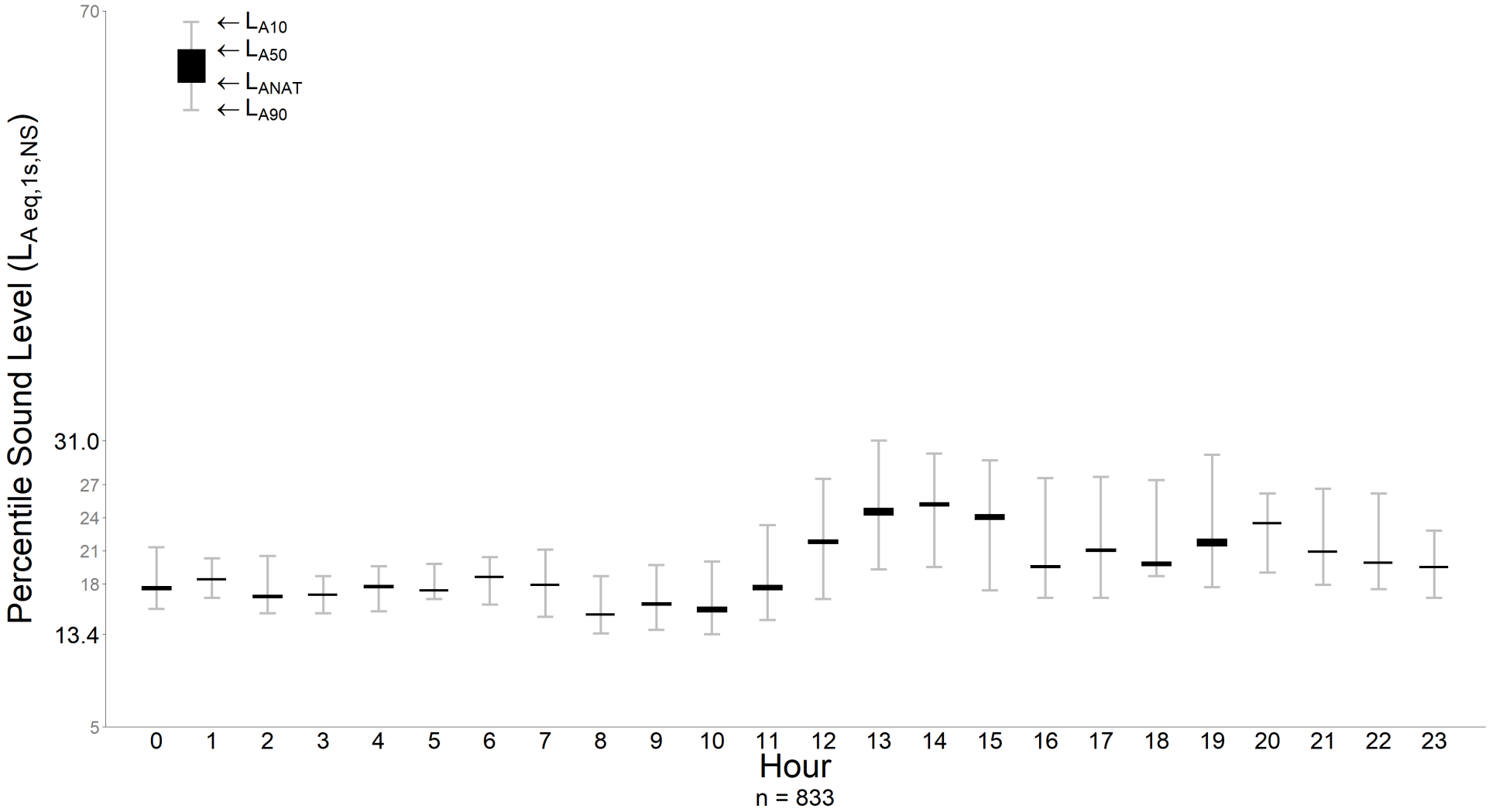


Figure 29. GAARNNUT: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

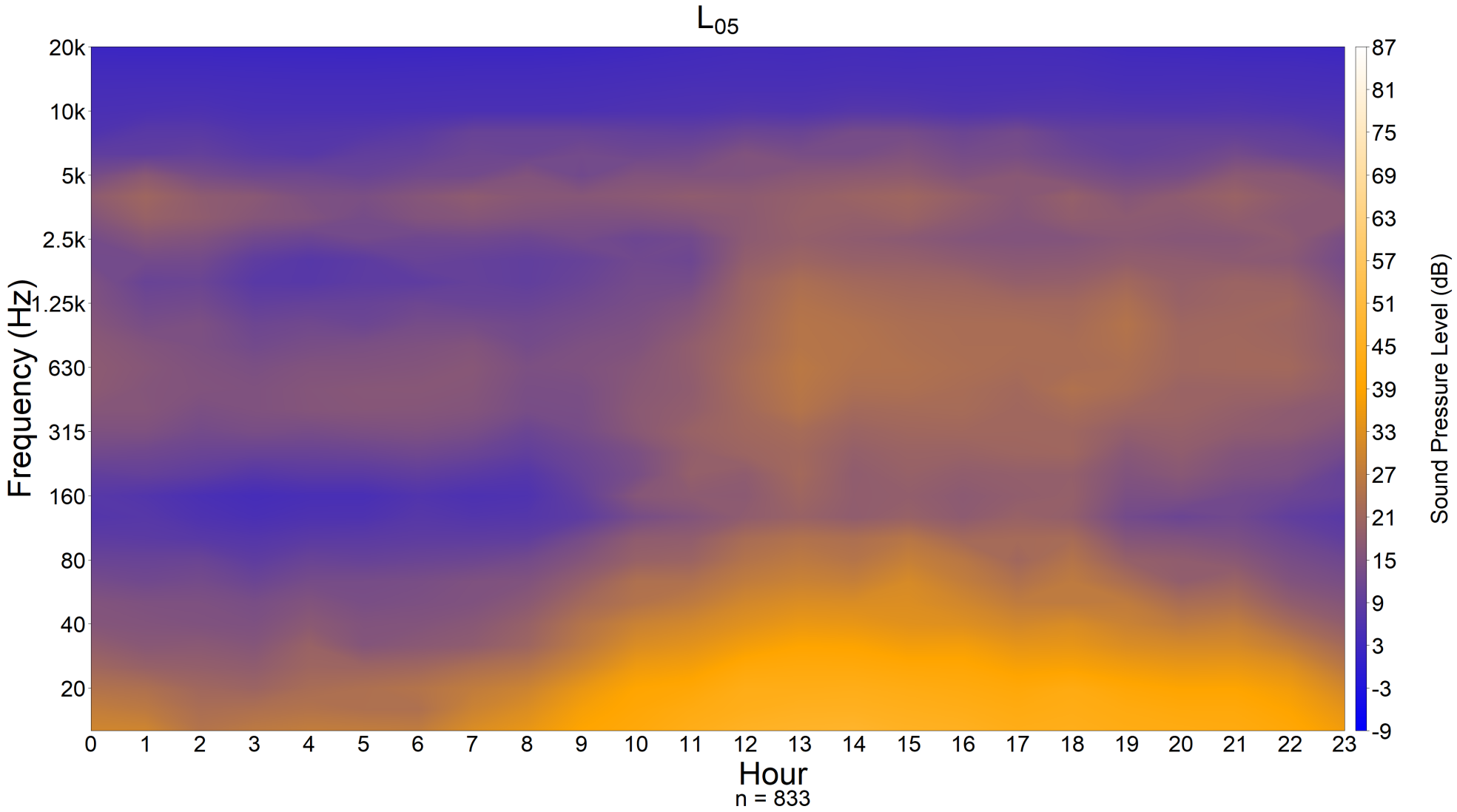


Figure 30. GAARNNUT: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

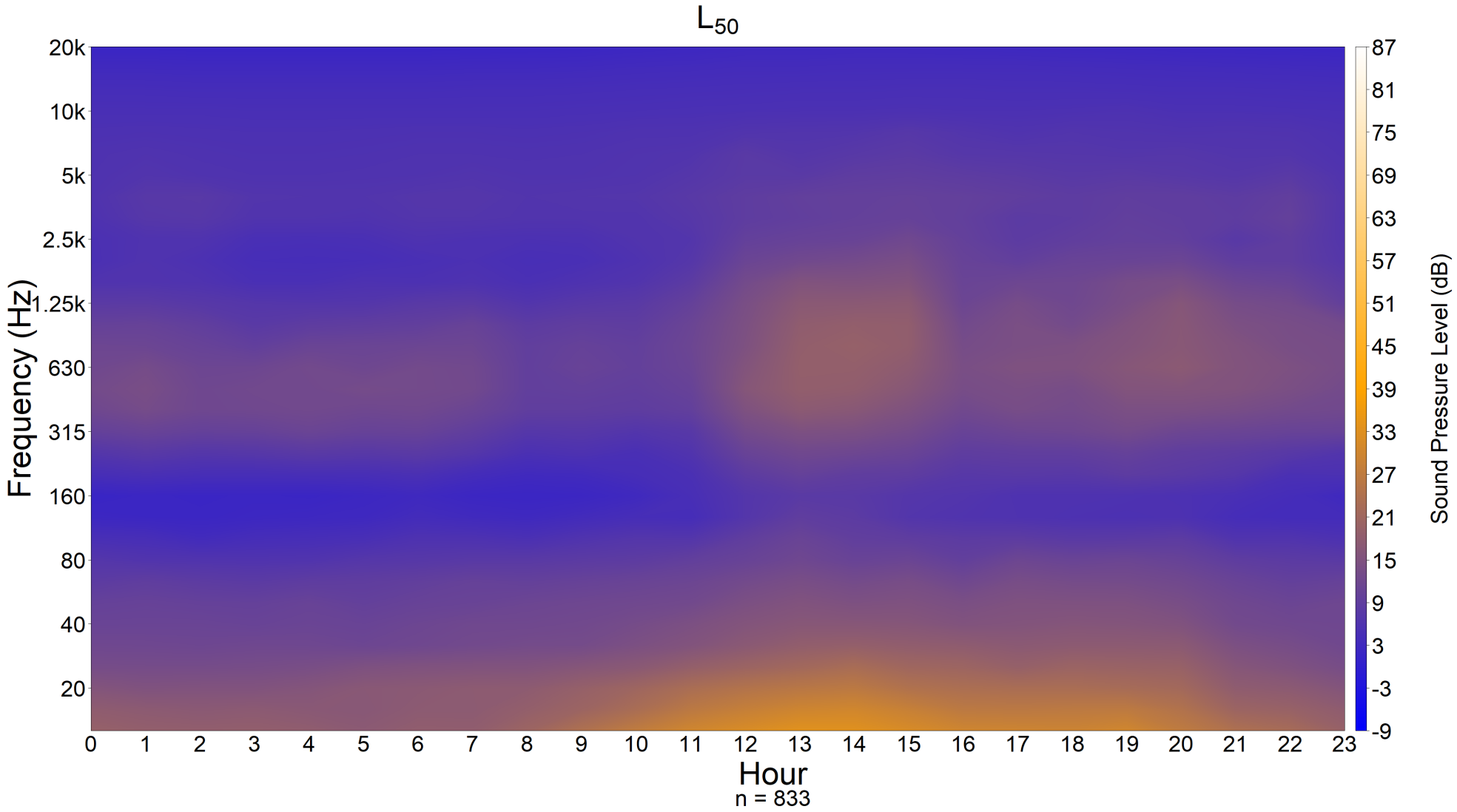


Figure 31. GAARNNUT: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

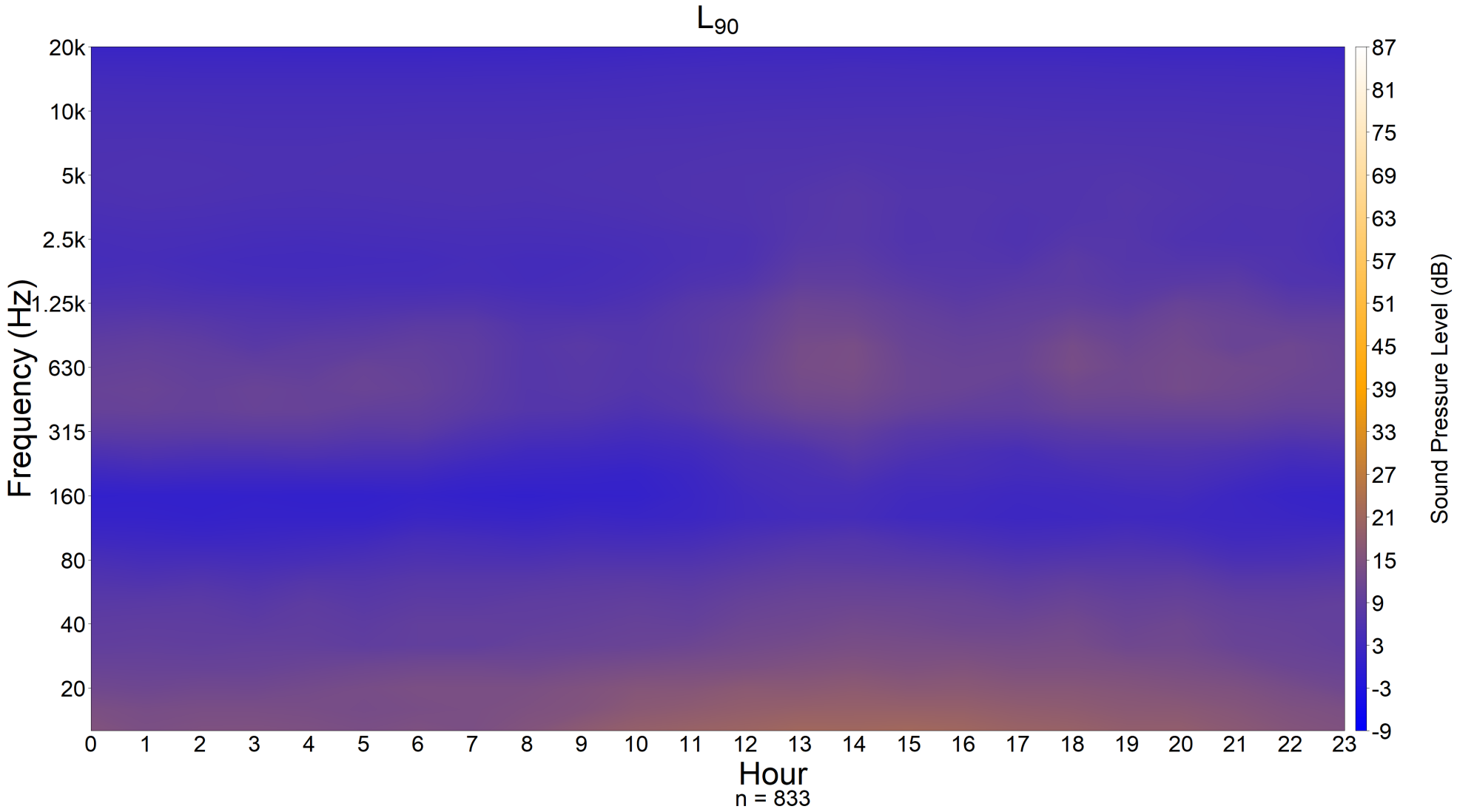


Figure 32. GAARNNUT: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

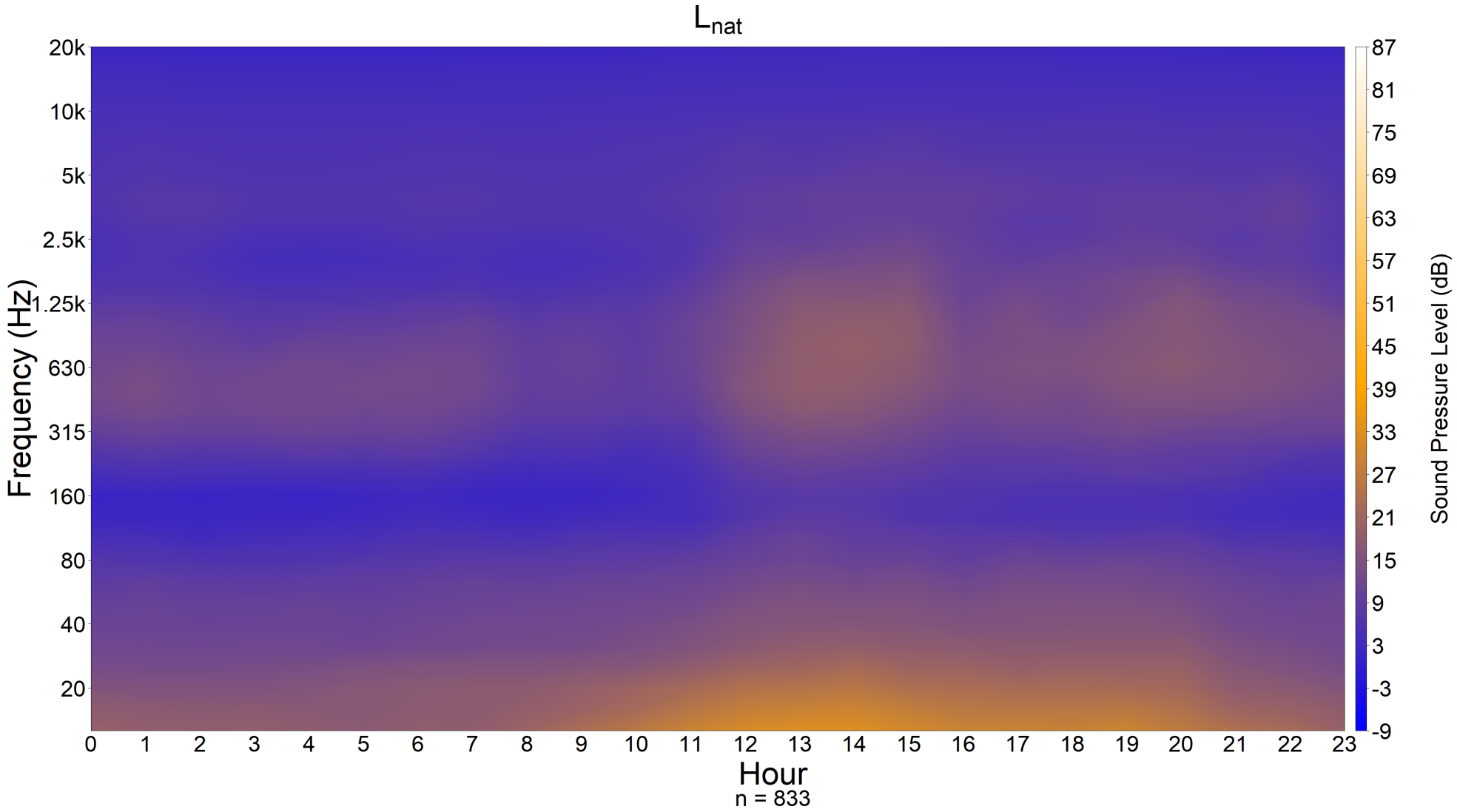


Figure 33. GAARNNUT: median and mean duration of noise−free intervals by hour of day, in Summer 2014.

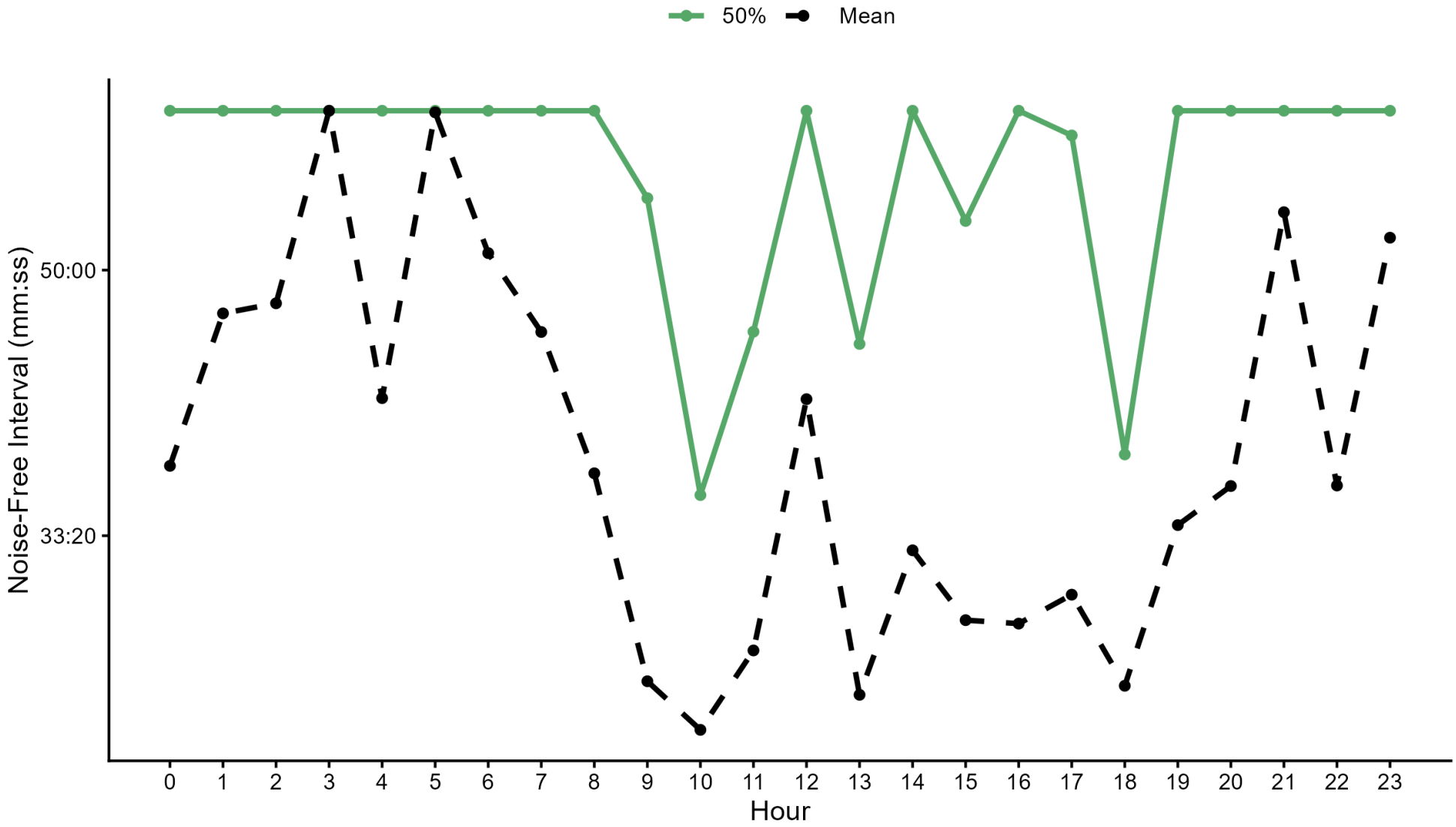


Figure 34. GAARNNUT: percent of time audible by sound source category (noise/natural), in Summer 2014.

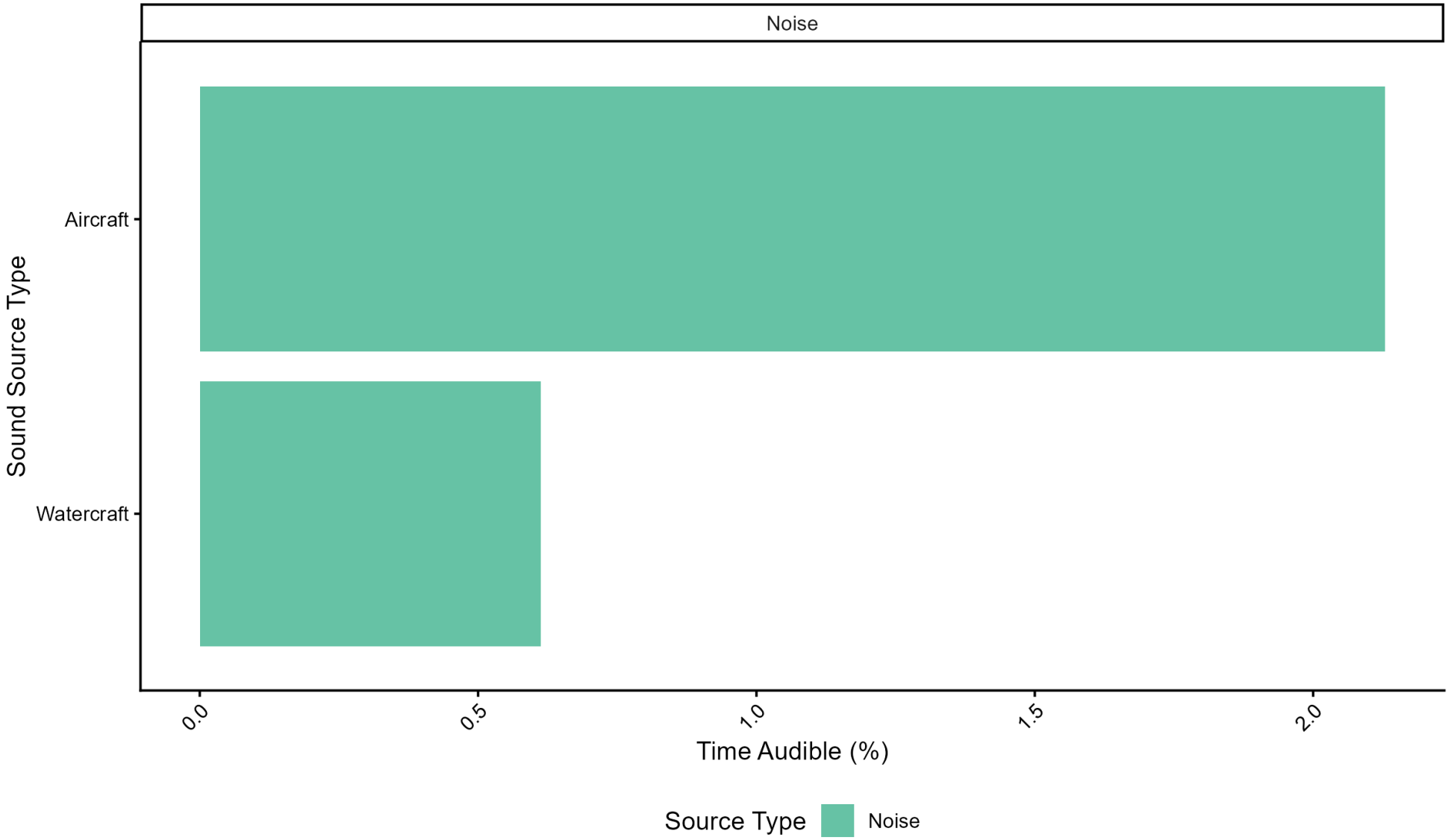


Figure 35. GAARNNUT: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2014.

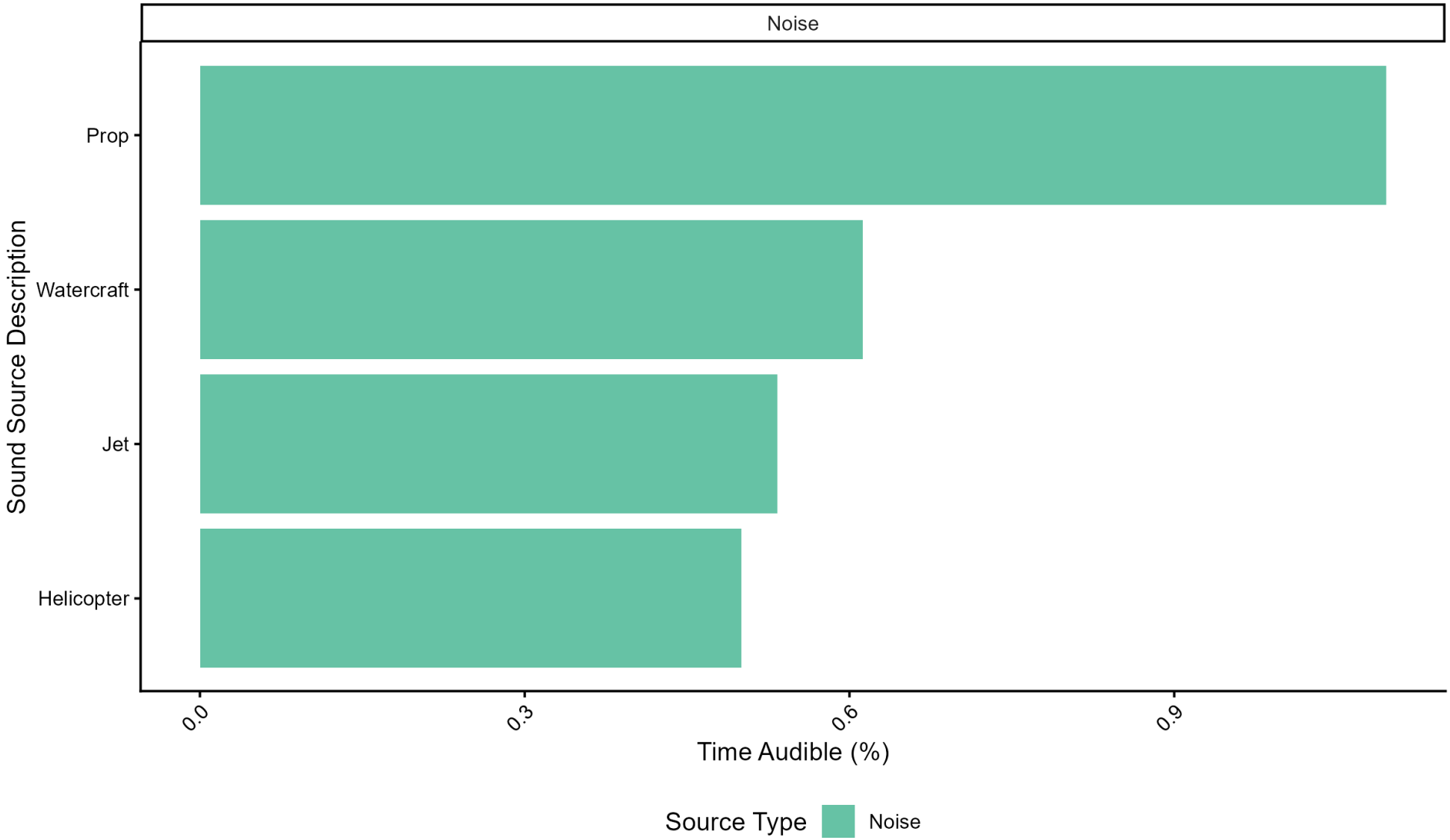


Figure 36. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARNNUT in Summer, 2014. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

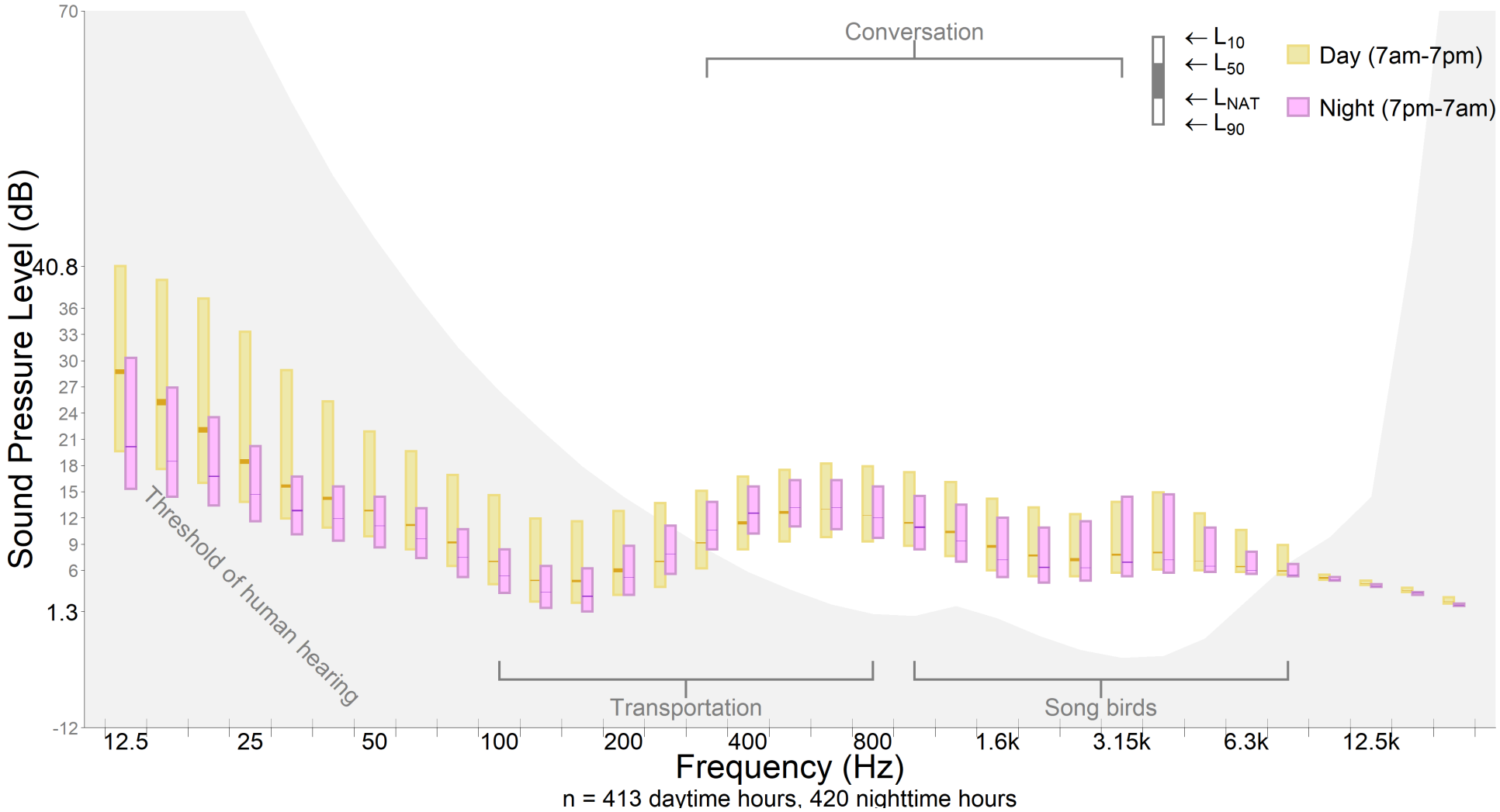


Figure 37. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARNUTU in Summer, 2014, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

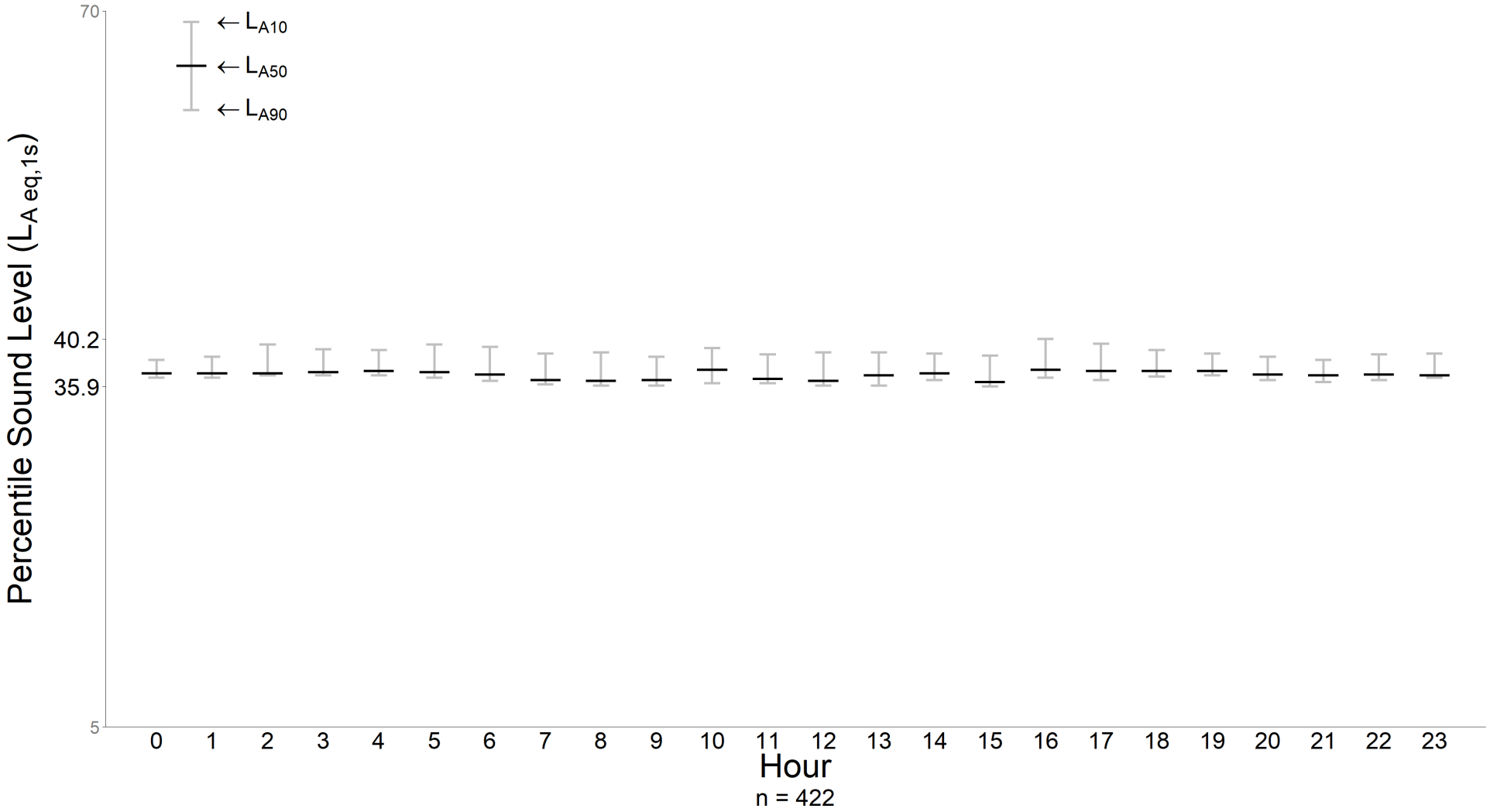


Figure 38. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARNUTU in Summer, 2014.

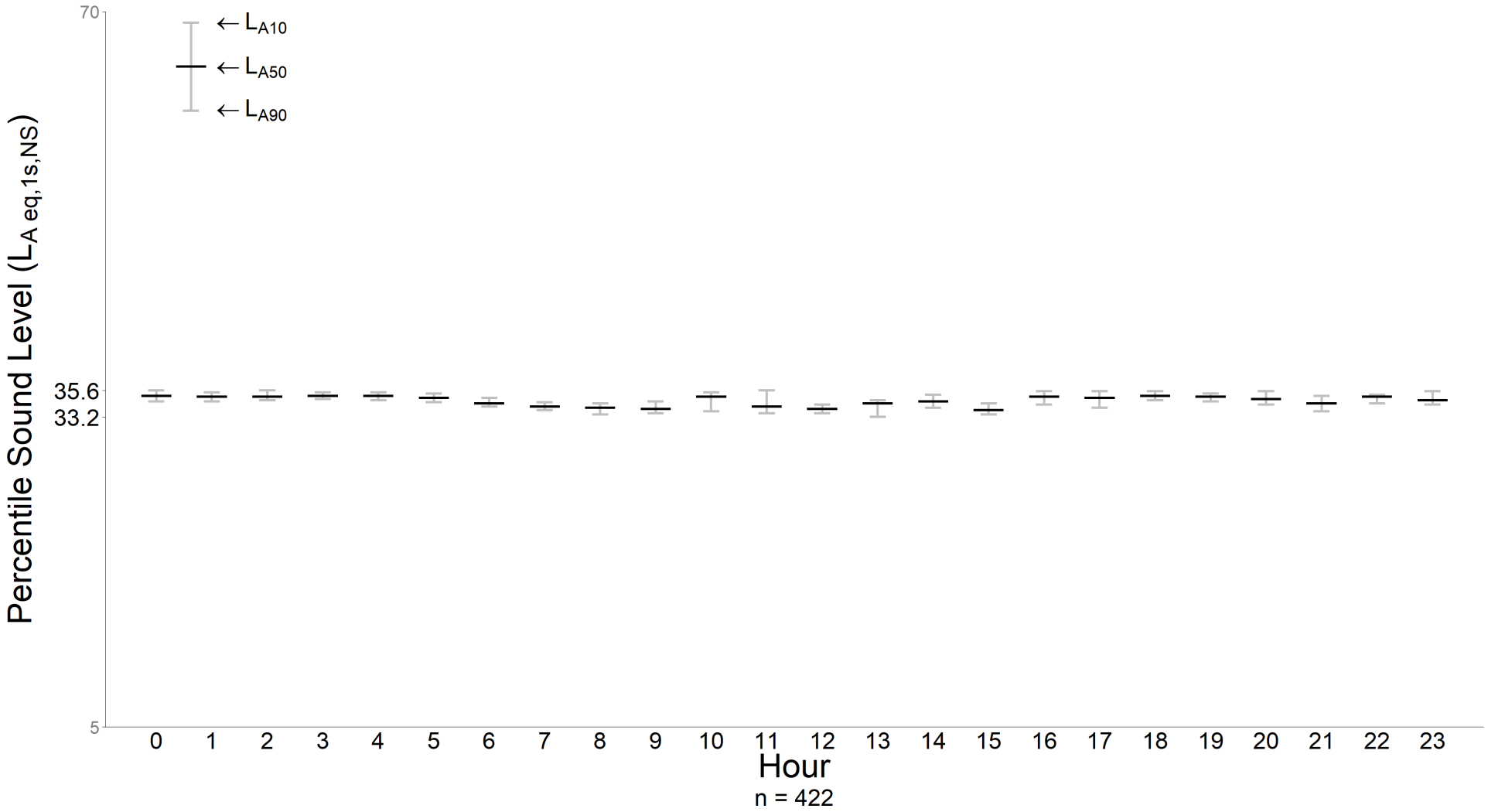


Figure 39. GAARNUTU: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

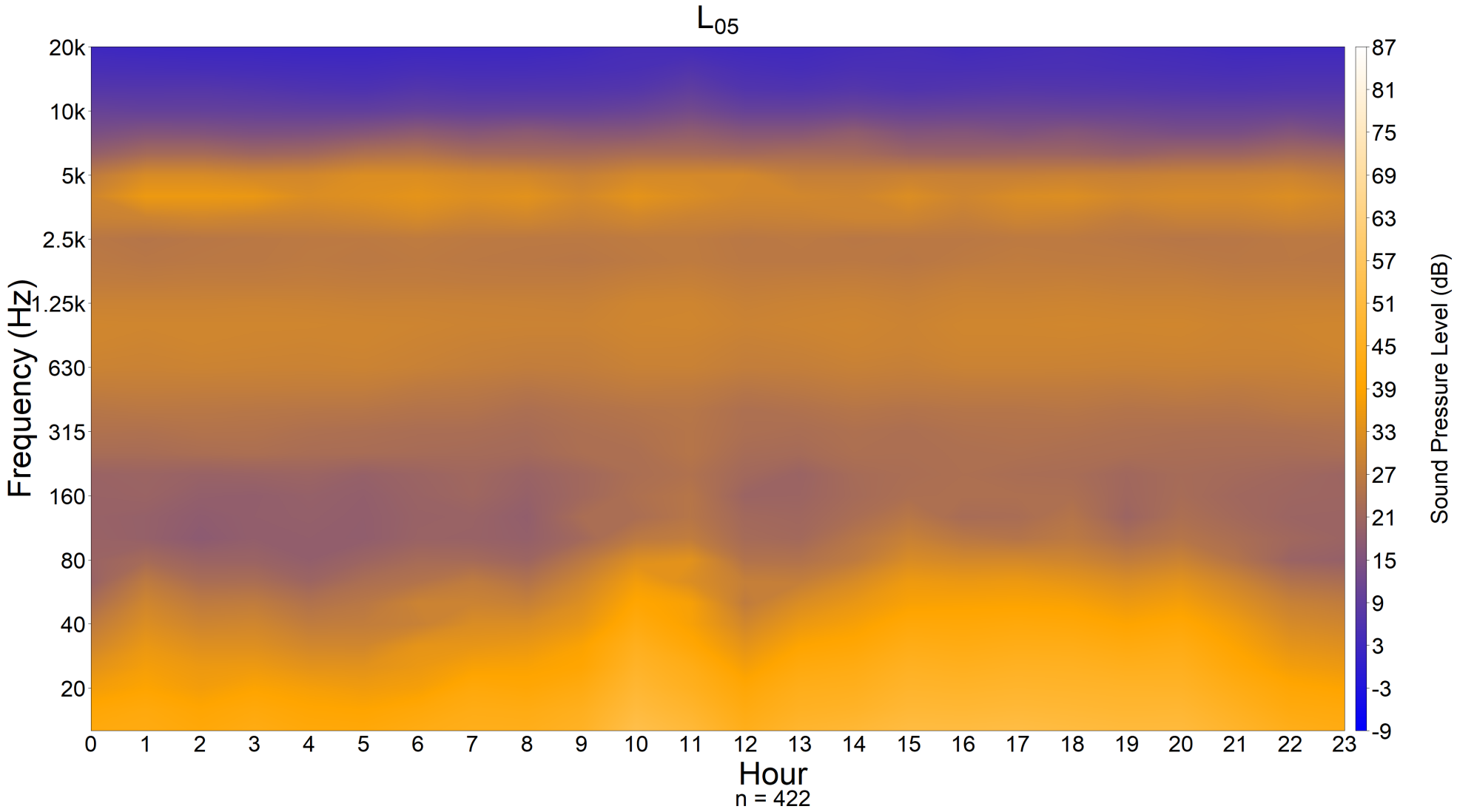


Figure 40. GAARNUTU: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

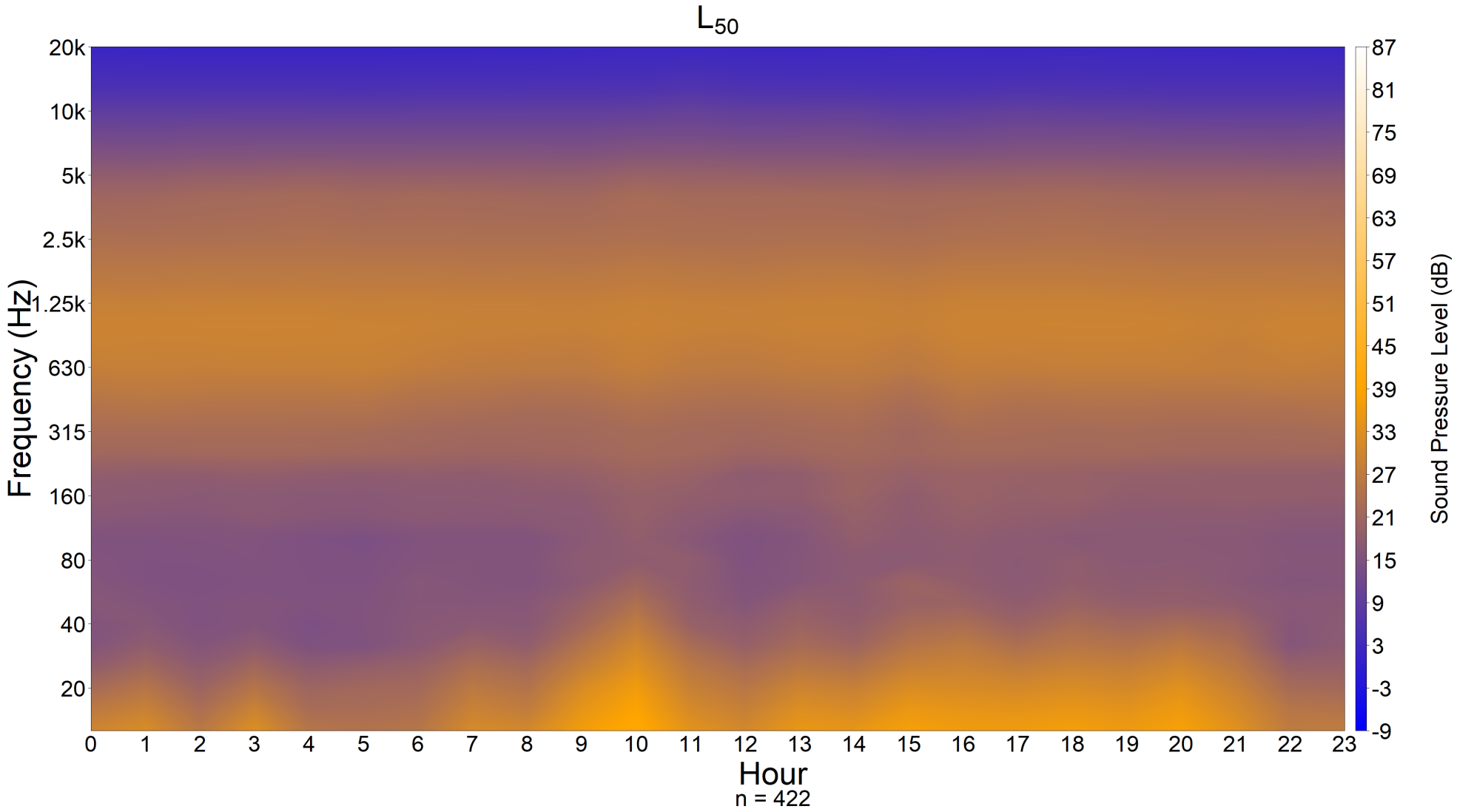


Figure 41. GAARNUTU: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

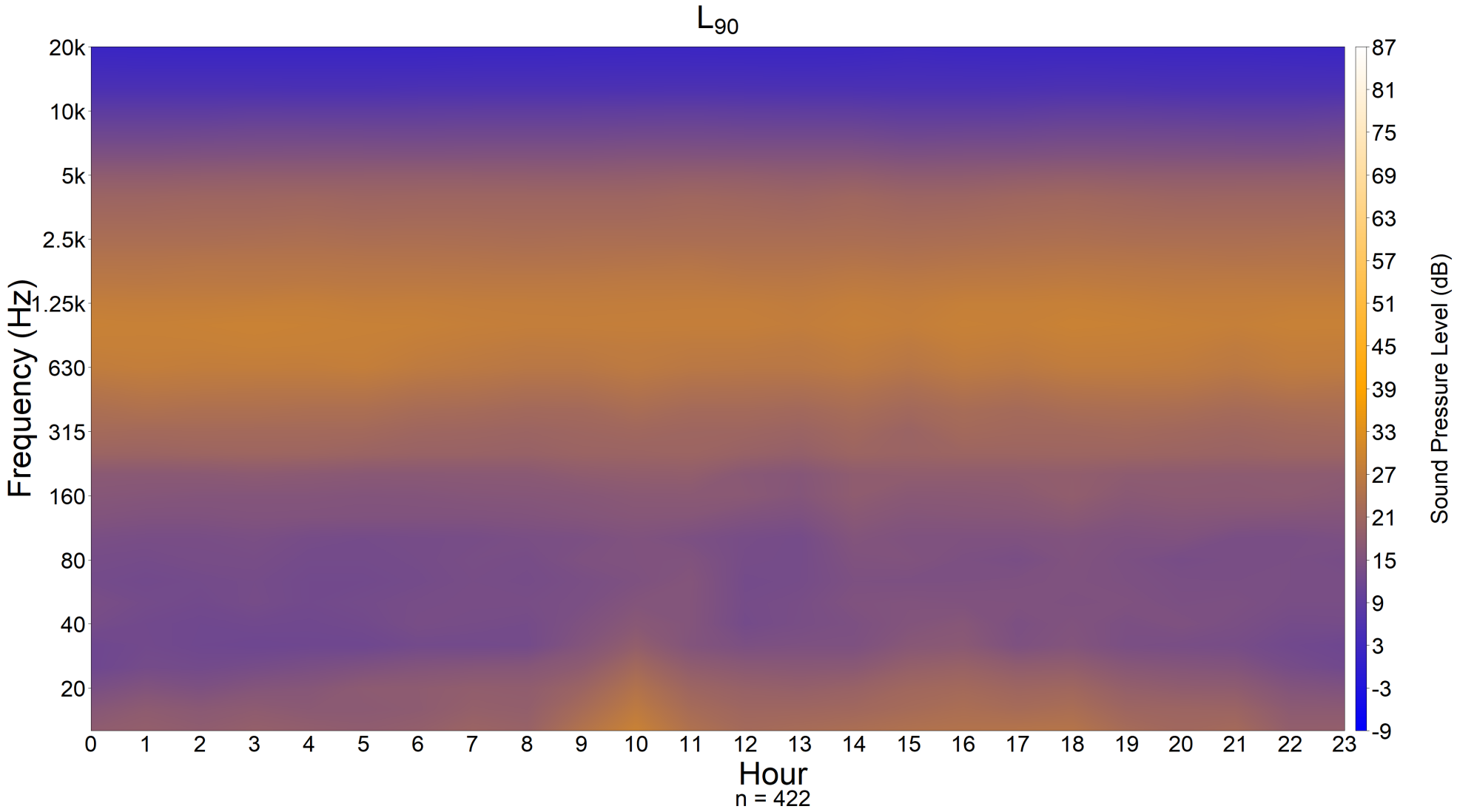


Figure 42. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARNUTU in Summer, 2014. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

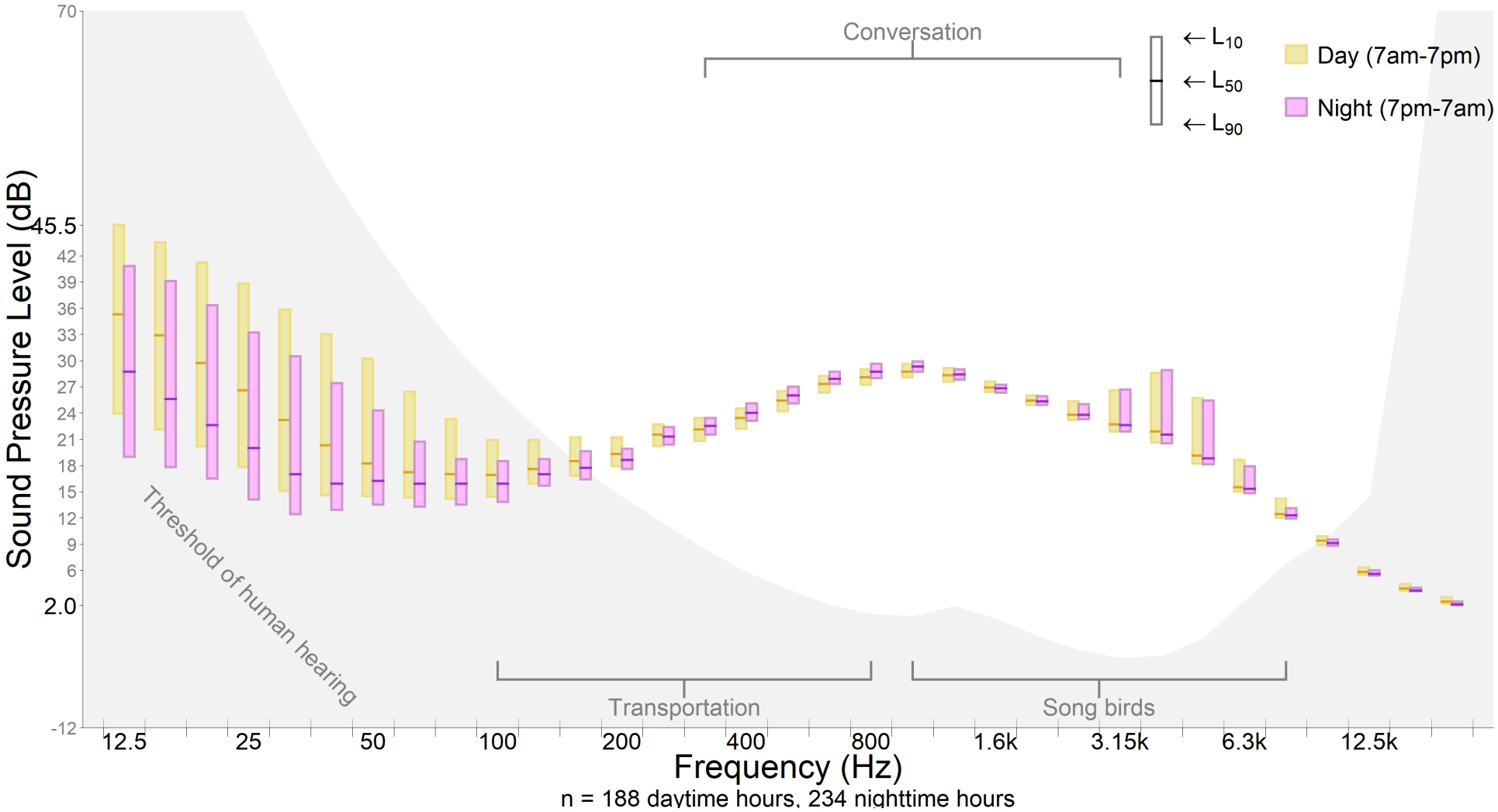


Figure 43. GAARNWAL: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2014.

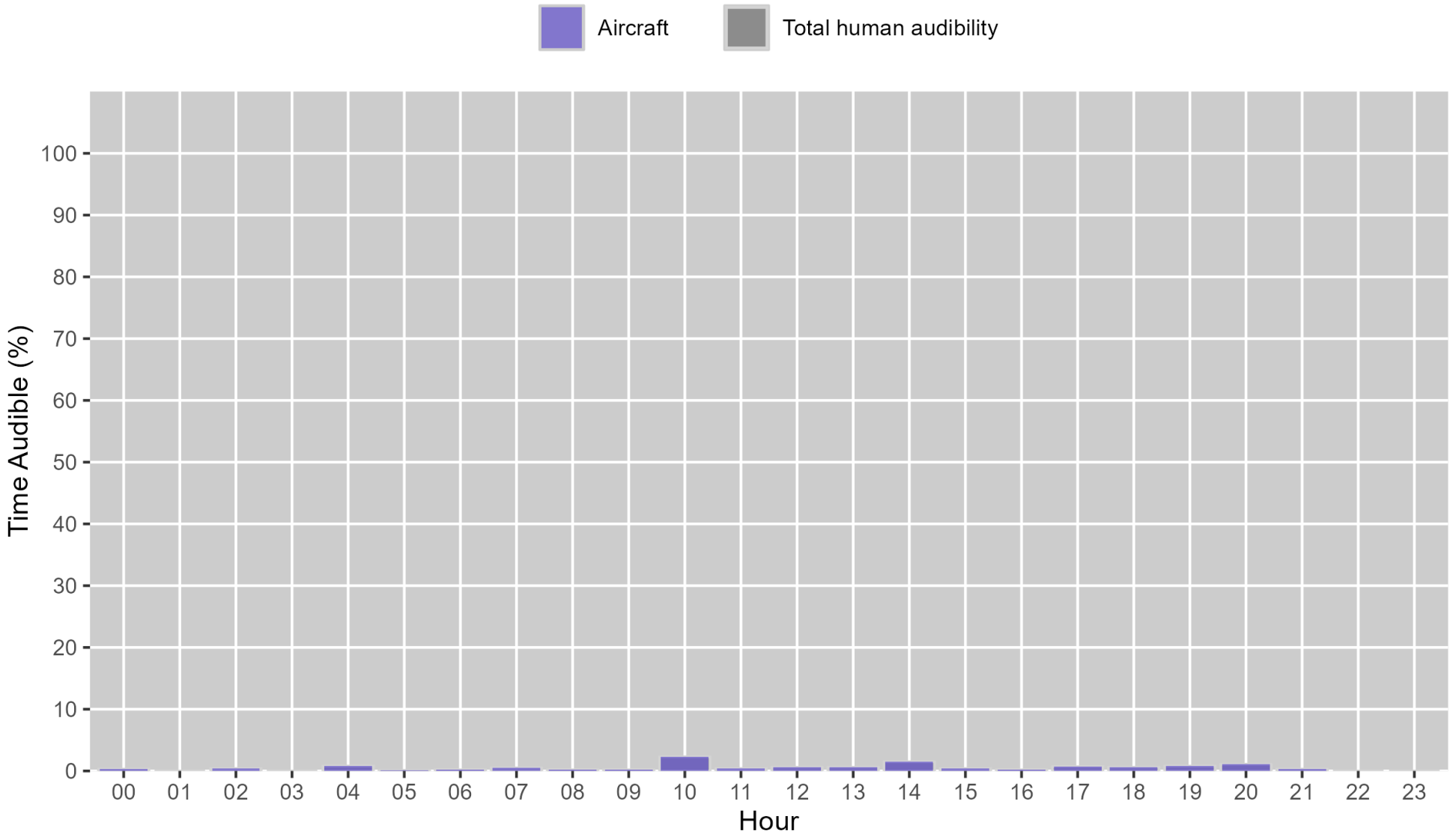


Figure 44. GAARNWAL: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2014.

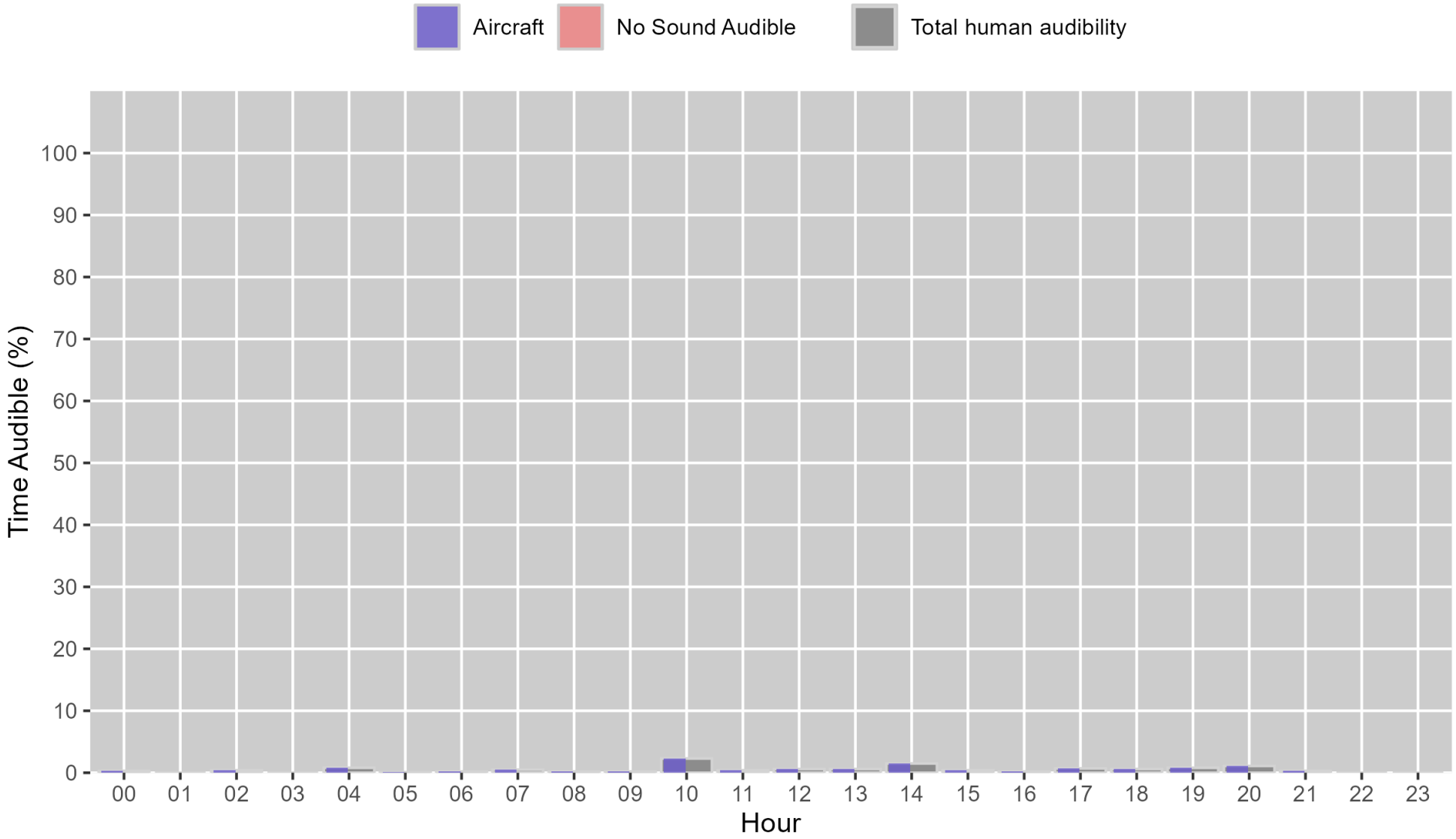


Figure 45. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARNWAL in Summer, 2014, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

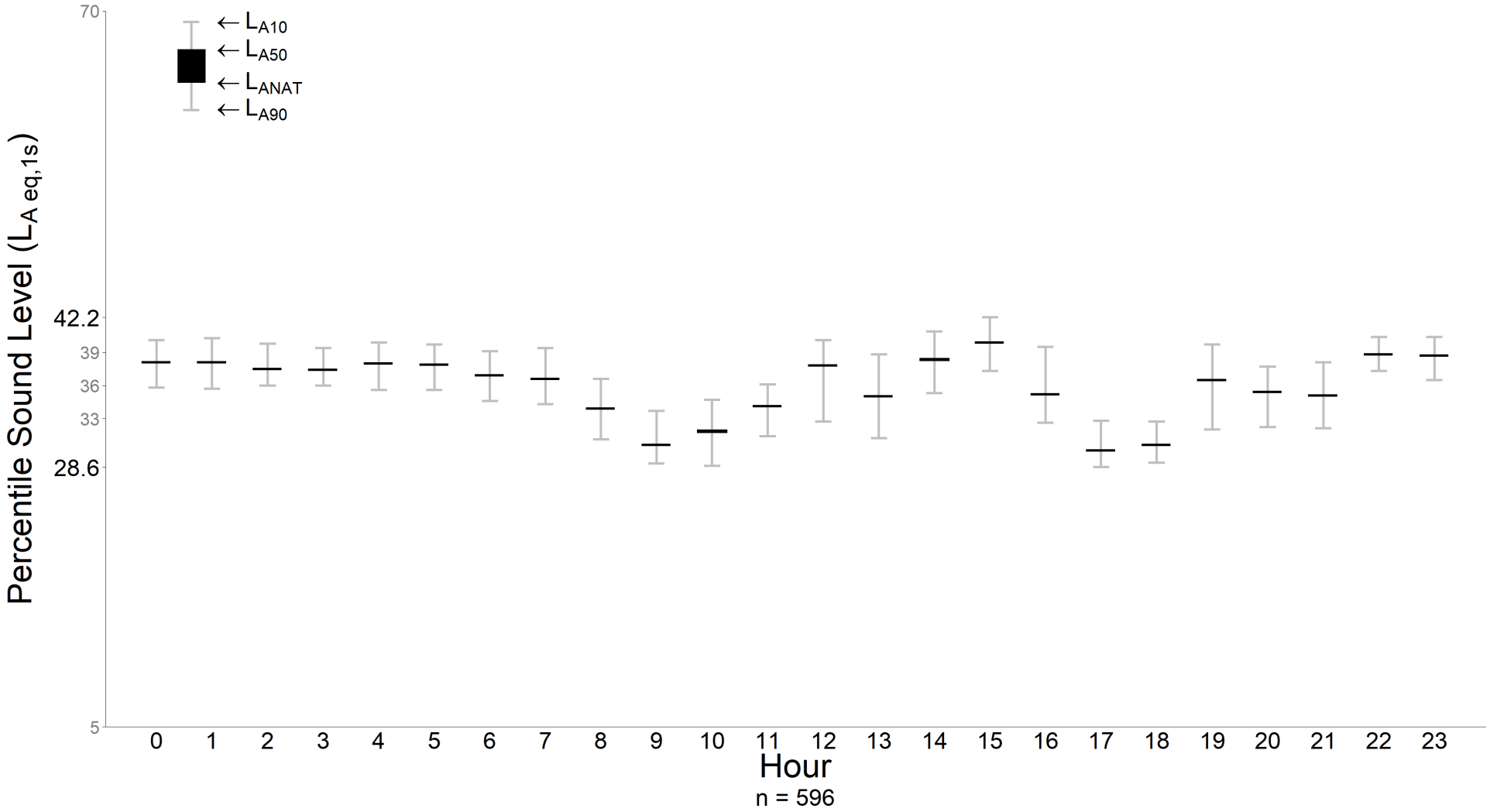


Figure 46. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARNWAL in Summer, 2014.

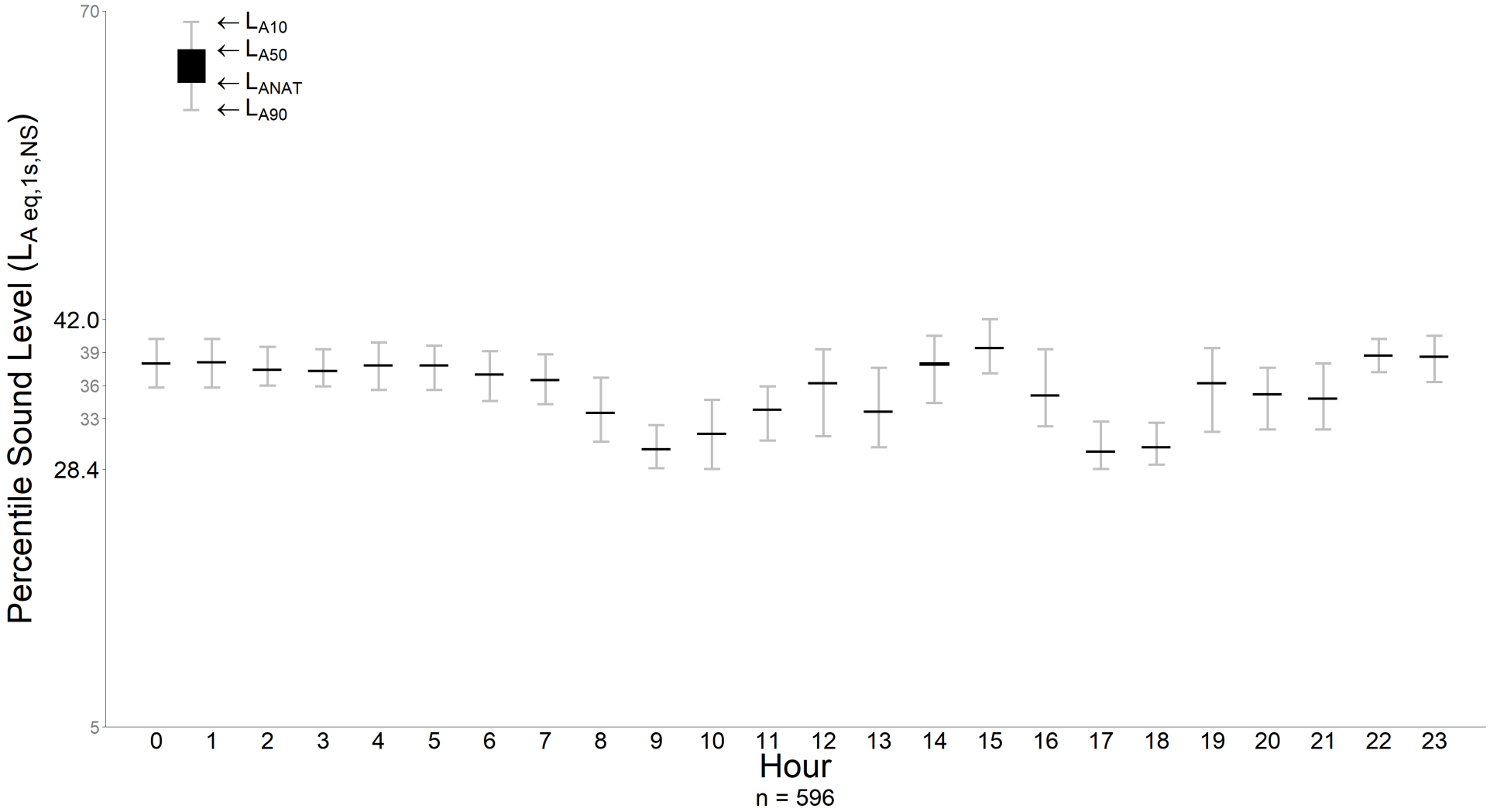


Figure 47. GAARNWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

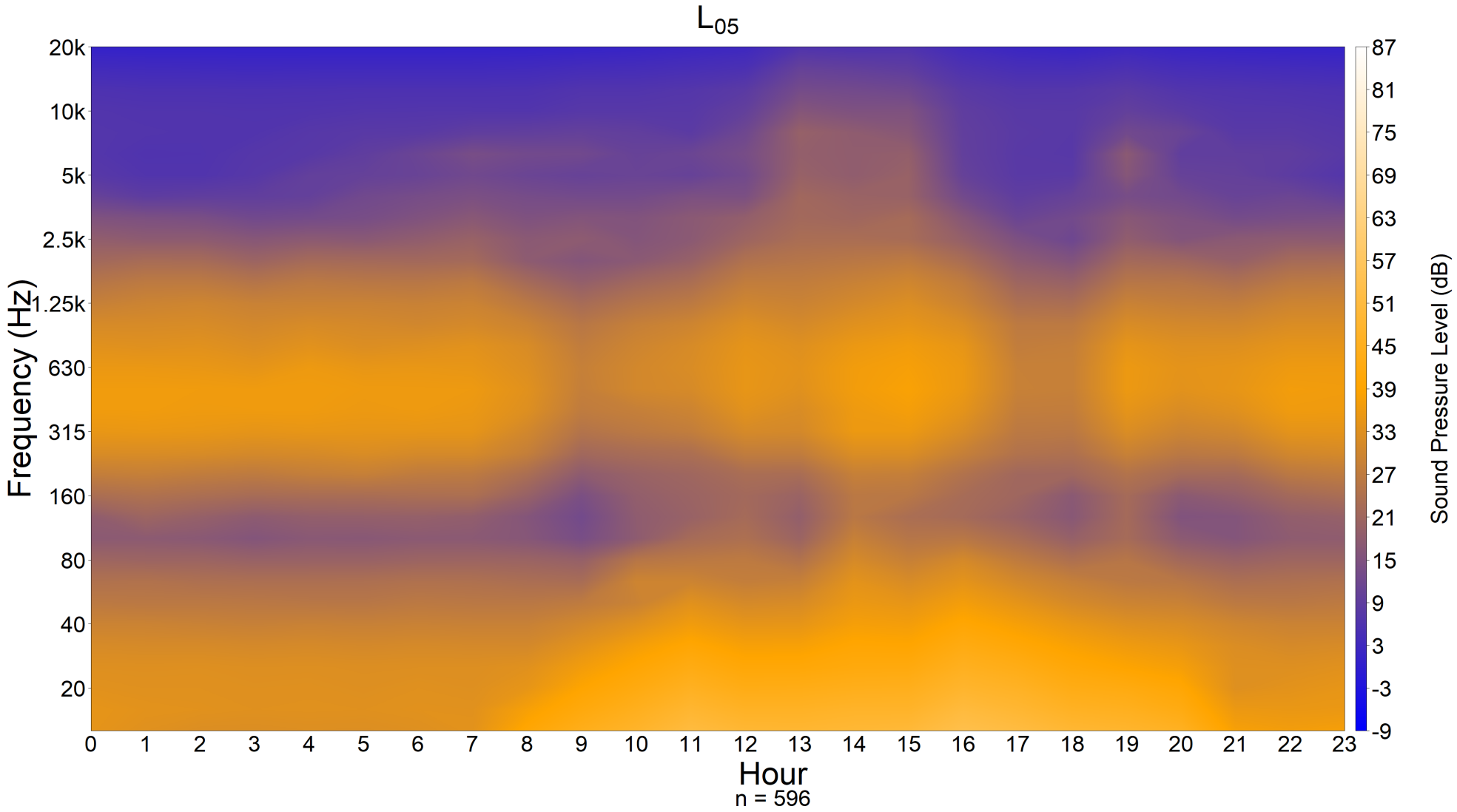


Figure 48. GAARNWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

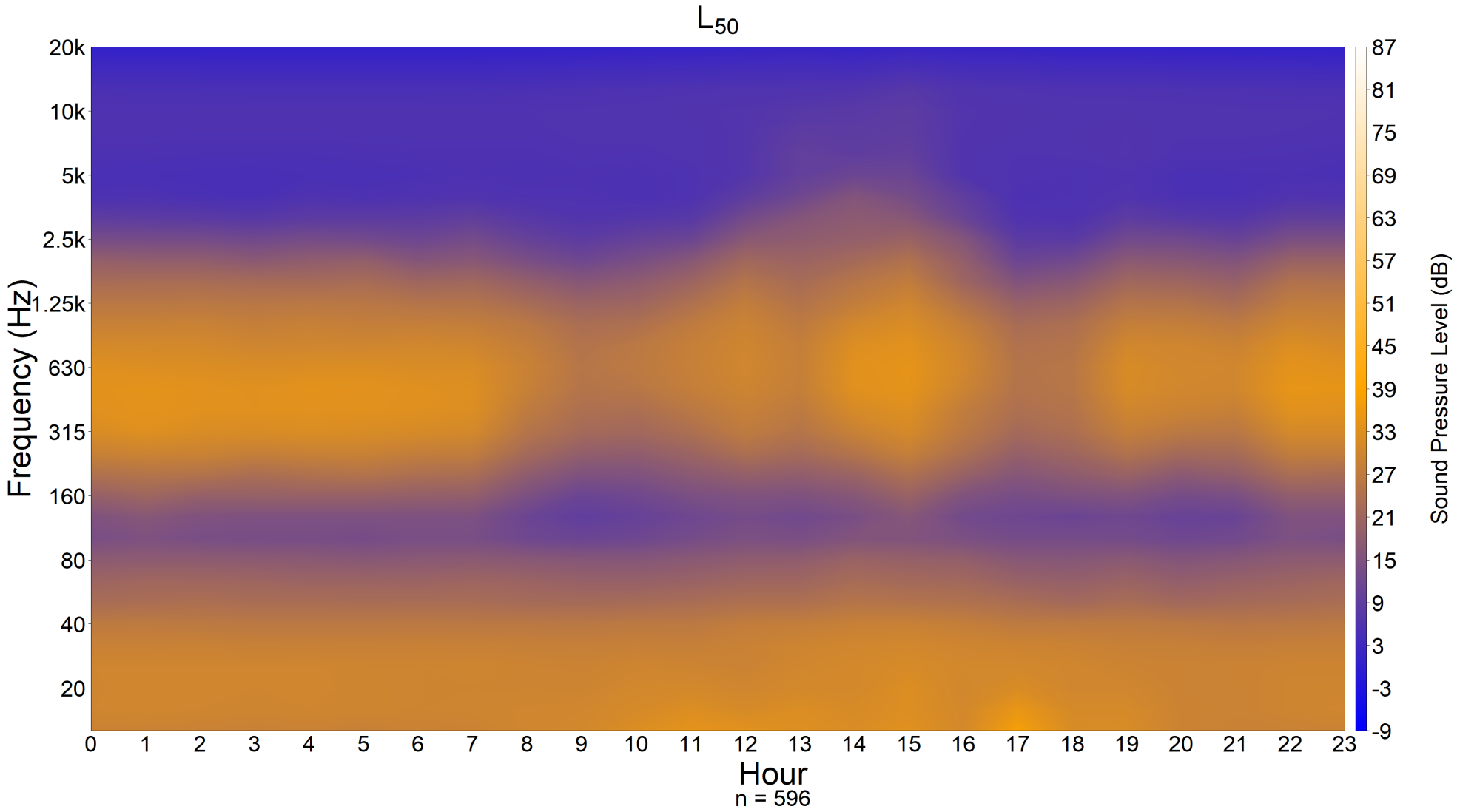


Figure 49. GAARNWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

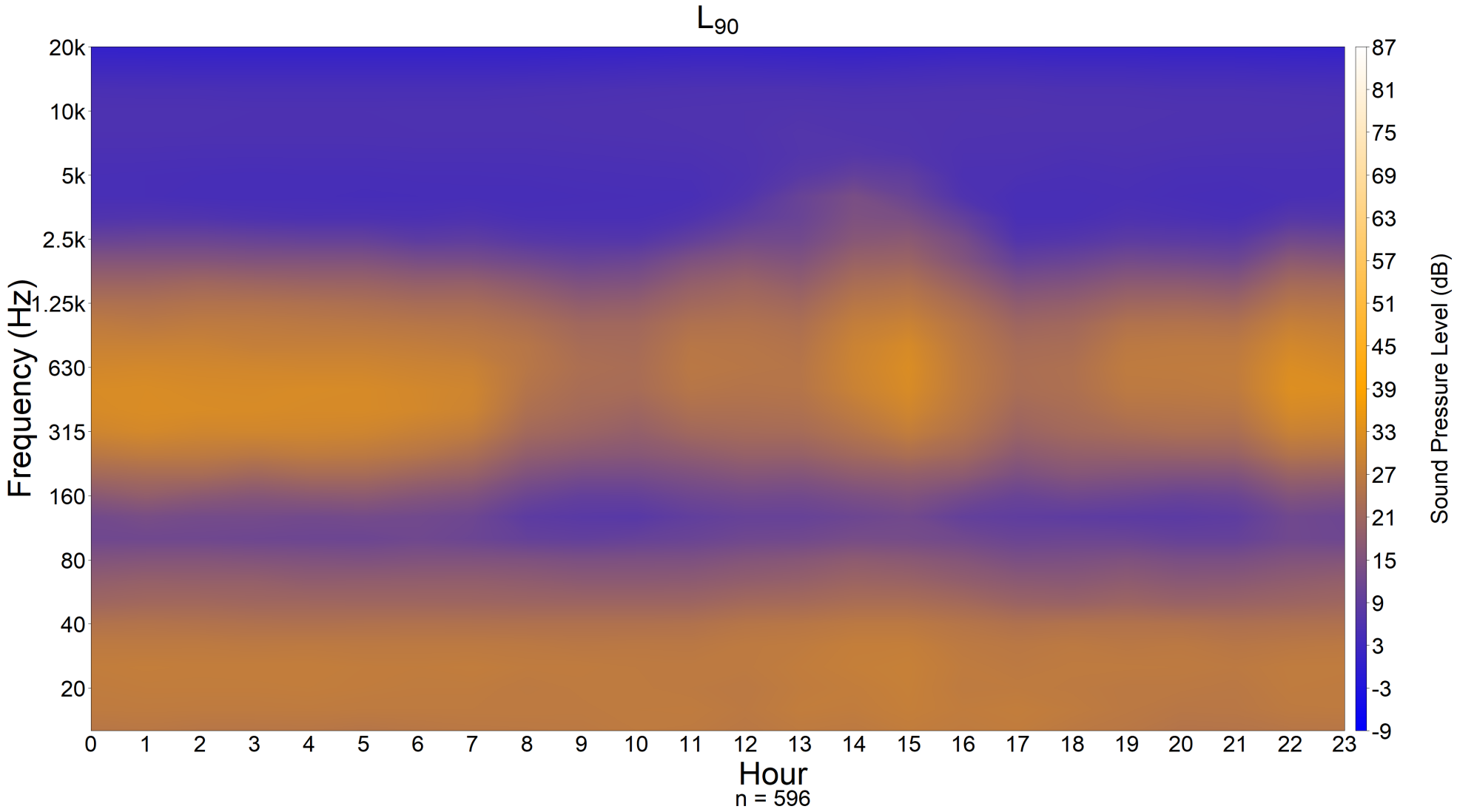


Figure 50. GAARNWAL: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

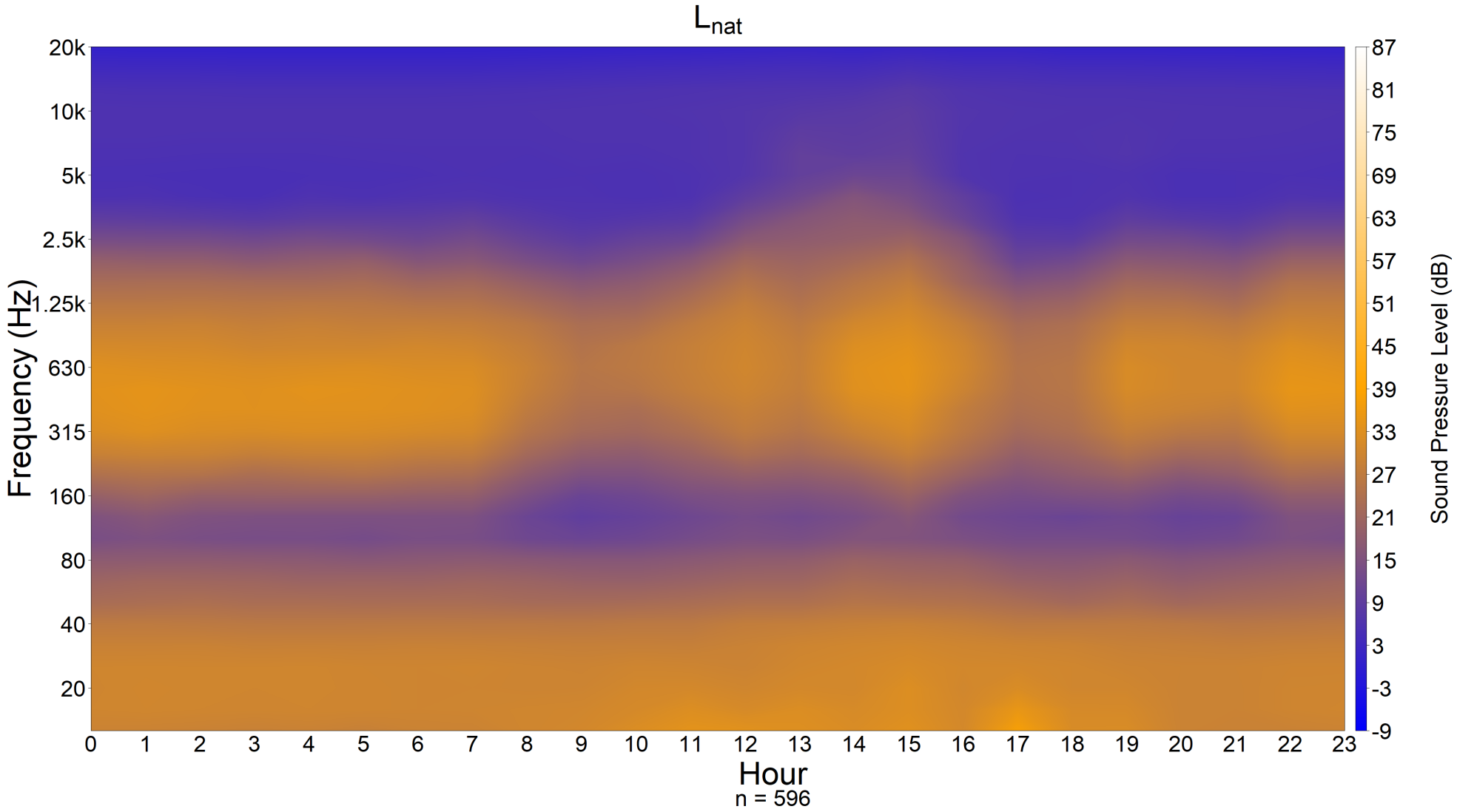


Figure 51. GAARNWAL: median and mean duration of noise−free intervals by hour of day, in Summer 2014.

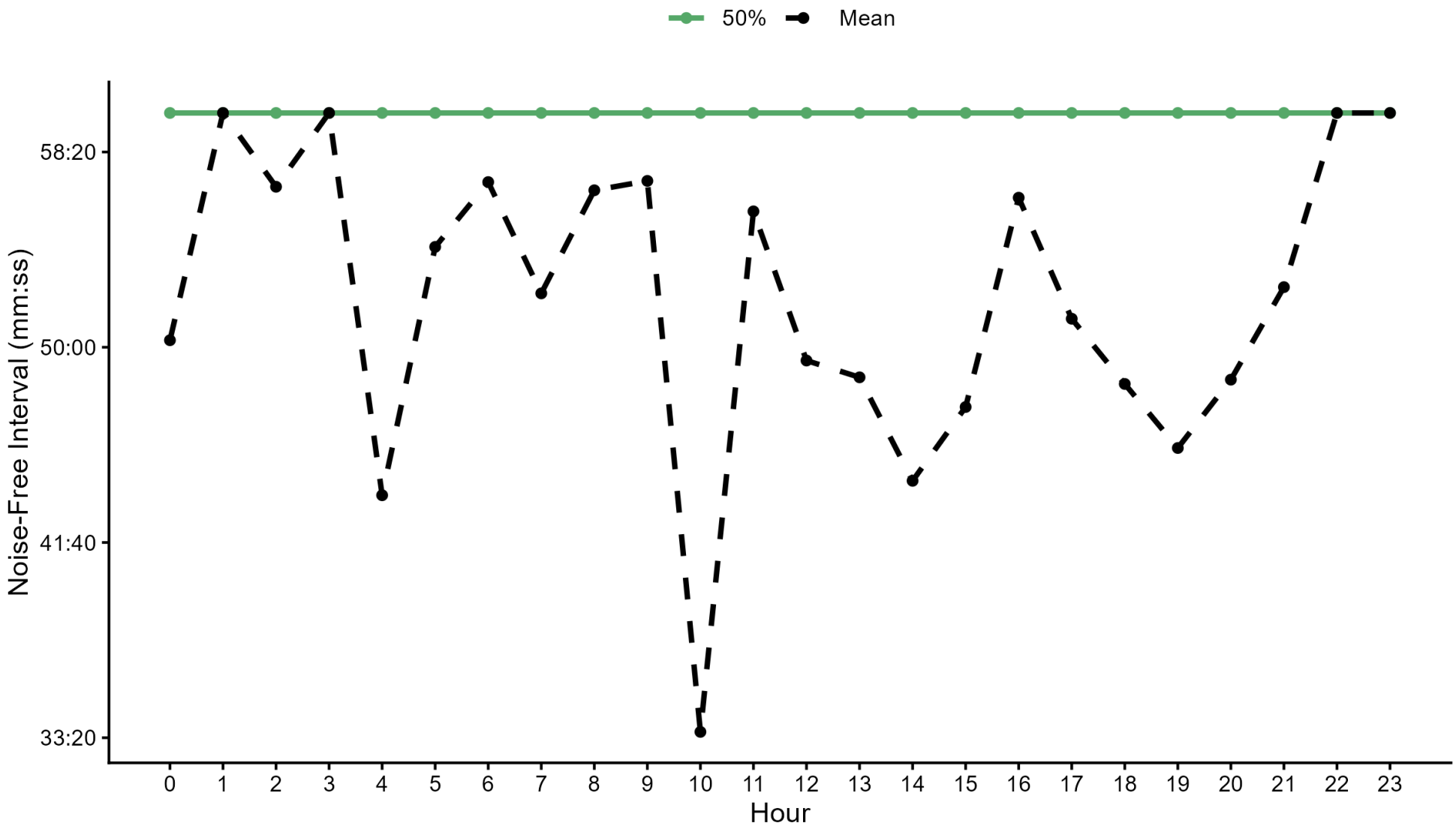


Figure 52. GAARNWAL: percent of time audible by sound source category (noise/natural), in Summer 2014.

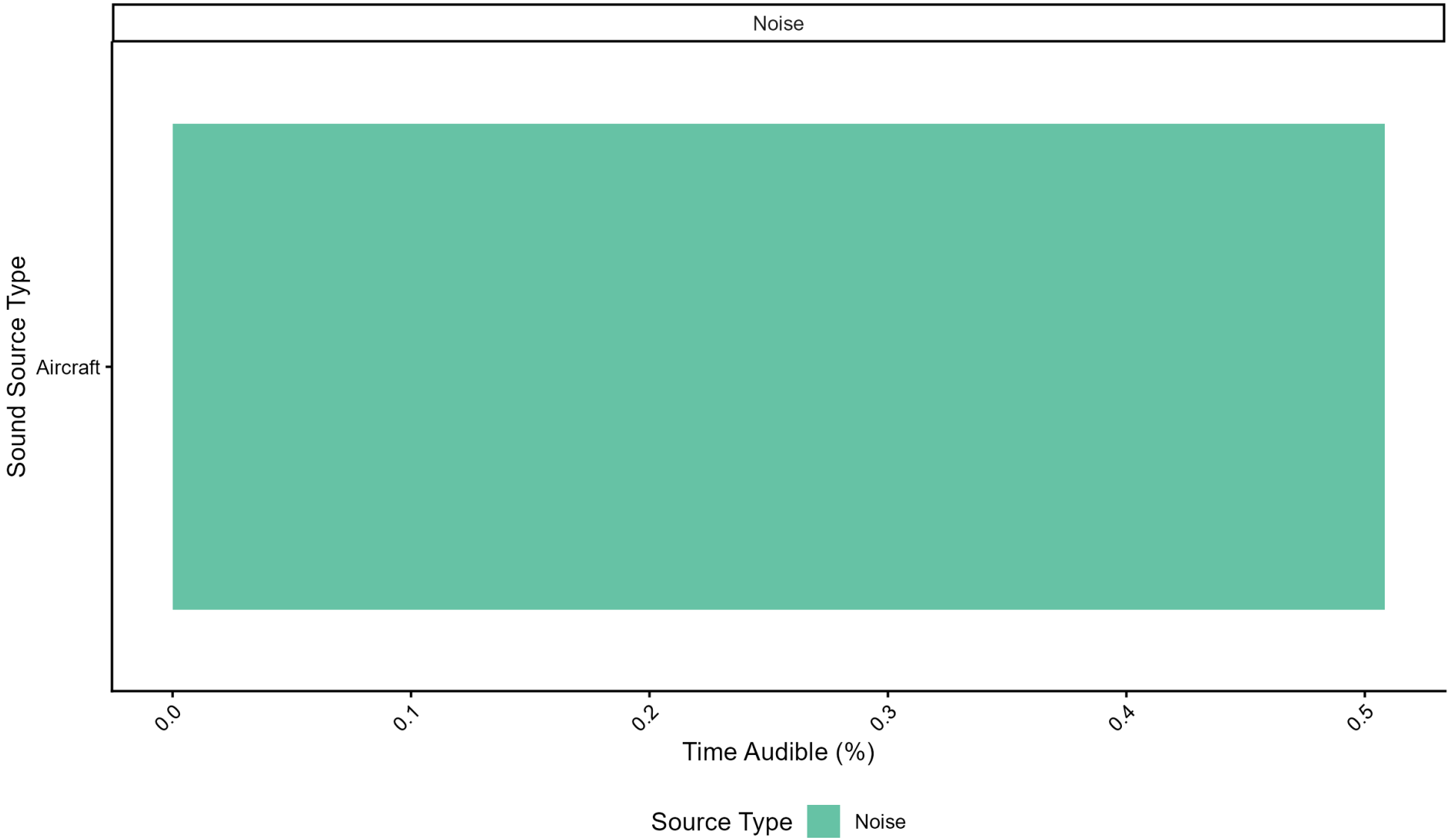


Figure 53. GAARNWAL: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2014.

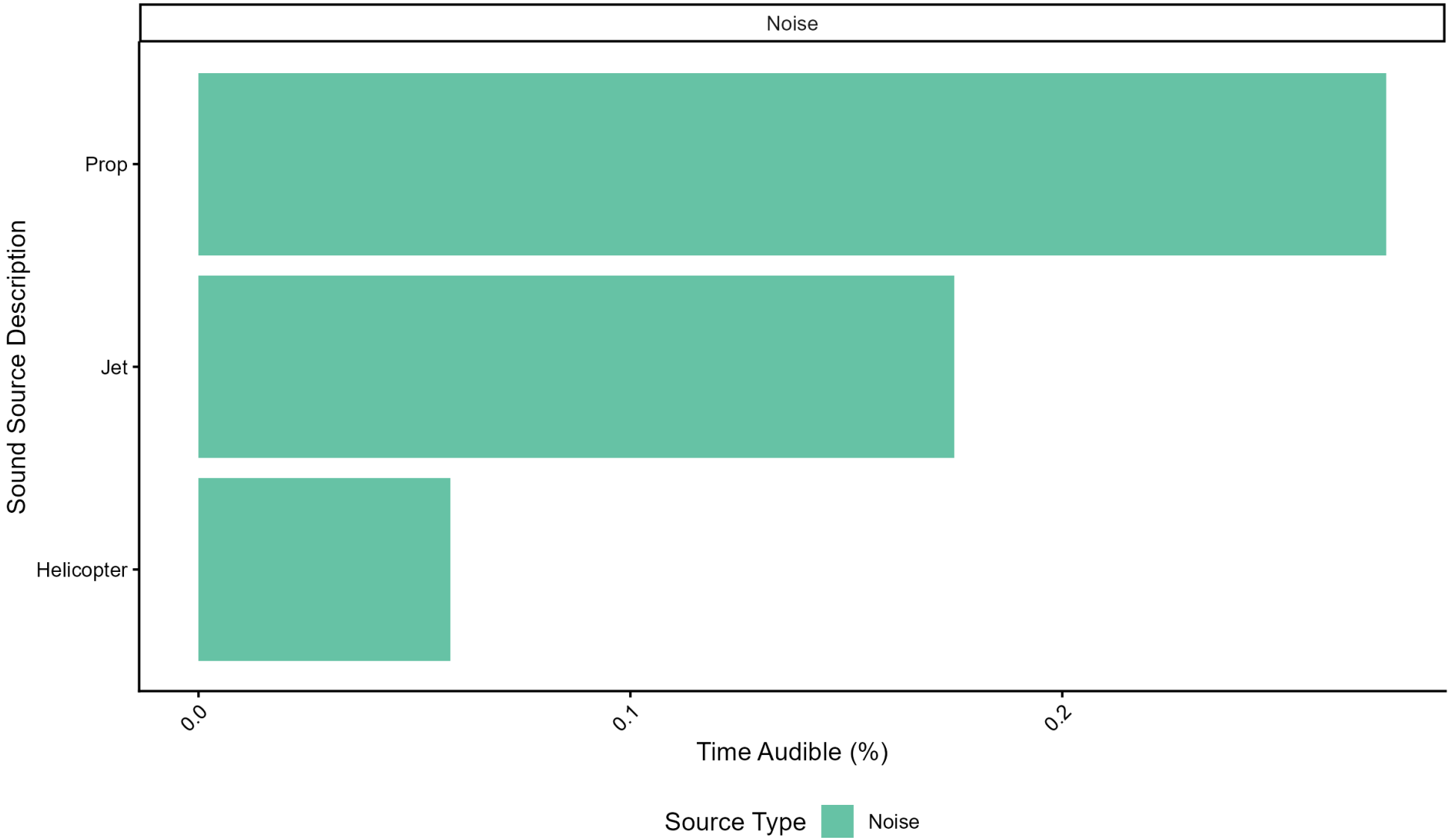


Figure 54. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARNWAL in Summer, 2014. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

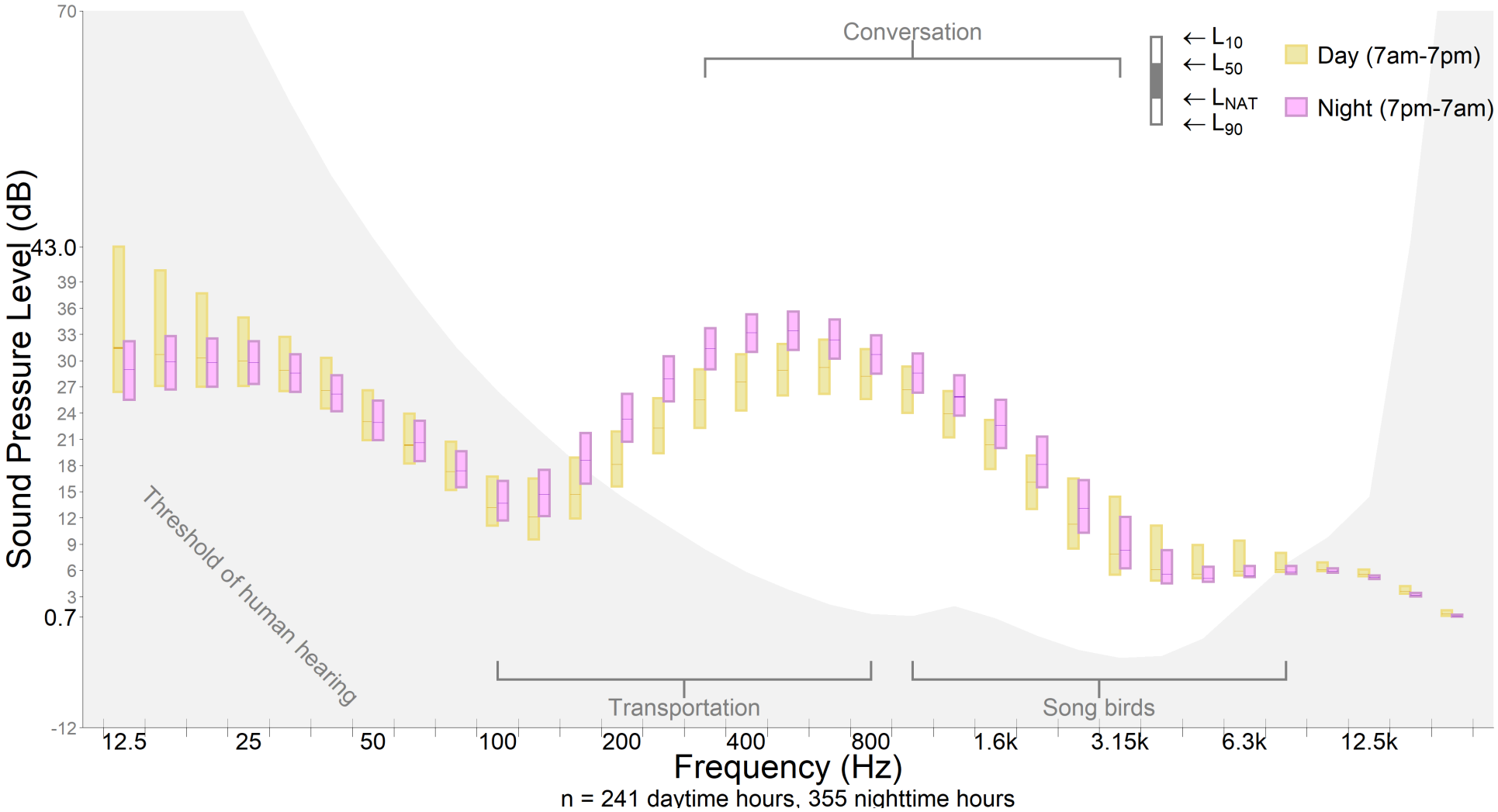


Figure 55. GAARSELB: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2014.

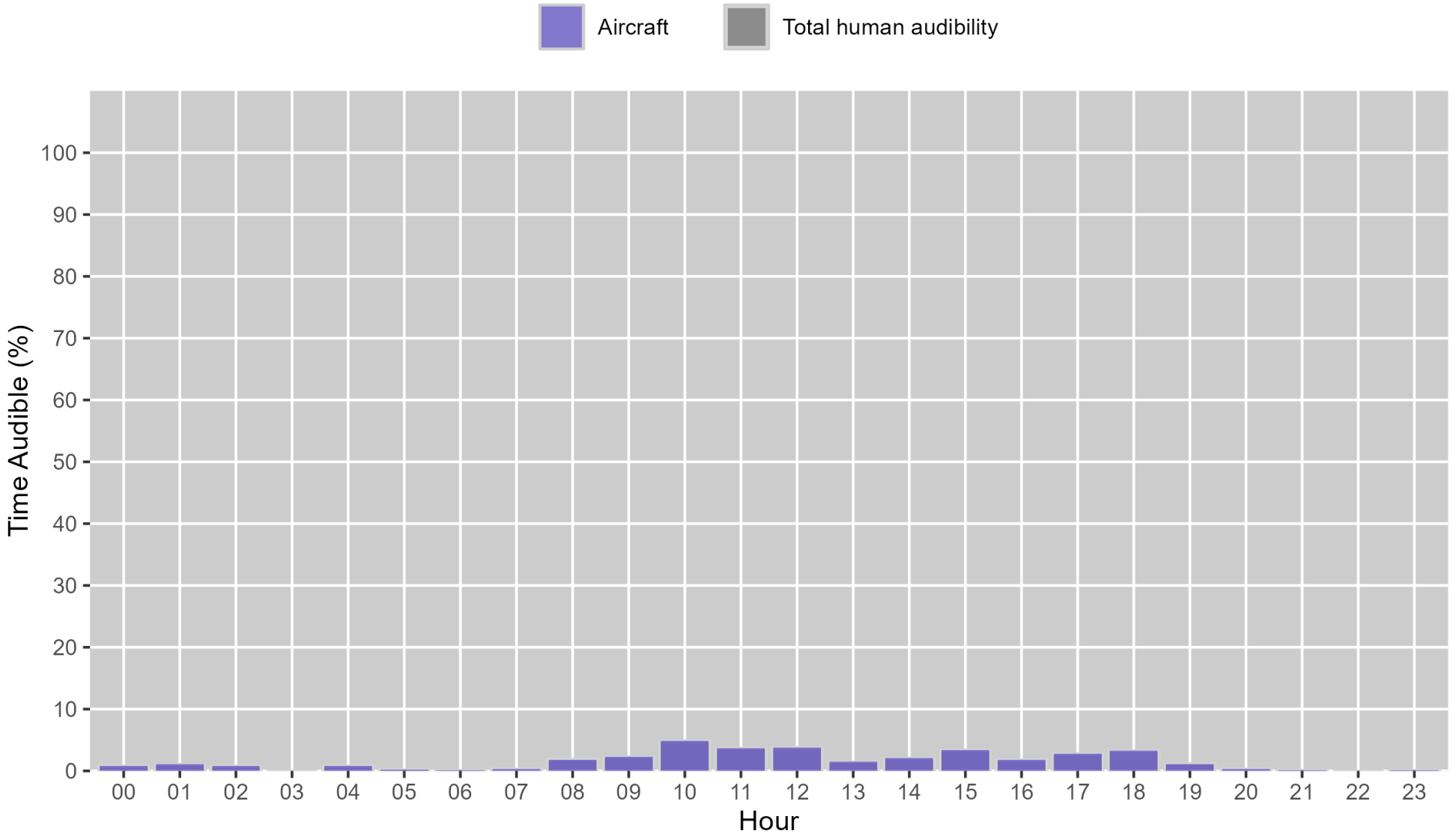


Figure 56. GAARSELB: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2014.

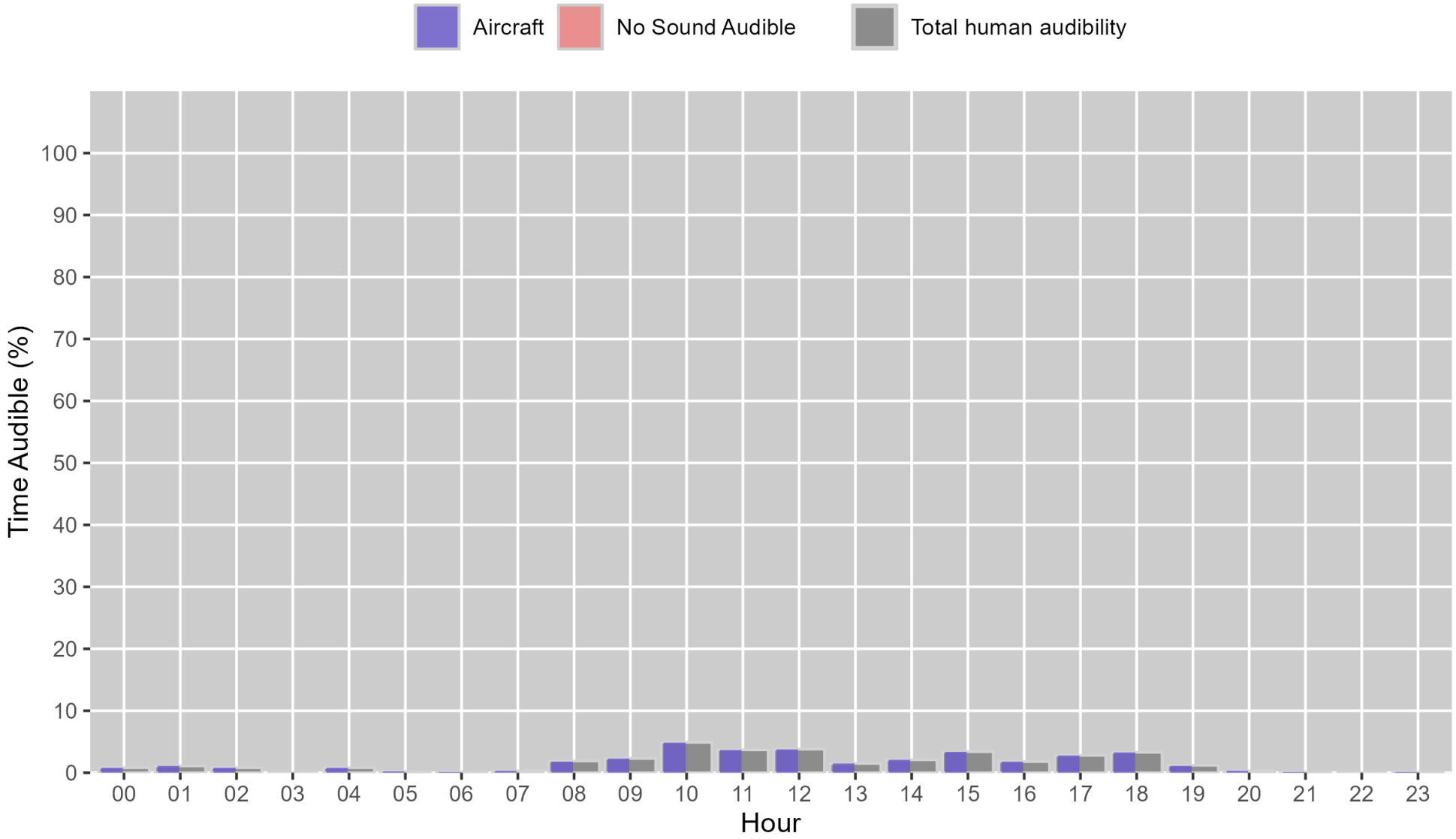


Figure 57. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARSELB in Summer, 2014, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

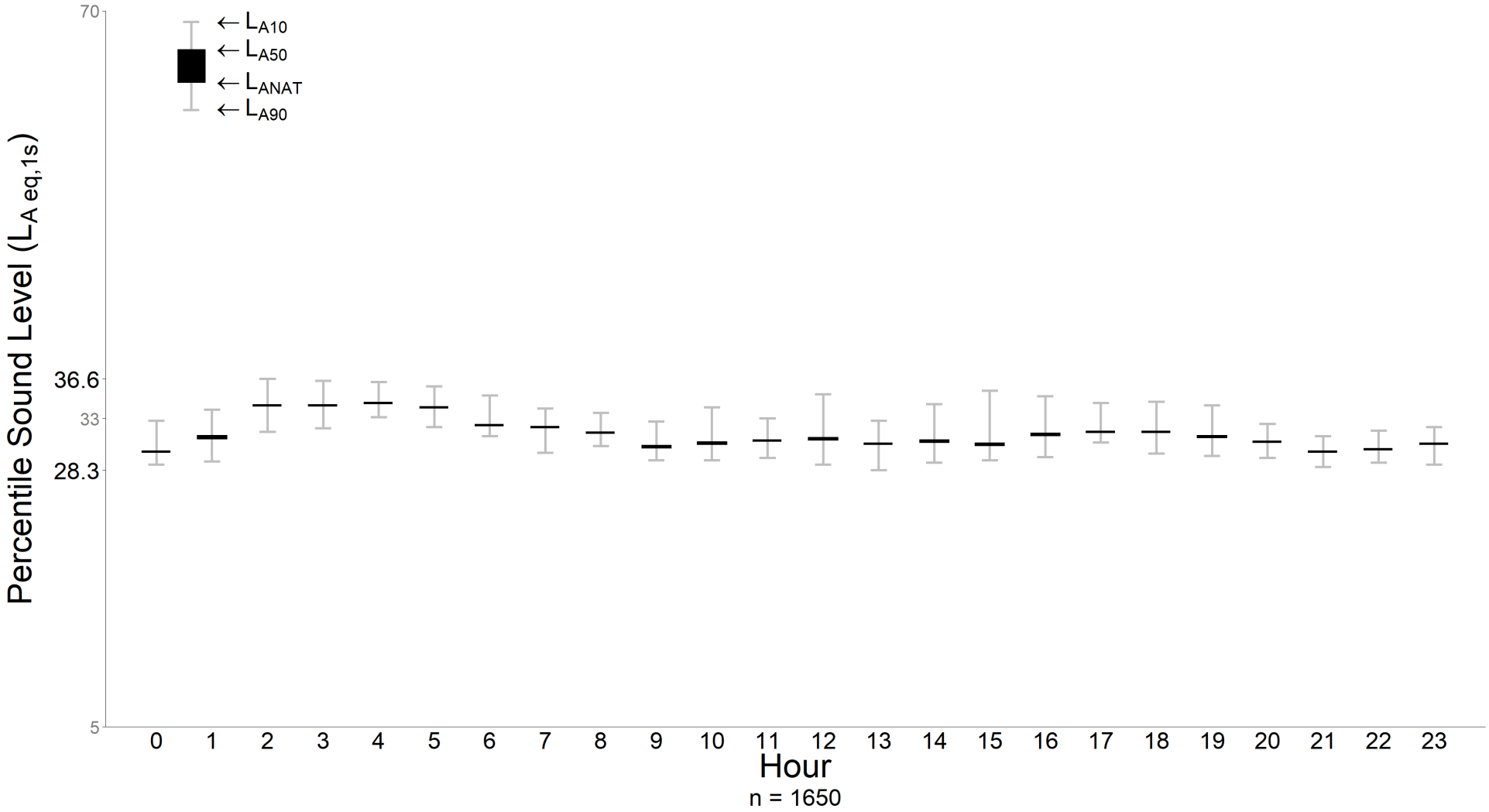


Figure 58. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARSELB in Summer, 2014.

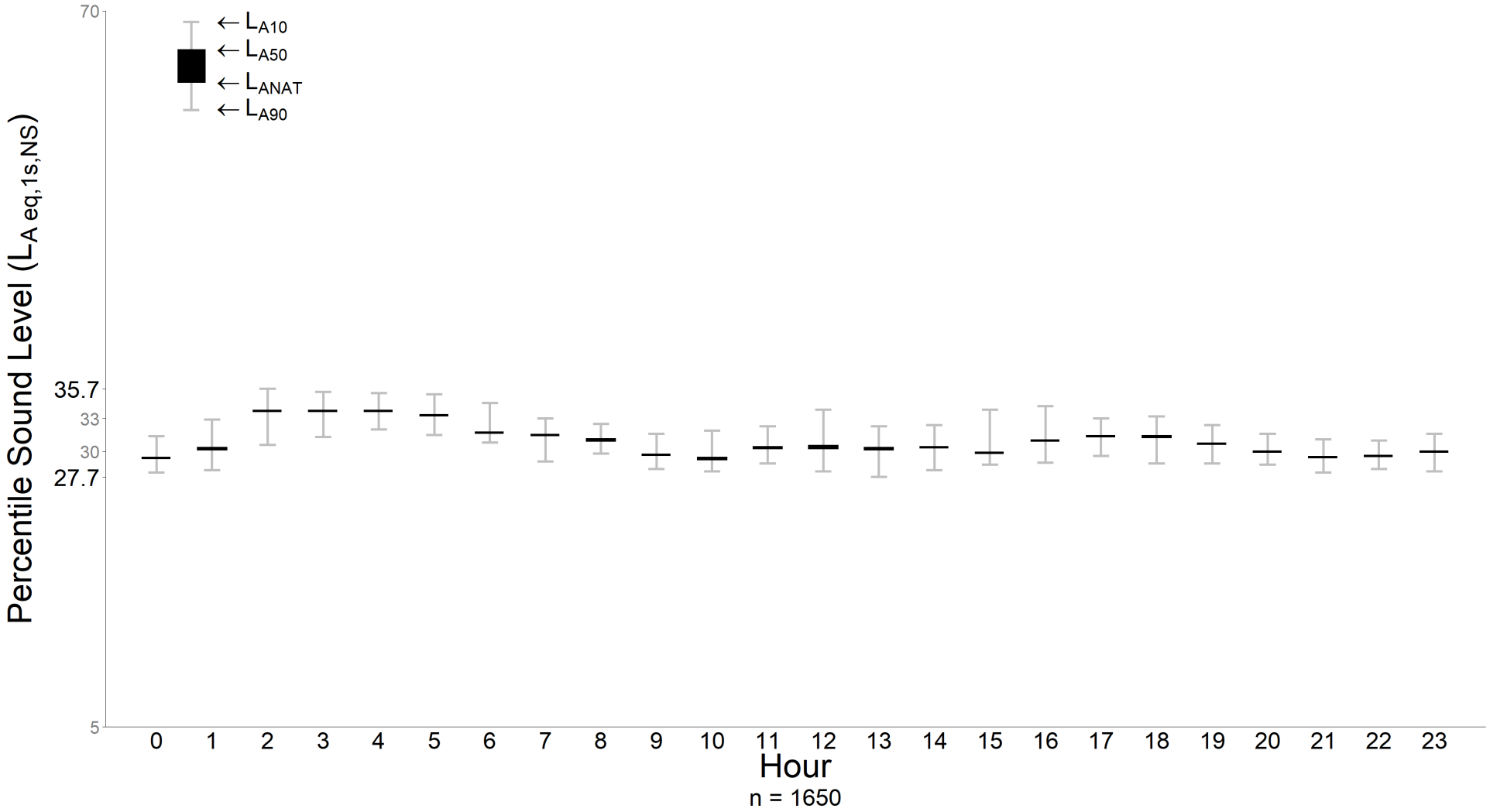


Figure 59. GAARSELB: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

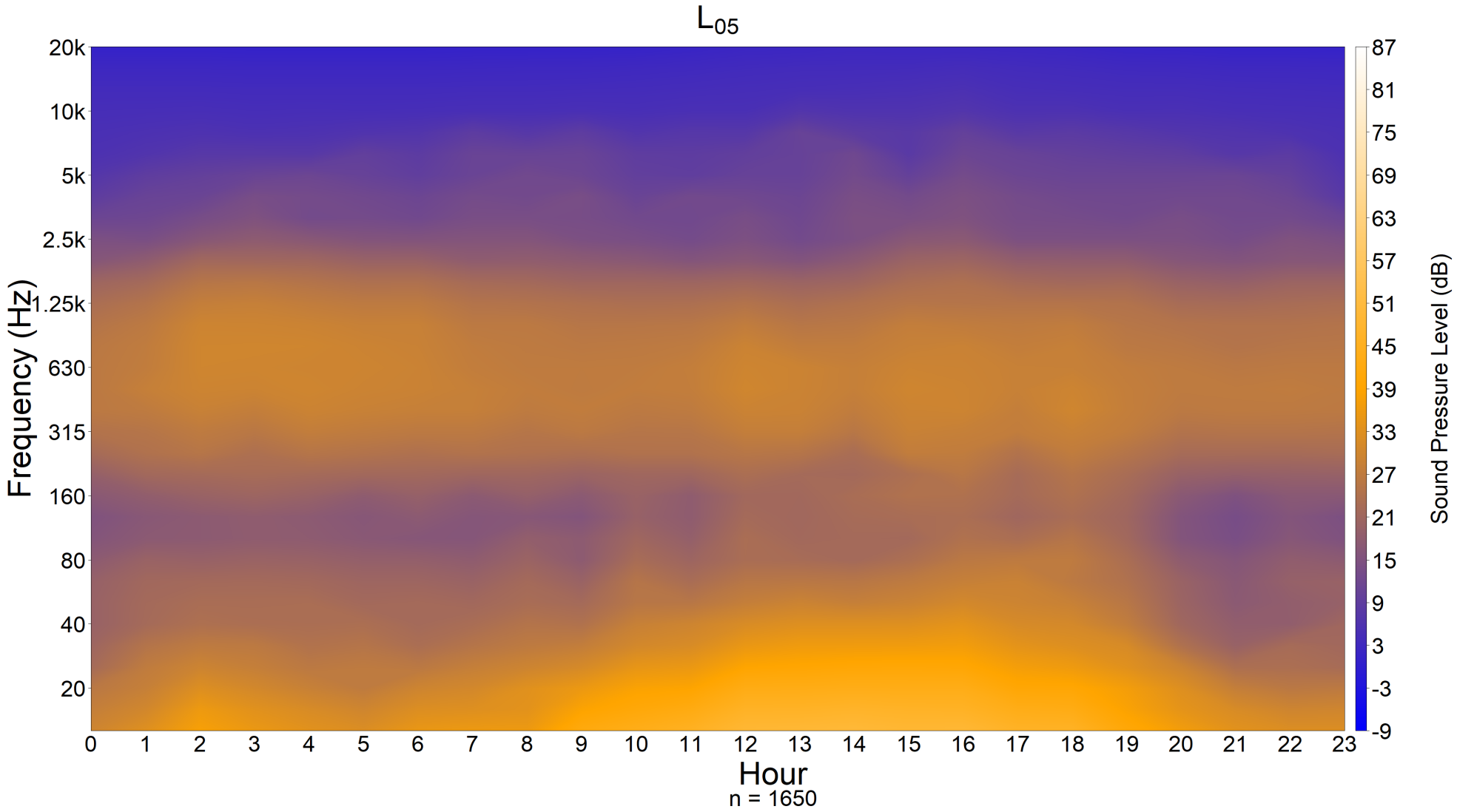


Figure 60. GAARSELB: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

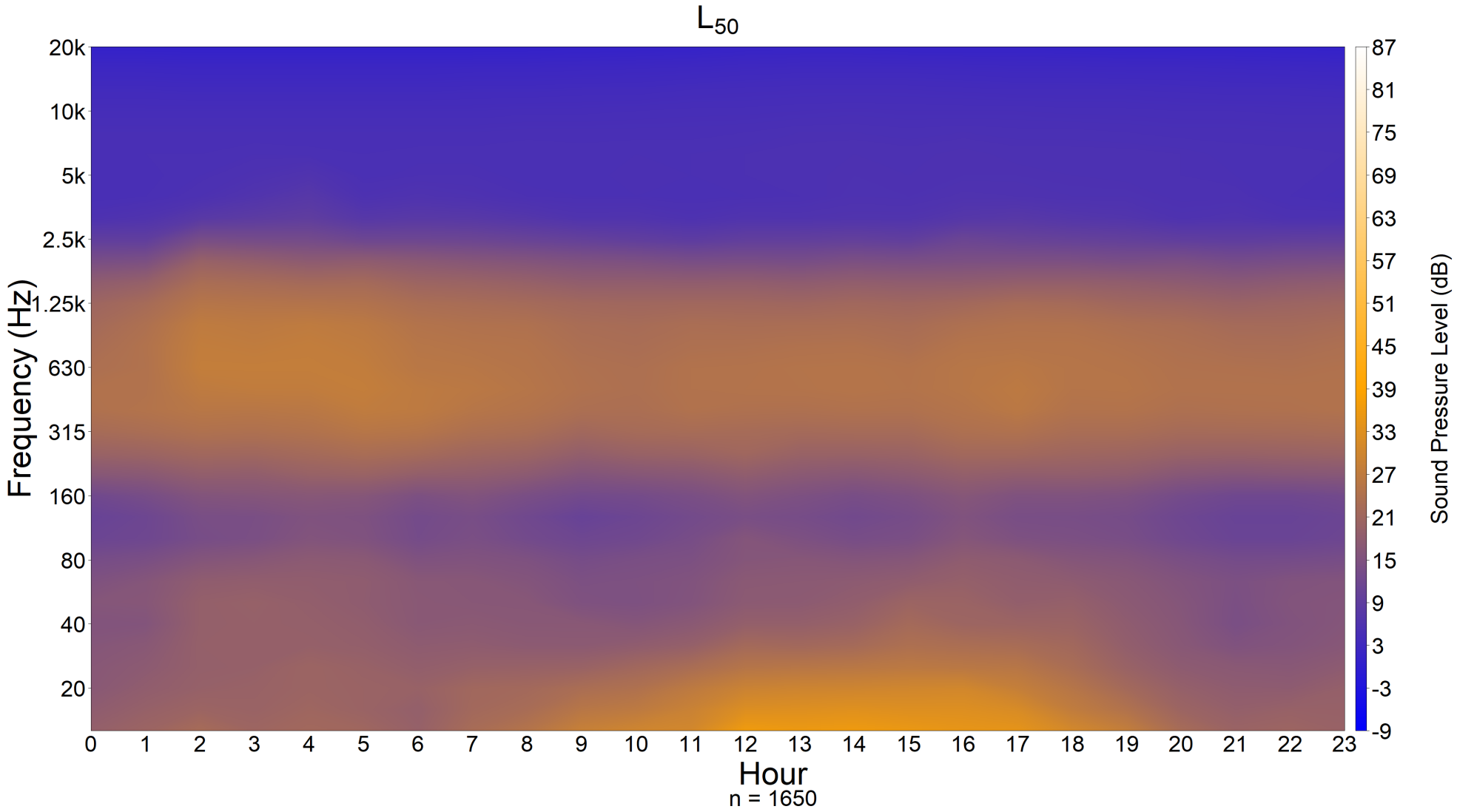


Figure 61. GAARSELB: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

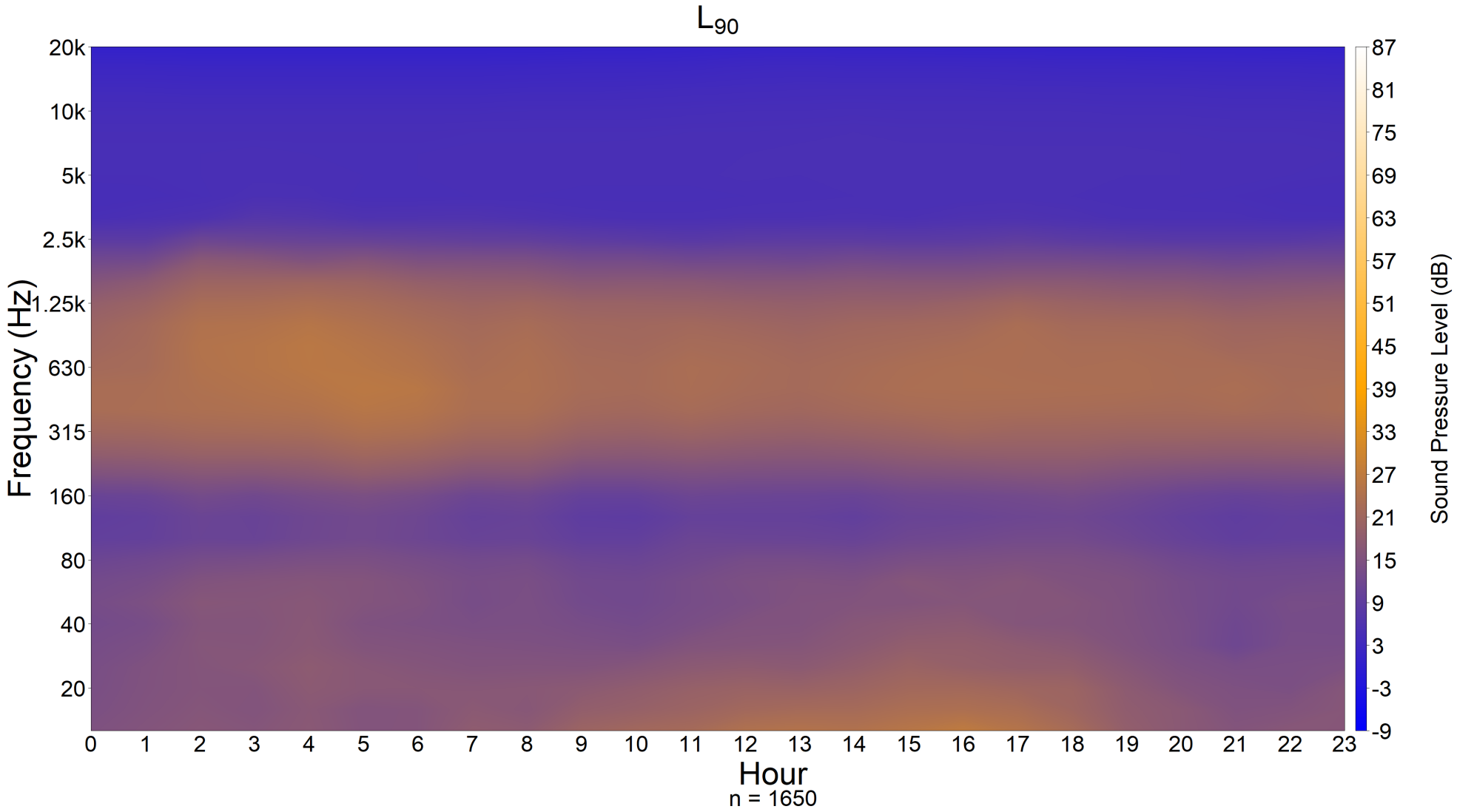


Figure 62. GAARSELB: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2014.

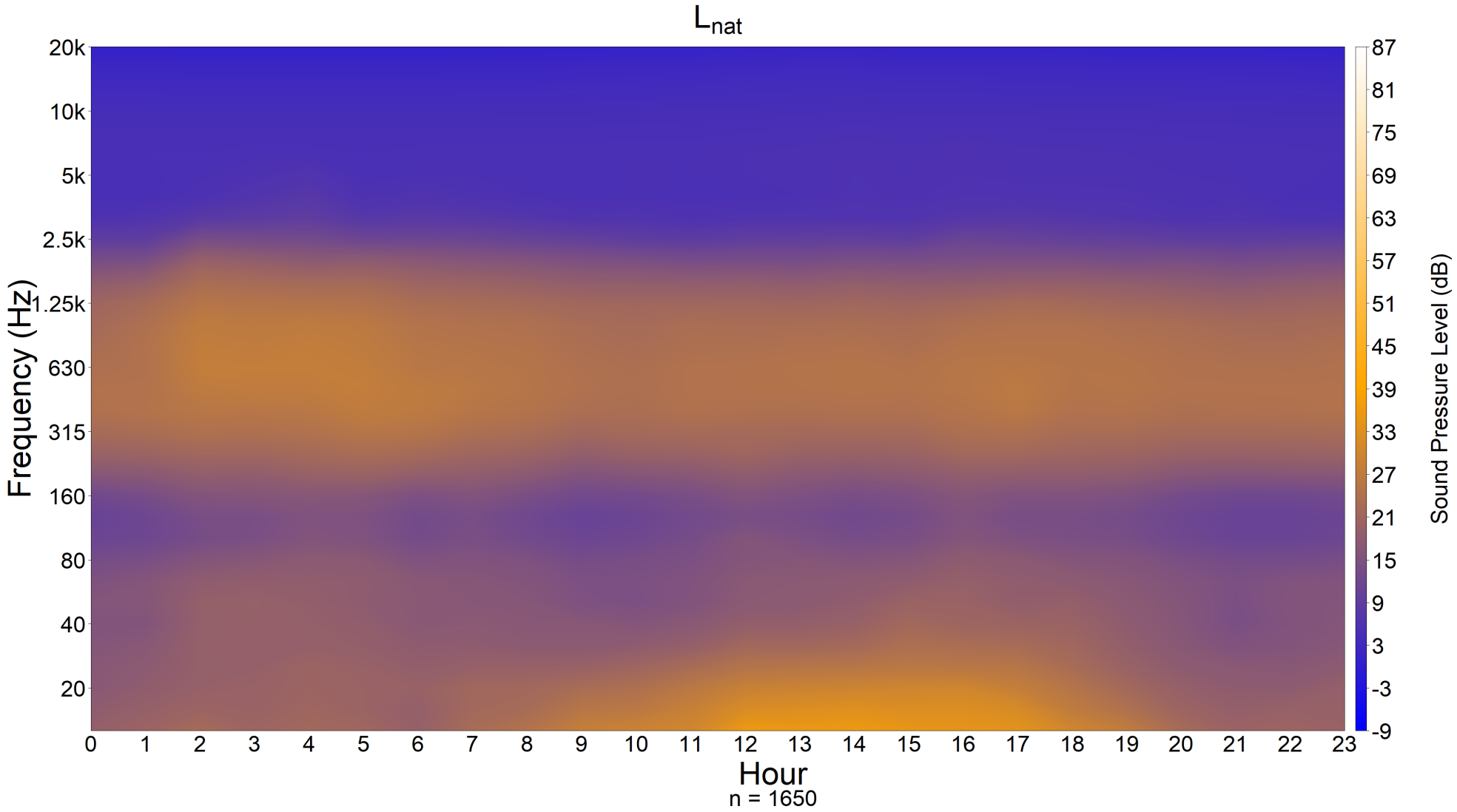


Figure 63. GAARSELB: median and mean duration of noise−free intervals by hour of day, in Summer 2014.

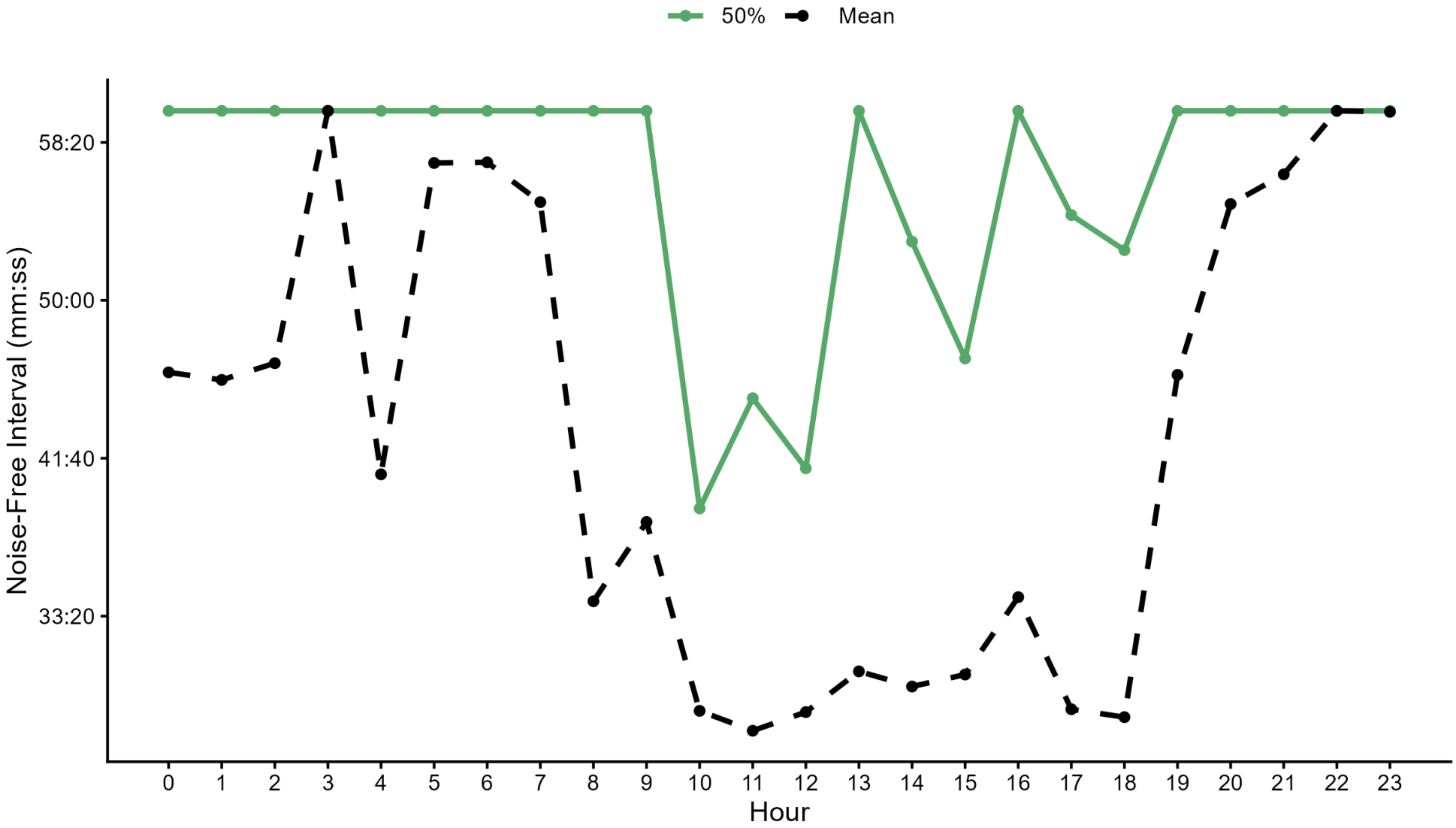


Figure 64. GAARSELB: percent of time audible by sound source category (noise/natural), in Summer 2014.

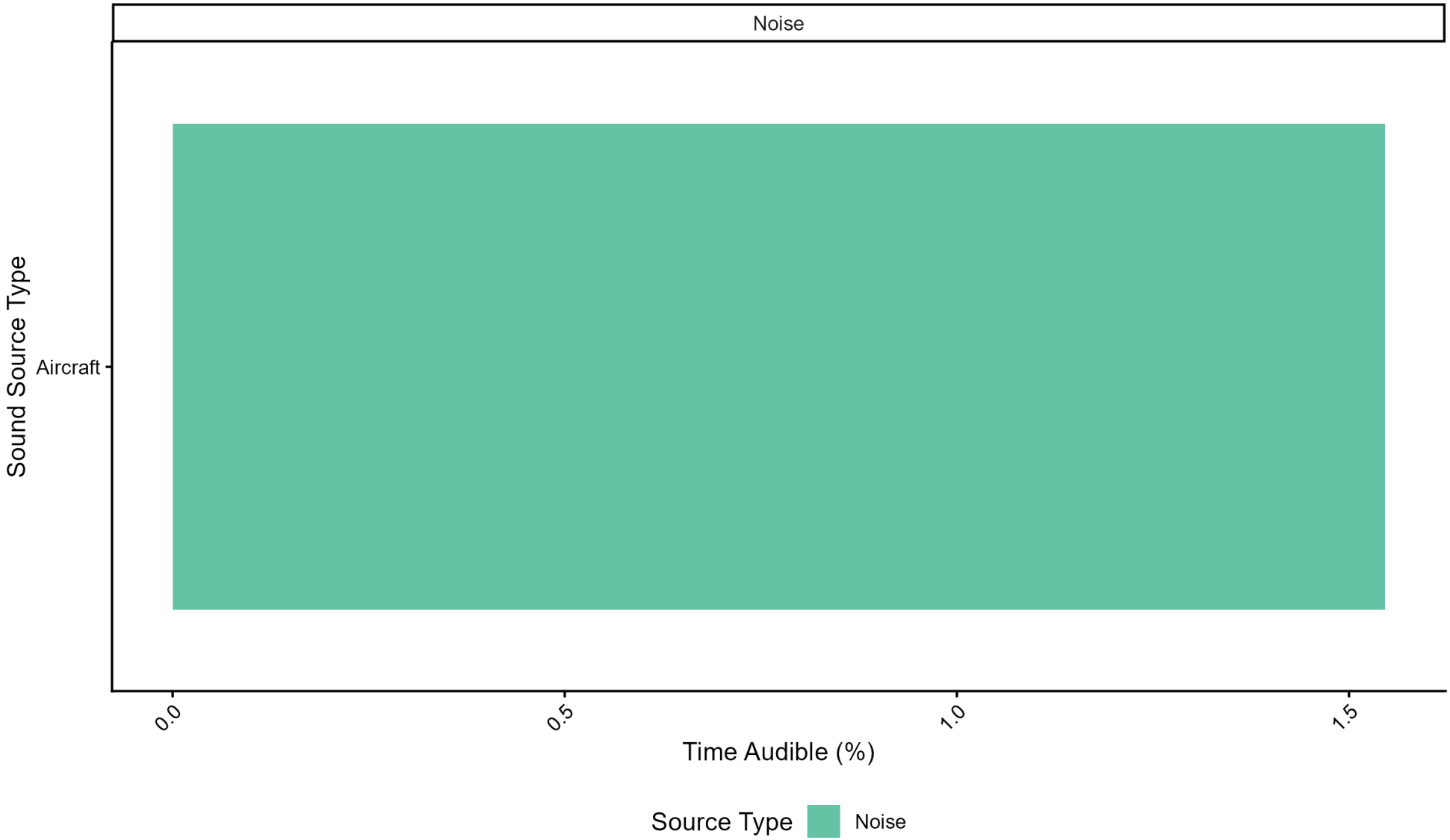


Figure 65. GAARSELB: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2014.

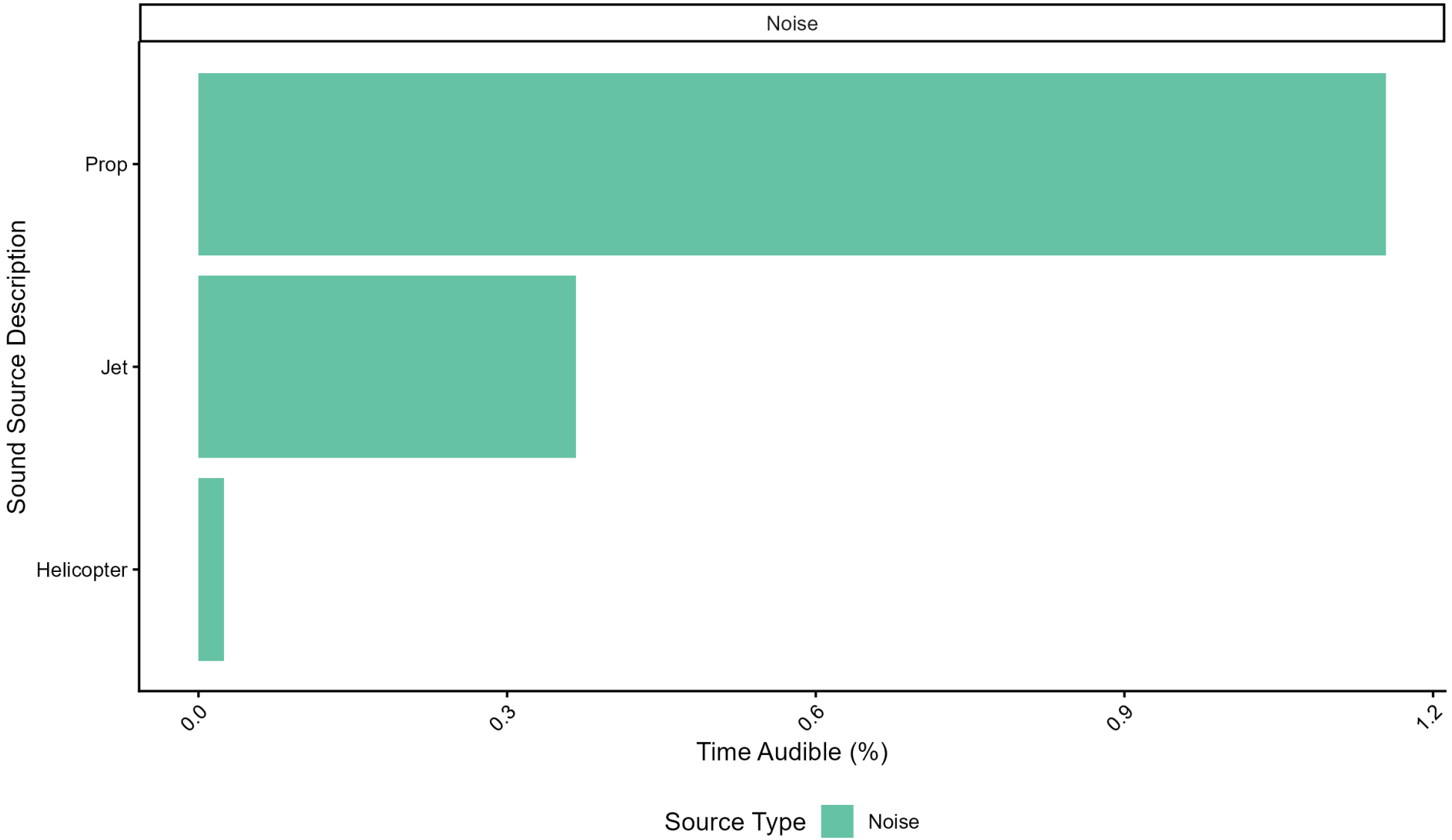


Figure 66. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARSELB in Summer, 2014. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

